

Book XI of Euclid's Elements

Formal Reconstruction — Technical Documentation

Jurek Róžański

Military University of Technology, Warsaw, Poland

CGJTEAMLAB

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Chapter 4

Proposition XI.4

4.1 Statement

Euclid XI.4 gives the basic criterion for perpendicularity to a plane: if a line through the common point of two intersecting lines of a plane is perpendicular to each of those two lines, then it is perpendicular to the plane itself.

The formal reconstruction is implemented in

```
CGJteamLab/Proposition11_4.lean
```

on top of the spatial infrastructure in

```
CGJteamLab/Hilbert3DAxioms.lean  
CGJteamLab/Hilbert3DInterface.lean
```

The Lean source is authoritative for theorem signatures, helper names, actual dependencies, and proof status.

4.2 Euclid XI.3 and the target notion

Book XI, Definition 3 says that a straight line is perpendicular to a plane when it makes right angles with every straight line which meets it and lies in that plane. The project encodes exactly this universal synthetic notion.

For two incidence lines:

```
def HilbertLinesPerpendicularAt  
  [H : HilbertIncidence Geo]  
  (l m : Geo.Line)  
  (O : Geo.Point) : Prop :=  
  H.OnLine O l /\  
  H.OnLine O m /\  
  exists A B : Geo.Point,  
    Ne A O /\  
    Ne B O /\
```

```

H.OnLine A l /\
H.OnLine B m /\
Not (PrimCollinear Geo A O B) /\
HilbertRightAngle Geo A O B

```

The explicit noncollinearity condition is important. It prevents a degenerate “self-perpendicular” witness and yields the reusable theorem

```

hilbert_linesPerpendicularAt_ne

```

showing that perpendicular lines are distinct.

For a line and a plane:

```

def HilbertLinePerpendicularPlaneAt
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l : Geo.Line)
  (pi : S.Plane)
  (O : Geo.Point) : Prop :=
H.OnLine O l /\
S.OnPlane O pi /\
forall m : Geo.Line,
  HilbertLineInPlane Geo m pi ->
  H.OnLine O m ->
  HilbertLinesPerpendicularAt Geo l m O

```

Thus the conclusion of XI.4 is not a metric abbreviation. It is literally the universal quantification required by Euclid XI.Def.3.

4.3 Classical configuration

Let m, n be two distinct straight lines of a plane π , meeting at O , and let l be a line through O perpendicular to both.

The desired conclusion is that l is perpendicular to every line $q \subset \pi$ through O .

4.4 Euclid’s classical proof

Euclid makes the argument constructive.

Take equal segments on the two given plane lines:

$$AE = EB = CE = ED,$$

where E is the common point of the two given lines. Draw an arbitrary straight line GEH through E in the plane, choose an arbitrary point F on the elevated perpendicular, and join the auxiliary segments.

The classical dependency chain is:

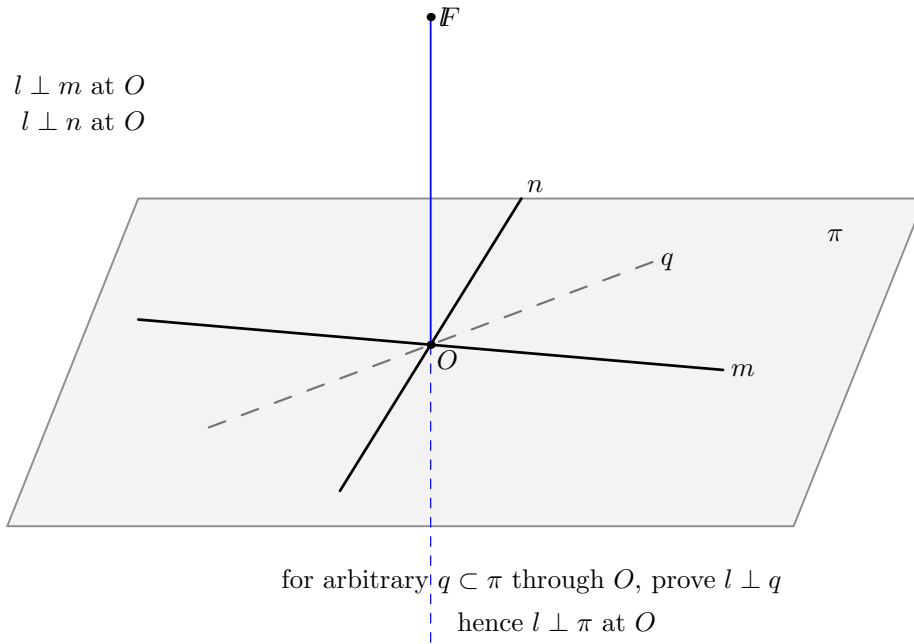


Figure 4.1: Classical configuration for Euclid XI.4. The ambient line l passes through O and is perpendicular at O to two distinct lines m, n lying in the plane π . The dashed line q represents an arbitrary line of π through O , which is the universal target of the conclusion.

1. Vertical angles at E are equal (I.15). Together with the four equal radial segments, I.4 gives

$$AD = CB$$

and a corresponding angle equality.

2. Applying I.26 to the planar transversal gives

$$GE = EH, \quad AG = BH.$$

3. Since FE is perpendicular to both original plane lines, I.4 applied to two right triangles gives

$$FA = FB, \quad FC = FD.$$

4. I.8 compares two spatial triangles and supplies the angle relation needed to transport the planar data to the arbitrary transversal.

5. A further I.4 gives

$$FG = FH.$$

6. Finally, from

$$GE = EH, \quad FE = FE, \quad FG = FH,$$

I.8 yields

$$\angle GEF = \angle HEF.$$

Since $G - E - H$, the adjacent equal angles are right angles.

7. The transversal GEH was arbitrary. Hence FE makes a right angle with every line through E in the plane. By XI.Def.3, $FE \perp \pi$.

This is a genuinely three-dimensional use of Book I congruence theory: several compared triangles are not assumed to lie in one common plane.

4.5 Formal theorem

The production theorem is:

```

theorem euclid_proposition_11_4
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (m n : PlaneLine Geo pi)
  (l : Geo.Line)
  (O : PlanePoint Geo pi)
  (hmn : Ne m n)
  (hPerpM :
    HilbertLinesPerpendicularAt Geo l m.1 O.1)
  (hPerpN :
    HilbertLinesPerpendicularAt Geo l n.1 O.1) :
  HilbertLinePerpendicularPlaneAt Geo l pi O.1

```

The theorem does not assume separately that $l \neq m$ or $l \neq n$. Those facts follow from the strengthened nondegenerate line–line perpendicularity predicate.

4.6 Spatial architecture

4.6.1 No ambient planar type-class instance

The ambient geometry is three-dimensional. The proof deliberately does *not* install a global instance

```
HilbertCongruence Geo
```

or

```
HilbertOrder Geo
```

on the whole space. Planar reasoning is performed only in explicit plane slices

$$\text{PlaneGeo}(\pi).$$

The corresponding points and lines are subtypes:

```

PlanePoint Geo pi
PlaneLine   Geo pi

```

and the interface provides bridges between induced planar facts and ambient facts, for example

```
planeGeo_congruent
planeGeo_angleCongruent_iff_ambient
planeGeo_sameRay_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_linesPerpendicularAt_iff_ambient
```

This separation is one of the central architectural features of the proof.

4.6.2 Spatial congruence

Euclid freely applies I.4 and I.8 to triangles in different planes. The project reconstructs this through the spatial congruence layer rather than by coordinates or by an ambient Euclidean metric.

The general spatial infrastructure includes:

```
hilbert_space_congruent_reflexive
hilbert_space_congruent_symmetry
hilbert_space_sas_remaining_angles
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

The target triangle may be placed in its own explicit plane and compared with an ambient source triangle. This is the formal counterpart of Euclid’s unqualified use of plane triangle congruence inside solid geometry.

4.7 The balanced cross

The first construction is the formal analogue of Euclid’s

$$AE = EB = CE = ED.$$

The helper

```
planeGeo_opposite_points_on_line_congruent_to
```

constructs opposite equal points on one line. Applying it twice yields

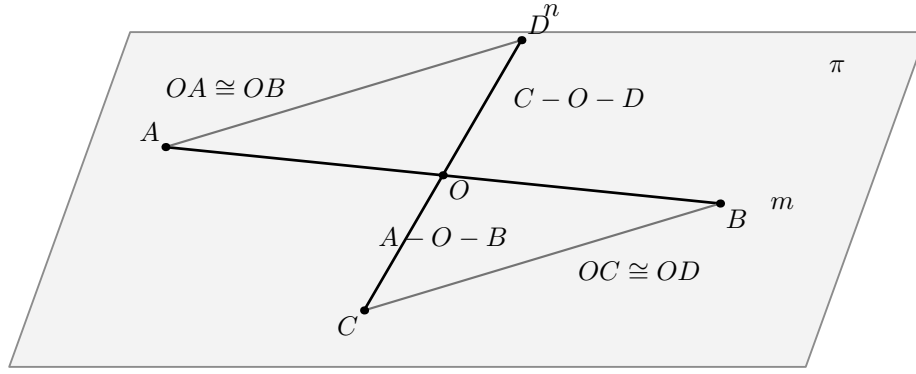
```
hilbert_XI4_balanced_cross_exists
```

and points A, B, C, D satisfying

$$A - O - B, \quad C - O - D$$

and

$$OA \cong OR, \quad OB \cong OR, \quad OC \cong OR, \quad OD \cong OR.$$



all objects in this figure lie in π

Figure 4.2: Balanced cross in π . Points A, B lie on m , points C, D lie on n , with $A - O - B$, $C - O - D$, $OA \cong OB$, and $OC \cong OD$. The connector segments AD and BC are the first pair used by Euclid's congruence argument.

The first triangle comparison is isolated by

```
planeGeo_XI4_cross_noncollinear
hilbert_XI4_balanced_cross_first_SAS
```

and proves the ambient consequences

$$AD \cong BC, \quad \angle OAD \cong \angle OBC.$$

This is the formal Hilbert version of Euclid's first use of I.4.

4.8 Fixed-transversal prototype

Before the proof was generalized to every transversal, a fixed connector configuration was developed. Its role is mathematically transparent: given

$$A - G - D, \quad B - J - C, \quad G - O - J,$$

the balanced cross implies

$$OG \cong OJ, \quad AG \cong BJ.$$

The relevant planar ingredients are

```
planeGeo_XI4_transversal_noncollinear
hilbert_XI4_transversal_ASA
```

corresponding to Euclid's I.26 step.

The spatial point $F \in l$, together with the two original perpendicularities, then gives

$$AF \cong BF, \quad CF \cong DF.$$

The proof packages this in

```

hilbert_XI4_spatial_perpendicular_bisector_equidistant
hilbert_XI4_spatial_opposite_pairs_equidistant

```

The spatial SSS/SAS chain is then:

```

hilbert_XI4_spatial_SSS_angle
hilbert_XI4_space_angle_sameRay_first_congruent
hilbert_XI4_spatial_SSS_angle_normalized
hilbert_XI4_spatial_SAS_transversal_side
hilbert_XI4_spatial_transversal_F_equidistant
hilbert_XI4_second_spatial_SSS_angle
hilbert_XI4_right_angle_of_adjacent_equal

```

Its mathematical content is

$$\begin{aligned}
& AD \cong BC, \quad AF \cong BF, \quad CF \cong DF \\
& \quad \Downarrow \\
& \quad \angle DAF \cong \angle CBF \\
& \quad \Downarrow \\
& \quad A - G - D, \quad B - J - C \\
& \quad \Downarrow \\
& \quad \angle GAF \cong \angle JBF \\
& \quad \Downarrow \\
& \quad AG \cong BJ, \quad AF \cong BF \\
& \quad \Downarrow \\
& \quad GF \cong JF \\
& \quad \Downarrow \\
& \quad OG \cong OJ, \quad OF \cong OF, \quad GF \cong JF \\
& \quad \Downarrow \\
& \quad \angle GOF \cong \angle JOF \\
& \quad \Downarrow \\
& \quad G - O - J \\
& \quad \Downarrow \\
& \quad \text{RightAngle}(G, O, F).
\end{aligned}$$

The last implication is exactly Euclid's definition of a right angle as two adjacent equal angles formed by a straight line.

4.9 The arbitrary transversal: the real formal difficulty

The public conclusion quantifies over *every* line of π through O . A conventional drawing suggests that such a transversal meets the two connector segments AD and BC , but that is false in general. This was the key positional issue in the formal proof.

The correct argument uses two possible connector families:

$$AD/BC \quad \text{or} \quad BD/AC.$$

For a transversal $q \neq m, n$, apply Pasch to the triangle ABD . The line q enters through AB at O , hence it exits either through AD or through BD .

This is isolated as

```
hilbert_XI4_transversal_pasch_first_exit
```

Next,

```
hilbert_XI4_opposite_point_on_transversal
```

constructs the point J opposite G on q with

$$G - O - J, \quad OG \cong OJ.$$

Two neutral landing lemmas then prove:

```
hilbert_XI4_first_exit_opposite_lands_on_BC
hilbert_XI4_second_exit_opposite_lands_on_AC
```

so that the opposite point lands on the correct paired connector. The combined package is

```
hilbert_XI4_arbitrary_transversal_connector_package
```

and returns

$$\begin{aligned} G, J \in q, \quad G - O - J, \quad OG \cong OJ, \\ \text{and either} \\ A - G - D, \quad B - J - C, \\ \text{or} \\ B - G - D, \quad A - J - C. \end{aligned}$$

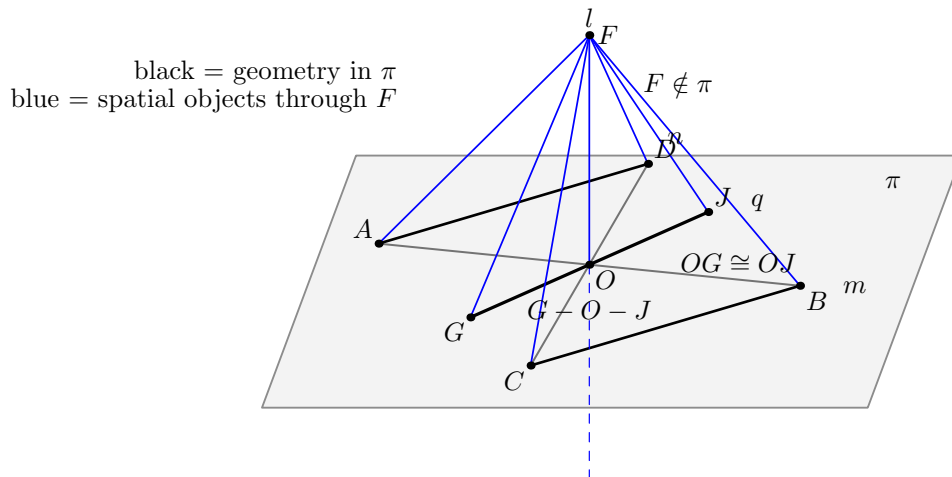


Figure 4.3: Arbitrary-transversal stage. For $q \subset \pi$ through O , Pasch selects one of the two connector families. The opposite point J is constructed with $G - O - J$ and $OG \cong OJ$. The spatial SSS/SAS chain then proves that $\angle GOF$ is right.

This entire stage uses incidence, order, Pasch, and congruence. It does not use the Euclidean parallel axiom.

4.10 Right angle on every transversal

With the connector package available, both branches reduce to the same spatial SSS/SAS scheme. The theorem

```
hilbert_XI4_arbitrary_transversal_right_angle
```

states, in substance:

$$\begin{aligned} q \subset \pi, \quad O \in q, \quad q \neq m, n, \\ F \notin \pi, \quad AF \cong BF, \quad CF \cong DF \end{aligned}$$

imply the existence of $G \in q$, $G \neq O$, such that

$$\text{RightAngle}(G, O, F).$$

The two exceptional cases $q = m$ and $q = n$ are discharged directly from the hypotheses of XI.4.

4.11 Why the point F is outside the plane

The core proof chooses an arbitrary nonvertex point F on l . To use spatial noncollinearity cleanly, it proves

$$F \notin \pi.$$

This is the theorem

```
hilbert_XI4_perpendicular_point_off_plane
```

The proof is neutral and conceptually important.

Assume $F \in \pi$. Then the line l , determined by O and F , is itself a line of π . Transfer the two ambient perpendicularities to **PlaneGeo** **pi**. Choose nonvertex points $A \in m$ and $C \in n$. Then both

$$\angle FOA \quad \text{and} \quad \angle FOC$$

are right.

The helper

```
hilbert_XI4_two_right_angles_same_first_arm_collinear
```

uses Hilbert III.4 uniqueness to prove that A, O, C are collinear. Line uniqueness then forces $m = n$, contradicting the hypothesis.

No parallel postulate is used.

4.12 Closing the universal line–plane statement

The core theorem

```
hilbert_XI4_line_perpendicular_plane_core
```

does the following:

1. choose $F \neq O$ on l ;
2. construct the balanced cross A, B, C, D in π ;
3. prove $F \notin \pi$;
4. prove $AF \cong BF$ and $CF \cong DF$;
5. let q be an arbitrary line of π through O ;
6. if $q = m$ or $q = n$, use the hypotheses;
7. otherwise invoke the arbitrary-transversal theorem and obtain a point $G \in q$ such that $\angle GOF$ is right;
8. construct the witness required by `HilbertLinesPerpendicularAt Geo 1 q 0`.

There is one subtle final representation step. The right angle obtained from the transversal theorem is oriented as

$$\angle GOF.$$

The line-line perpendicularity witness is more convenient with the order

$$\angle FOG.$$

The proof does not use an ambient planar congruence instance to swap the arms. Since l and q intersect at O , they determine an explicit plane ρ . Inside `PlaneGeo rho` the right angle is swapped using the ordinary planar congruence API, and the result is transported back to the ambient geometry.

This preserves the 3D architecture: all planar congruence reasoning occurs inside an explicit plane slice.

4.13 Complete formal DAG

The proof architecture can be summarized as follows.

```

two distinct plane lines m,n through O
+ l perpendicular to m and n at O
-> balanced cross A-O-B, C-O-D
-> AD = BC and angle OAD = angle OBC
-> choose F != O on l
-> F notin pi
-> AF = BF, CF = DF
-> arbitrary q in pi through O
-> if q=m or q=n: hypothesis
-> otherwise Pasch in triangle ABD
-> exit through AD or BD
-> opposite J on q, G-O-J, OG = OJ
-> landing on BC or AC
-> ASA: AG = BJ (or BG = AJ)
-> spatial SSS/SAS: GF = JF
-> spatial SSS: angle GOF = angle JOF
-> G-O-J -> RightAngle GOF
-> l perpendicular q at O
-> forall q in pi through O, l perpendicular q
-> HilbertLinePerpendicularPlaneAt Geo l pi O.

```

4.14 Classical dependencies versus Lean dependencies

Role	Euclid	Lean reconstruction
Four equal radial segments	cut off AE, EB, CE, ED equal	opposite-point construction on each of the two lines inside <code>PlaneGeo pi</code>
Vertical angles	I.15	developed vertical-angle and angle-congruence infrastructure
First triangle congruence	I.4	balanced-cross SAS helper
Transversal equalities	I.26	Pasch, transversal ASA, and the two landing branches
Distances from F	I.4 on right triangles	spatial perpendicular-bisector/equidistance helper
Cross-plane triangle comparison	I.8 and I.4	general spatial SSS/SAS API through explicit carrier planes
Arbitrary line in the plane	“drawn at random”	universal q ; cases $q = m$, $q = n$, or the Pasch branch
Line perpendicular to plane	XI.Def.3	the universal line-plane perpendicularity predicate

The Lean DAG is longer because it must prove all incidence, order, noncollinearity, carrier-plane, and orientation facts that the classical diagram leaves implicit. The geometric argument itself remains Euclidean XI.4.

4.15 Axiomatic status

The public theorem assumes the spatial Hilbert hierarchy

```

HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo

```

```
HilbertSpaceOrder
HilbertSpaceCongruence
```

No `HilbertEuclideanPlane` instance occurs in the theorem signature. In particular, the proof of XI.4 does not use Hilbert Group IV, the Euclidean parallel axiom.

No continuity or Archimedean axiom is introduced by XI.4.

The final Lean audit is:

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[propept,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this stage. XI.4 introduces no new proposition-specific axiom.

4.16 Formalization notes

4.16.1 The proof is synthetic

There are no coordinates, vectors, scalar products, normal vectors, trigonometric functions, angle measures, or real-valued lengths.

All claims are expressed through:

- incidence of points, lines, and planes;
- strict betweenness;
- same-ray and side-of-line relations;
- segment and angle congruence;
- Hilbert right angles;
- Pasch;
- planar and spatial SAS/SSS/ASA arguments.

4.16.2 The point F is not a normal-vector surrogate

The point F is merely a nonvertex point chosen on the candidate line l . It is used to form ordinary synthetic triangles such as AOF , BOF , GOF , and JOF . The proof never assigns coordinates to F and never interprets OF as a vector.

4.16.3 Why PlaneGeo is essential

Some arguments are intrinsically planar: Pasch, same-side uniqueness of an angle copy, same-ray normalization, and ordinary Hilbert angle construction. Other arguments compare triangles in different planes. The proof therefore alternates explicitly between:

$$\text{ambient 3D geometry} \longleftrightarrow \text{PlaneGeo}(\pi) \text{ or } \text{PlaneGeo}(\rho).$$

This is not implementation noise. It records exactly where Euclid is using plane geometry inside a solid-geometric proof.

4.16.4 Hidden positional data

The formal proof makes explicit all of the following facts:

- the two given plane lines are genuinely distinct;
- perpendicularity witnesses are nondegenerate;
- the four balanced points lie on the intended carriers;
- $A - O - B$ and $C - O - D$ are strict orders;
- the auxiliary point F is outside π ;
- a general transversal need not meet the fixed pair AD, BC ;
- Pasch selects one of two connector families;
- the opposite point on the transversal lies on the paired connector;
- every spatial triangle used by SSS or SAS is noncollinear;
- the final arm swap is performed in the explicit plane through l and the arbitrary transversal.

These facts are precisely the information supplied visually, but not logically, by a conventional diagram.

4.16.5 Why the two connector families matter

A tempting but incorrect formalization is to assume that every line through O meets both segments AD and BC . The diagram can make this appear plausible, but it is not true.

The final proof instead analyzes the first exit of the transversal from triangle ABD . This produces exactly the dichotomy required by the geometry:

$$(AD, BC) \text{ or } (BD, AC).$$

This is one of the most significant proof-design corrections in the XI.4 development.

4.17 Reusable infrastructure produced by XI.4

The proposition required one general correction to the 3D interface: line-line perpendicularity now carries an explicit noncollinearity witness, so that distinctness of the carriers is a theorem rather than an external hypothesis.

The general 3D interface also exposes the bridges

```
HilbertLinesPerpendicularAt
hilbert_linesPerpendicularAt_ne
planeGeo_linesPerpendicularAt_iff_ambient
HilbertLinePerpendicularPlaneAt
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
HilbertLinePerpendicularPlaneAt.incidence
```

The remaining declarations beginning with `hilbert_XI4_` are kept in `Proposition11_4.lean` because they package the concrete Euclidean XI.4 configuration.

4.18 Dependency summary

Hilbert spatial incidence + order + congruence
 $\Downarrow$
 PlaneGeo slices and ambient/plane bridges
 $\Downarrow$
 nondegenerate line-line perpendicularity
 $\Downarrow$
 balanced cross $A - O - B, C - O - D$
 $\Downarrow$
 $AD \cong BC, \angle OAD \cong \angle OBC$
 $\Downarrow$
 $F \in l, F \neq O, F \notin \pi$
 $\Downarrow$
 $AF \cong BF, CF \cong DF$
 $\Downarrow$
 Pasch for arbitrary $q \subset \pi$ through O
 $\Downarrow$
 two connector branches + opposite point J
 $\Downarrow$
 $GF \cong JF, \angle GOF \cong \angle JOF$
 $\Downarrow$
 $\text{RightAngle}(G, O, F)$
 $\Downarrow$
 $l \perp q$ for every $q \subset \pi$ through O
 $\Downarrow$
`euclid_proposition_11_4.`

4.19 Status

Production theorem	<code>Geometry.euclid_proposition_11_4</code>
Production file	<code>CGJteamLab/Proposition11_4.lean</code>
General 3D interface	<code>CGJteamLab/Hilbert3DInterface.lean</code>
Coordinates/vectors	no
Euclidean parallel axiom (Group IV)	no
Continuity	no new use
New XI.4-specific axiom	no
Full project build	clean
Axiom audit	completed

4.20 Conclusion

Euclid XI.4 is the first major test of the project’s spatial Hilbert architecture. The proposition begins with only two line-line perpendicularities at a common point and must end with a universal statement about every line of a plane through that point.

The formal proof shows that this transition can be reconstructed synthetically. The essential tools are not coordinates or vector orthogonality, but the same ingredients visible in Euclid’s proof: equal-segment constructions, vertical angles, triangle congruence, an arbitrary transversal, and the definition of a perpendicular to a plane.

Lean exposes two pieces of structure that the classical diagram conceals. First, the arbitrary transversal requires a genuine Pasch argument with two connector families. Second, the repeated uses of Book I congruence in solid geometry require an explicit mechanism for comparing triangles carried by different planes.

Once these points are made explicit, the final statement is exactly Euclid XI.Def.3: the candidate line is perpendicular to every line of the given plane through the common point. In that sense the proof is not a modern replacement of XI.4 but a detailed synthetic reconstruction of its logical content.

Chapter 5

Proposition XI.5

5.1 Introduction

Proposition XI.5 is the first Book XI proposition whose conclusion is a genuinely spatial coplanarity statement forced by perpendicularity.

A line l passes through a point O and is perpendicular there to three lines m, n, p . Two of these lines, m and n , are assumed distinct. The conclusion is that m, n, p lie in one plane.

The formal reconstruction is not a line-by-line transcription of Euclid's proof. The current spatial interface makes it possible to use Proposition XI.4 twice and thereby obtain a short contradiction argument.

5.2 Euclid's Statement

Euclid's Proposition XI.5 states:

If a straight line is set up at right angles to three straight lines which meet one another at their common point of section, then the three straight lines lie in one plane.

In Euclid's notation the common perpendicular is AB , and the three lines through the common point B are BC , BD , and BE .

5.3 Classical Configuration

Euclid assumes, for contradiction, that two of the three lines, say BD and BE , lie in a reference plane while BC lies in a more elevated plane.

He then considers the plane through AB and BC . Its intersection with the reference plane is a line BF . Thus AB, BC, BF lie in one plane.

At the same time, since AB is perpendicular to BD and BE , Proposition XI.4 implies that AB is perpendicular to the reference plane. Therefore AB is also perpendicular to BF .

The contradiction is that, in the plane through AB and BC , the lines BC and BF would both form a right angle with AB at the same point.

5.4 Classical Proof

The local classical dependency pattern is:

$$\text{XI.3} \longrightarrow \text{intersection line } BF,$$

$$\text{XI.4} \longrightarrow AB \perp (\text{reference plane}),$$

followed by Euclid's Definition XI.3, which turns line-plane perpendicularity into perpendicularity to every line of the plane meeting the perpendicular.

Thus the classical proof is a contradiction proof based on the intersection of two planes and on the impossibility of two distinct perpendiculars to the same line at the same point in one plane.

5.5 What is genuinely three-dimensional here?

The essential spatial content is not the elementary fact about right angles. It is the management of planes.

The proof must establish:

- a plane through two intersecting lines;
- a second plane through another pair of intersecting lines;
- that these two planes are distinct;
- a line of intersection of the two planes;
- containment of relevant lines in the appropriate planes.

The current Lean proof does not open a new planar `PlaneGeo` argument of its own. Instead, its planar geometric content is inherited through the already proved Proposition XI.4.

5.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_5
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
```

```

[_HSC : HilbertSpaceCongruence
  (Geo := Geo) (H := H) (S := S)]
(1 m n p : Geo.Line)
(0 : Geo.Point)
(hmn : Ne m n)
(hPerpM :
  HilbertLinesPerpendicularAt Geo 1 m 0)
(hPerpN :
  HilbertLinesPerpendicularAt Geo 1 n 0)
(hPerpP :
  HilbertLinesPerpendicularAt Geo 1 p 0) :
exists pi : S.Plane,
  HilbertLineInPlane Geo m pi /\
  HilbertLineInPlane Geo n pi /\
  HilbertLineInPlane Geo p pi

```

The conclusion is constructive at the level of incidence: Lean returns an explicit plane π containing all three lines.

The hypothesis `hmn : Ne m n` makes explicit a point that is usually left implicit in the classical diagram. The two intersecting lines m, n must be distinct in order to determine the reference plane.

5.7 Spatial / Planar Architecture Used

The proposition uses the ambient three-dimensional layer:

- `HilbertSpacePrimitive` for planes and point–plane incidence;
- `HilbertSpaceIncidence` for spatial plane constructions;
- `HilbertLineInPlane` for line–plane containment;
- `HilbertLinesPerpendicularAt` for spatial line–line perpendicularity;
- `HilbertLinePerpendicularPlaneAt` for line–plane perpendicularity.

The ambient geometry does not install the planar order and congruence classes as global instances on the spatial object. The architectural distinction is:

```

-- not global ambient instances:
HilbertOrder Geo
HilbertCongruence Geo

-- spatial layers:
HilbertSpaceOrder
HilbertSpaceCongruence

```

Planar order and congruence enter only through the established `PlaneGeo` machinery used inside Proposition XI.4.

5.8 Proof Architecture of the Reconstruction

Let m and n be the first two lines perpendicular to l at O .

5.8.1 Step 1: construct the reference plane

Since $m \neq n$ and both pass through O , the theorem

```
hilbert_plane_through_two_intersecting_lines
```

constructs a plane π containing m and n .

5.8.2 Step 2: apply Proposition XI.4

The lines m and n are packaged as `PlaneLines` of π , and O as a `PlanePoint` of π .

Proposition XI.4 then yields

$$l \perp \pi \quad \text{at } O.$$

In Lean this is the call to

```
euclid_proposition_11_4
```

with the two given perpendicularities $l \perp m$ and $l \perp n$.

5.8.3 Step 3: split on whether p already lies in π

If

$$p \subset \pi,$$

the theorem is finished immediately.

Assume therefore

$$p \not\subset \pi.$$

Since $l \perp p$, the lines l and p are distinct. They meet at O , so they determine another plane σ .

5.8.4 Step 4: intersect the two planes

The planes π and σ must be distinct. If $\pi = \sigma$, then $p \subset \sigma$ would imply $p \subset \pi$, contrary to the case assumption.

The theorem

```
hilbert_plane_intersection_line
```

therefore gives a line q through O contained in both planes:

$$q \subset \pi, \quad q \subset \sigma.$$

Since $l \perp \pi$ and q is a line of π through O ,

$$l \perp q.$$

This is obtained directly from

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

5.8.5 Step 5: the second use of XI.4

Now p and q both lie in σ , and

$$l \perp p, \quad l \perp q.$$

Suppose $p \neq q$. Applying Proposition XI.4 inside σ gives

$$l \perp \sigma.$$

But $l \subset \sigma$ by construction. This is impossible.

Hence

$$p = q.$$

Since $q \subset \pi$, it follows that $p \subset \pi$, and therefore

$$m, n, p \subset \pi.$$

5.9 Lean Formalization

The small proposition-local helper is:

```
theorem hilbert_linePerpendicularPlaneAt_not_in_plane
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l : Geo.Line)
  (pi : S.Plane)
  (O : Geo.Point)
```

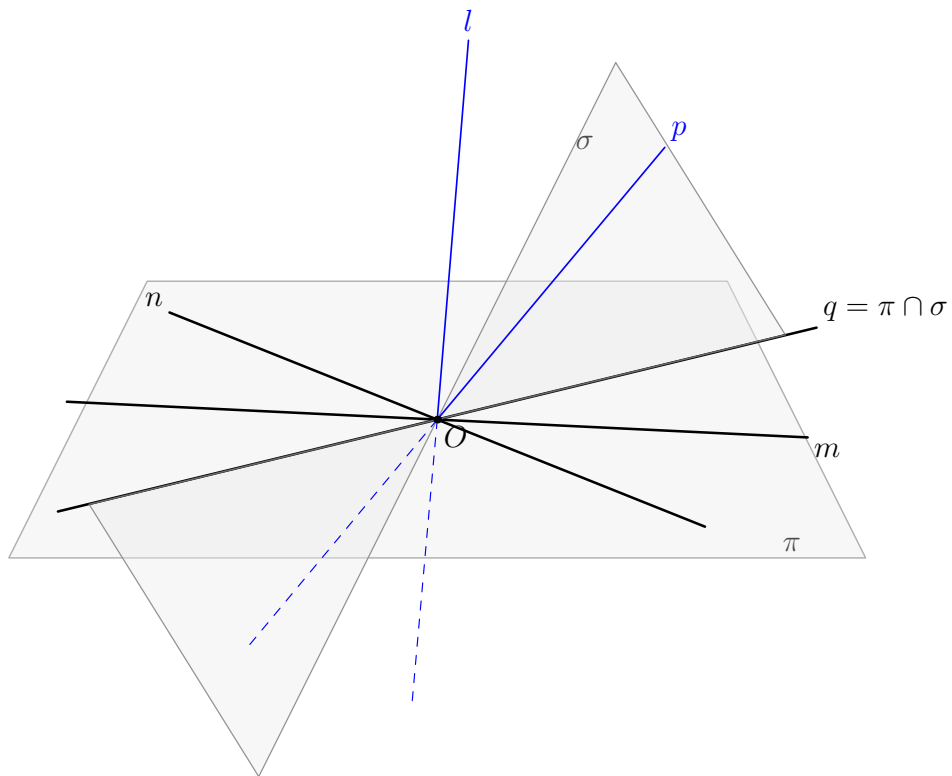


Figure 5.1: The formal contradiction argument for Proposition XI.5. The black lines m, n, q lie in the reference plane π . The spatial lines l, p are blue. Under the temporary assumption $p \not\subset \pi$, the lines l, p determine the second plane σ , and $q = \pi \cap \sigma$. If $p \neq q$, a second application of XI.4 would make l perpendicular to σ , although $l \subset \sigma$.

```

(hPerp :
  HilbertLinePerpendicularPlaneAt Geo l pi 0) :
Not (HilbertLineInPlane Geo l pi) := by

intro hlp

have hSelf :
  HilbertLinesPerpendicularAt Geo l l 0 :=
  HilbertLinePerpendicularPlaneAt.perpendicular_to_line
  (Geo := Geo)
  hPerp
  hlp
  hPerp.1

exact
(hilbert_linesPerpendicularAt_ne
  Geo l l 0 hSelf) rfl

```

This helper is purely definitional in geometric content. If a line perpendicular to a plane were itself contained in the plane, then the universal line–plane perpendicularity condition would make the line perpendicular to itself. The predicate `HilbertLinesPerpendicularAt` is nondegenerate, so this is impossible.

The theorem body then consists of two plane constructions, one plane-intersection construction, and two calls to XI.4.

5.10 Hidden Formal Data

Several facts visible in a classical spatial diagram must be supplied explicitly in Lean.

- O lies in the plane determined by m, n .
- The ambient lines m, n must be packaged as `PlaneLine Geo pi`.
- Their distinctness must be transported to the subtype level.
- If $p \not\subset \pi$, the plane through l, p must be constructed explicitly.
- The two planes must be proved distinct before their intersection line can be used.
- The intersection line q must carry proofs that it lies in both π and σ .
- For the second XI.4 call, p, q and O are repackaged as `PlaneLine` and `PlanePoint` objects of σ .

These are mathematical incidence obligations, not merely syntactic casts. They encode information that Euclid’s diagram leaves implicit.

5.11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and XI.4.

The Lean proof imports Proposition XI.4 and uses the current spatial intersection theorem

```
hilbert_plane_intersection_line
```

instead of importing a separate formal Proposition XI.3.

Thus the two dependency graphs are not identical.

The actual Lean Book XI dependency is:

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}}.$$

Proposition XI.5 is then the coplanarity tool required by the classical route to Proposition XI.6.

5.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by the XI.5 theorem body.

The inherited planar Hilbert and Book Zero infrastructure enters transitively through Proposition XI.4. In particular, XI.4 internally uses induced plane geometry, while XI.5 itself is primarily a spatial incidence argument.

The new helper `hilbert_linePerpendicularPlaneAt_not_in_plane` does not add a geometric axiom; it follows from the existing definitions and nondegeneracy of line–line perpendicularity.

5.13 Axiomatic Status

The audit command is:

```
import CGJteamLab.Proposition11_5

#print axioms Geometry.euclid_proposition_11_5
```

The reported dependency set is:

```
[propept,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Therefore:

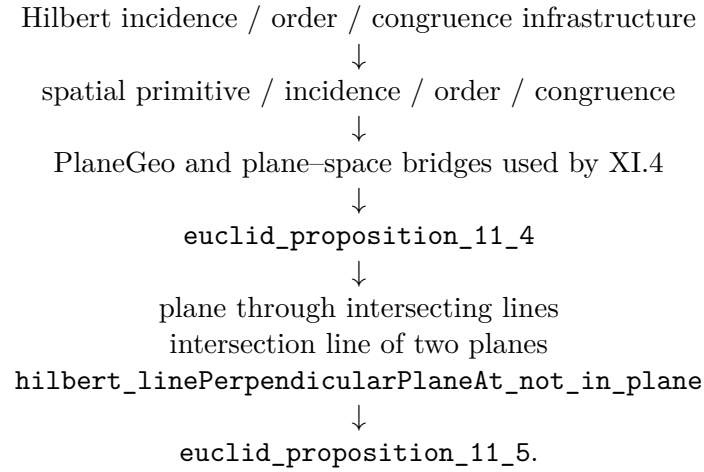
- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;

- new proposition-local helper: yes;
- new Hilbert3DAxioms declaration: no;
- new PlaneGeo bridge theorem: no;
- sorry/admit: no.

The inherited dependency `Geometry.bookZero_nullSegment1` is the same accepted dependency already present in Proposition XI.4.

5.14 Dependency Summary

The actual formal dependency chain is:



At the current checkpoint the production theorem compiles locally and its axiom audit has been run successfully. A full repository `lake build` is a separate checkpoint and should be recorded only after it is run.

5.15 Conclusion

Proposition XI.5 identifies the plane perpendicular to a given line at a given point as the unique carrier of all lines through that point perpendicular to the line, in the precise three-line form stated by Euclid.

The formal proof is shorter than the classical local proof graph because the present 3D interface already contains an intersection-line theorem for two planes and because XI.4 can be applied a second time to exclude the hypothetical second plane.

No coordinates, vectors, scalar products, continuity principle, or new spatial axiom enter the proof.

Chapter 6

Proposition XI.6

6.1 Introduction

Proposition XI.6 is the first proposition in the present Book XI development whose conclusion is a genuinely spatial parallelism statement. Two ambient lines l and m are perpendicular to the same plane π , at distinct feet B and D . Euclid concludes that the two lines are parallel.

In three-dimensional geometry this conclusion contains two logically different assertions. The lines must first be shown to be *coplanar*; only then can planar geometry prove that they do not meet. This distinction is essential in the formal reconstruction, because two disjoint ambient lines may be skew.

The current Lean proof follows the metric core of Euclid’s construction closely:

$$\text{auxiliary perpendicular and equal segment} \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5.}$$

After XI.5 has supplied the missing common plane, the proof changes deliberately to an induced planar geometry and closes the disjointness argument by the neutral alternate-angle criterion.

6.2 Euclid’s Statement

Euclid’s Proposition XI.6 states:

If two straight lines be at right angles to the same plane, the straight lines will be parallel.

In the classical configuration the two perpendiculars are AB and CD , meeting the reference plane at B and D . The formal theorem makes the nondegenerate case explicit by assuming

$$B \neq D.$$

6.3 Classical Configuration

Join the feet B and D . In the reference plane construct through D a line DE perpendicular to BD , and choose E so that

$$DE \cong AB.$$

Join BE , AE , and AD .

Because AB is perpendicular to the reference plane, it is perpendicular to every line of that plane through B , in particular to BD and BE . Hence

$$\angle ABD, \quad \angle ABE$$

are right angles.

Likewise CD is perpendicular to the reference plane at D , so it is perpendicular to both DB and DE .

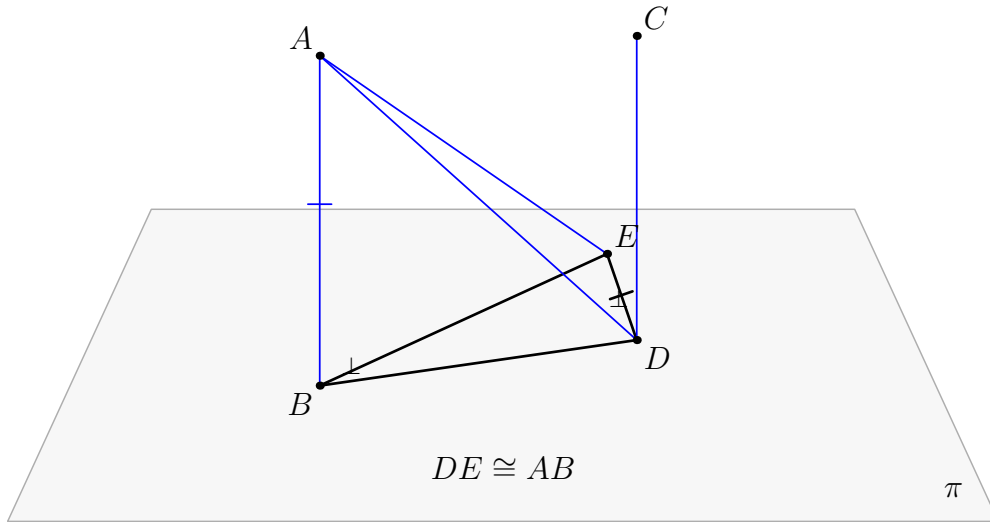


Figure 6.1: The classical configuration of Proposition XI.6. The black geometry lies in the reference plane π : B, D, E , the transversal BD , and the auxiliary perpendicular DE . The blue lines and connectors leave π . The construction $DE \cong AB$ prepares the SAS comparison of BAD and DEB ; the resulting equality $AD \cong EB$ prepares the SSS comparison of DEA and BAE .

6.4 Classical Proof

The classical metric part consists of two triangle-congruence steps.

First,

$$AB \cong DE, \quad BD \cong DB,$$

and the included angles

$$\angle ABD, \quad \angle EDB$$

are both right. Proposition I.4 therefore gives

$$AD \cong EB.$$

Second, the triangles DEA and BAE have

$$DE \cong BA, \quad EA \cong AE, \quad DA \cong BE.$$

Proposition I.8 gives

$$\angle EDA \cong \angle ABE.$$

Since $\angle ABE$ is right, $\angle EDA$ is right; hence

$$ED \perp DA.$$

Now at the point D the line ED is perpendicular to the three lines

$$DB, \quad DA, \quad DC.$$

Proposition XI.5 therefore places those three lines in one plane. Proposition XI.2 identifies the triangle plane determined by the intersecting lines, so AB is brought into the same plane as BD and DC . Finally

$$\angle ABD = \angle BDC = 90^\circ,$$

and Proposition I.28 yields

$$AB \parallel CD.$$

A compact classical dependency graph is therefore

$$\begin{array}{c}
 \text{construct } DE \perp BD, \quad DE \cong AB \\
 \Downarrow \\
 \text{I.4} \\
 \Downarrow \\
 AD \cong EB \\
 \Downarrow \\
 \text{I.8} \\
 \Downarrow \\
 ED \perp DA \\
 \Downarrow \\
 \text{XI.5} \\
 \Downarrow \\
 DB, DA, DC \text{ coplanar} \\
 \Downarrow \\
 \text{XI.2} \\
 \Downarrow \\
 AB, BD, DC \text{ coplanar} \\
 \Downarrow \\
 \text{I.28} \\
 \Downarrow \\
 AB \parallel CD.
 \end{array}$$

The construction of the auxiliary perpendicular and equal segment is part of the earlier Book I construction machinery. The substantive local congruence steps are I.4 and I.8.

6.5 What is genuinely three-dimensional here?

The difficult point is not the planar theorem that two perpendiculars to one transversal are parallel. That statement cannot even be applied until one knows that the two candidate parallel lines lie in one plane.

In ambient three-space the information

$$l \perp BD, \quad m \perp BD$$

does not by itself constitute a parallelism proof: without a coplanarity theorem the lines could still be skew.

The formal proof therefore separates XI.6 into

$$\boxed{\text{coplanarity}} \quad + \quad \boxed{\text{planar disjointness}}.$$

The first box is the genuinely spatial content and is exactly where Proposition XI.5 is used.

6.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_6
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (l m : Geo.Line)
  (B D : Geo.Point)
  (hBD : Ne B D)
  (hLperp :
    HilbertLinePerpendicularPlaneAt
      Geo l pi B)
  (hMperp :
    HilbertLinePerpendicularPlaneAt
      Geo m pi D) :
  HilbertSpaceLinesParallel Geo l m

```

The target relation is part of the reusable spatial interface `Hilbert3DInterface`:

```

def HilbertSpaceLinesParallel
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l m : Geo.Line) : Prop :=
exists sigma : S.Plane,
  HilbertLineInPlane Geo l sigma /\
  HilbertLineInPlane Geo m sigma /\
  HilbertLinesDisjoint Geo l m

```

Thus “parallel in space” means exactly:

1. there is a plane containing both lines;
2. the two lines are disjoint.

The first clause is not redundant. If spatial parallelism were defined only by ambient disjointness, skew lines would incorrectly satisfy the definition.

6.7 Spatial / Planar Architecture Used

The proof uses three layers which must remain distinct.

6.7.1 Ambient three-space

The ambient objects and relations include

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

In particular, segment congruence between configurations carried by different planes is handled through `HilbertSpaceCongruence`.

6.7.2 Induced plane geometry

Whenever a planar theorem is needed, the relevant ambient plane σ or π is converted into

```
PlanePoint
PlaneLine
PlaneGeo
```

and ordinary planar incidence, order, right-angle, side-of-line, and triangle theorems are used there.

There is deliberately no global ambient instance

```
HilbertOrder Geo
HilbertCongruence Geo
```

for the three-dimensional space.

6.7.3 Plane-space bridges

The proof repeatedly uses bridges such as

```
planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_primCollinear_to_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_sameRay_iff_ambient
```

These are not cosmetic casts. They state which geometric fact is being used in which plane slice.

6.8 Proof Architecture of the Reconstruction

6.8.1 Step 1: Euclid's auxiliary construction in the reference plane

Choose a point $A \neq B$ on the first normal l . Inside the reference plane π , construct the line

$$d = BD.$$

Through D construct a line e perpendicular to d , and lay off on e a point E such that

$$DE \cong AB.$$

This is packaged by

```
hilbert_XI6_auxiliary_perpendicular_equal_segment
```

The line d and the auxiliary line e are constructed in `PlaneGeo pi`. The segment copied onto DE , however, may have the spatial endpoint A ; that copy is therefore performed by `HilbertSpaceCongruence.segment_construction`.

6.8.2 Step 2: the first spatial SAS

Since $l \perp \pi$ at B and $d \subset \pi$ passes through B ,

$$l \perp d.$$

The auxiliary construction gives

$$e \perp d$$

at D .

The helper

```
hilbert_XI6_space_linesPerpendicularAt_right_angle_of_points
```

normalizes these line–line perpendicularities to the concrete right angles

$$\angle ABD, \quad \angle EDB.$$

The spatial right-angle comparison then supplies

$$\angle ABD \cong \angle EDB.$$

With

$$BA \cong DE, \quad BD \cong DB,$$

the spatial SAS theorem gives

$$AD \cong EB.$$

This whole block is packaged by

```
hilbert_XI6_first_SAS
```

and uses

```
hilbert_space_sas_third_side_and_angle
```

for the actual triangle comparison.

6.8.3 Step 3: the second spatial SSS and the new right angle

The next comparison is between the triangles DEA and BAE . The three side relations are

$$DE \cong BA, \quad EA \cong AE, \quad DA \cong BE.$$

The spatial SSS theorem gives

$$\angle EDA \cong \angle ABE.$$

Because $AB \perp \pi$ and $BE \subset \pi$, the angle ABE is right. The rightness is transported to EDA , yielding

$$ED \perp DA.$$

This block is packaged by

```
hilbert_XI6_second_SSS_right_angle
```

The proof must create explicit carrier planes for the relevant triangles. The right-angle transport is handled by

```
hilbert_XI6_space_right_angle_transport_in_plane
```

rather than by installing a forbidden ambient `HilbertCongruence Geo` instance.

6.8.4 Step 4: Proposition XI.5 supplies coplanarity

At D , the line $e = DE$ is now perpendicular to

$$d = DB, \quad a = DA, \quad m.$$

The last perpendicularity follows because $m \perp \pi$ and $e \subset \pi$.

Proposition XI.5 is applied exactly here:

```
euclid_proposition_11_5
```

It produces a plane σ containing

$$d, \quad a, \quad m.$$

Since $A \in a$ and $B \in d$, both A and B lie in σ . The line $l = AB$ therefore also lies in σ .

The result is isolated as

```
hilbert_XI6_normals_to_same_plane_coplanar
```

with conclusion

```
exists sigma : S.Plane,
  HilbertLineInPlane Geo l sigma /\
  HilbertLineInPlane Geo m sigma
```

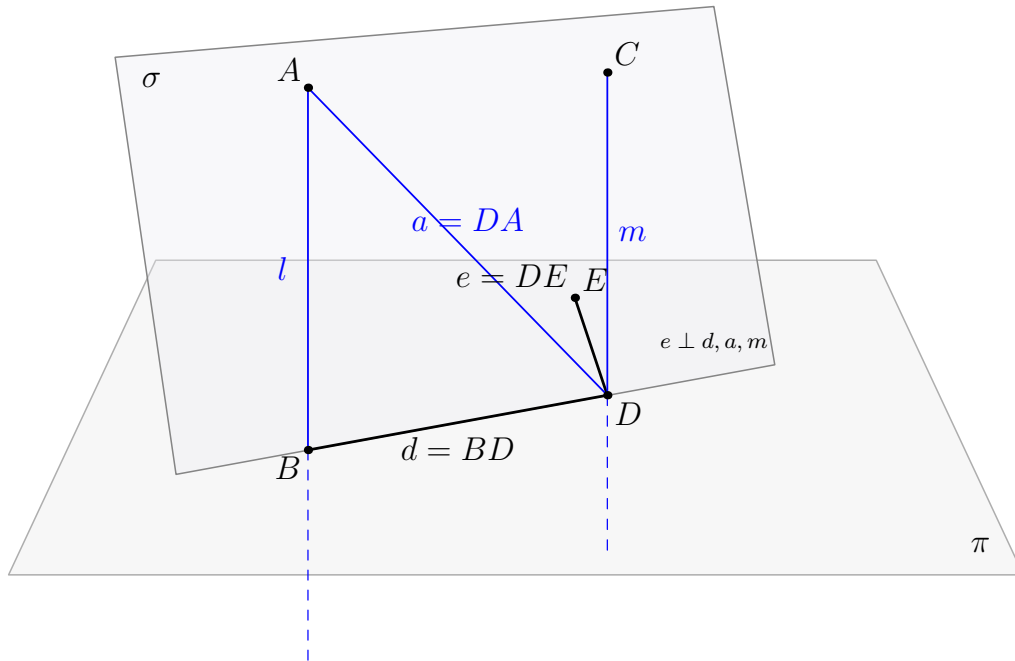


Figure 6.2: The genuinely spatial step in the formal XI.6 proof. The black lines $d = BD$ and $e = DE$ lie in the reference plane π . The blue lines $a = DA$, $m = DC$, and $l = AB$ leave π . At D , the line e is perpendicular to d, a, m , so XI.5 produces the second plane σ containing d, a, m . Because σ contains A and B , it also contains l . The lines l and m are therefore coplanar.

6.8.5 Step 5: the remaining problem is planar

After Step 4, both l and m lie in σ . The common transversal $d = BD$ lies in σ as well, and

$$l \perp d \text{ at } B, \quad m \perp d \text{ at } D.$$

The rest of the proof is carried out entirely inside

PlaneGeo Geo sigma.

The helper

```
hilbert_XI6_coplanar_normals_disjoint
```

chooses nonfoot points on l and m , establishes the required side-of-line orientation relative to d , compares the two right angles, and applies

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

to obtain planar parallelism.

The final conversion proves that the actual ambient carriers l and m are disjoint.

6.8.6 Step 6: assemble spatial parallelism

The public theorem now has exactly the two pieces required by `HilbertSpaceLinesParallel`:

$$l, m \subset \sigma$$

and

$$\text{HilbertLinesDisjoint}(l, m).$$

Therefore

$$\text{HilbertSpaceLinesParallel}(l, m).$$

6.9 Lean Formalization

The final theorem body is short because the geometric work has been isolated into two major helpers. First,

```
hilbert_XI6_normals_to_same_plane_coplanar
```

returns a plane `sigma` containing both ambient normals. Then

```
hilbert_XI6_coplanar_normals_disjoint
```

proves that their ambient carriers are disjoint. The public theorem packages exactly these three fields of `HilbertSpaceLinesParallel`: containment of l , containment of m , and disjointness.

The production module imports

```
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition27
```

The first import supplies the decisive spatial coplanarity theorem. The second supplies the neutral alternate-angle machinery used in the final induced plane.

6.10 Hidden Formal Data

The classical diagram suppresses several obligations which the Lean proof must make explicit.

- The feet B, D are distinct.
- The point A selected on l is outside the reference plane π .
- The auxiliary point E is nondegenerate and lies both on e and in π .
- Every triangle used in SAS or SSS carries an explicit noncollinearity proof.
- Spatial triangle comparisons require an explicit target carrier plane.
- Right-angle transport between ambient configurations is performed through an explicit `PlaneGeo`, not through an ambient planar congruence instance.
- Before XI.5 can be applied, the three lines d, a, m must be proved distinct in the required places and must all pass through D .
- The plane σ returned by XI.5 must be shown to contain both A and B , so that the whole carrier $l = AB$ lies in σ .
- In the final planar proof, the chosen points on l and m must be oriented on opposite sides of the transversal d . The proof performs an explicit `SameSide/OppositeSide` case split.
- Point-pair parallelism inside `PlaneGeo` `sigma` must finally be converted into disjointness of the original ambient carriers.

These obligations are geometric. They are the incidence, order, and plane-membership data that a conventional three-dimensional drawing usually leaves implicit.

6.11 Relationship with Earlier Book XI Propositions

Proposition XI.5 is a direct formal dependency:

$$\boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

XI.5 itself depends on XI.4, and XI.6 also reuses several XI.4 normalization helpers for perpendicular lines and right angles. Consequently the effective Book XI chain is

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

The classical proof also invokes XI.2 when it moves from the three coplanar lines through D to the plane containing the triangle ABD . The Lean proof does not need a separate formal call to Proposition XI.2: the spatial incidence API directly constructs and tracks the relevant carrier plane.

6.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The metric part of the formal proof reproduces the roles of Euclid I.4 and I.8 through the spatial SAS and SSS infrastructure:

```
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

Right angles and their transport are handled synthetically; no angle measure is introduced.

The classical final step is I.28. The production Lean file instead imports `Proposition27` and uses the established neutral alternate-angle theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

inside the explicitly constructed plane σ . Thus the formal module dependency and the classical Euclid proposition-number dependency are not identical, even though they close the same planar configuration.

Book Zero enters only transitively through the established Hilbert geometric infrastructure. No new Book Zero axiom or theorem is added by XI.6.

6.13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_6

#print axioms Geometry.euclid_proposition_11_6
```

and Lean reports

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same reported axiom profile as Propositions XI.4 and XI.5.

Therefore:

- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;
- new proposition-local helper theorems: yes;
- new `Hilbert3DAxioms` declaration: no;
- reusable `Hilbert3DInterface` declaration: `HilbertSpaceLinesParallel`, promoted from the XI.6 proposition module for reuse by XI.7 and later spatial results;

- new PlaneGeo bridge theorem: no;

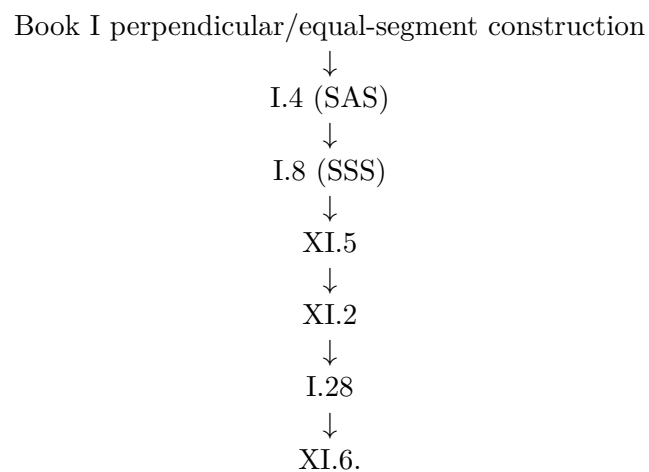
- sorry/admit: no.

The inherited `Geometry.bookZero_nullSegment1` dependency is not specific to XI.6.

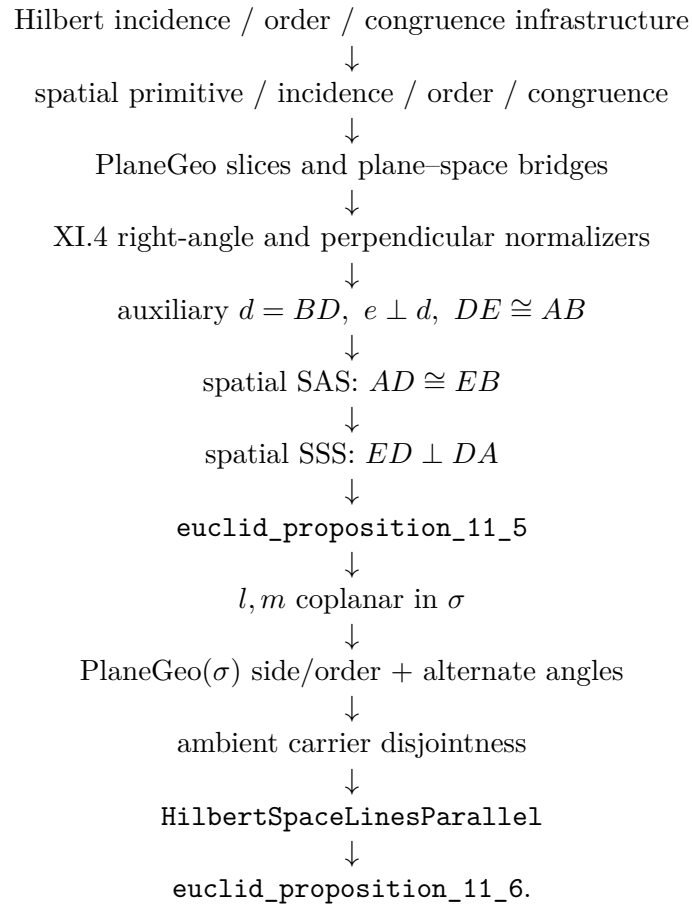
The production theorem compiled successfully, its axiom audit was run successfully, and the full repository `lake build` was reported clean at this checkpoint.

6.14 Dependency Summary

The classical local chain is



The actual formal chain is



6.15 Conclusion

Proposition XI.6 is a useful boundary theorem for the three-dimensional architecture. Its metric core is recognizably Euclidean: construct an auxiliary perpendicular and an equal segment, use SAS, then SSS, and obtain a third perpendicular at D .

The genuinely spatial step is XI.5. It converts three perpendicularities through one point into the common plane in which the final parallelism argument is allowed to take place. Lean makes this logical boundary explicit rather than allowing the diagram to hide it.

The final definition of spatial parallelism records the same distinction: coplanarity is proved first and disjointness second. No coordinates, vectors, scalar products, numerical angle measure, continuity principle, new spatial axiom, or Euclidean parallel axiom is introduced.

Chapter 7

Proposition XI.7

7.1 Introduction

Proposition XI.7 is an incidence proposition about spatial parallelism. Two ambient straight lines are parallel, points are chosen on the two lines, and Euclid asserts that the straight line joining the chosen points lies in the same plane as the parallels.

The formal reconstruction is substantially shorter than XI.4–XI.6. The reason is structural. In the current spatial interface, `HilbertSpaceLinesParallel` already contains an explicit common carrier plane. Once the two chosen points are shown to lie in that plane, Hilbert’s spatial incidence axiom I.6 places their joining line in the same plane.

Thus the formal proof is not a line-by-line translation of Euclid’s *reductio*. It proves the same geometric target through the stronger and more explicit incidence organization adopted by Hilbert.

7.2 Euclid’s Statement

Euclid’s Proposition XI.7 states:

If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB and CD be parallel straight lines. Choose E on AB and F on CD . The claim is that the joining straight line EF lies in the plane of AB and CD .

7.3 Classical Configuration

The classical configuration begins with the plane containing the two parallel lines AB and CD . The points E and F are arbitrary points on the two lines.

The proposition is genuinely spatial not because the final figure contains a line outside the plane, but because the assertion excludes that possibility.

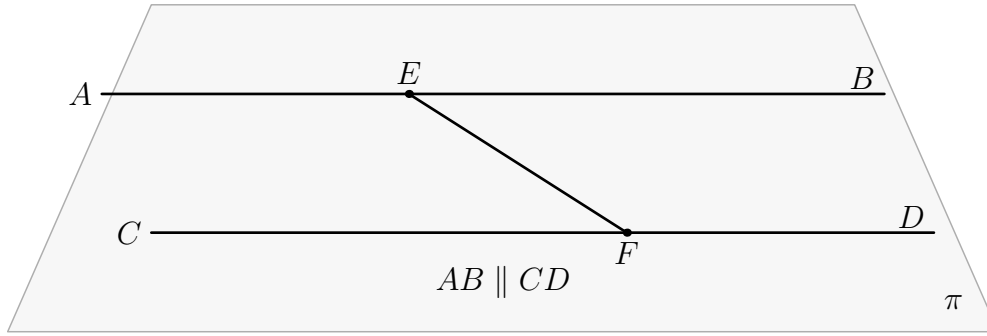


Figure 7.1: The configuration of Proposition XI.7. The two parallel lines AB and CD lie in the reference plane π , with $E \in AB$ and $F \in CD$. The proposition asserts that the joining line EF also lies in π . All final geometric objects in the figure are therefore drawn in black; no object in the proved configuration leaves the plane.

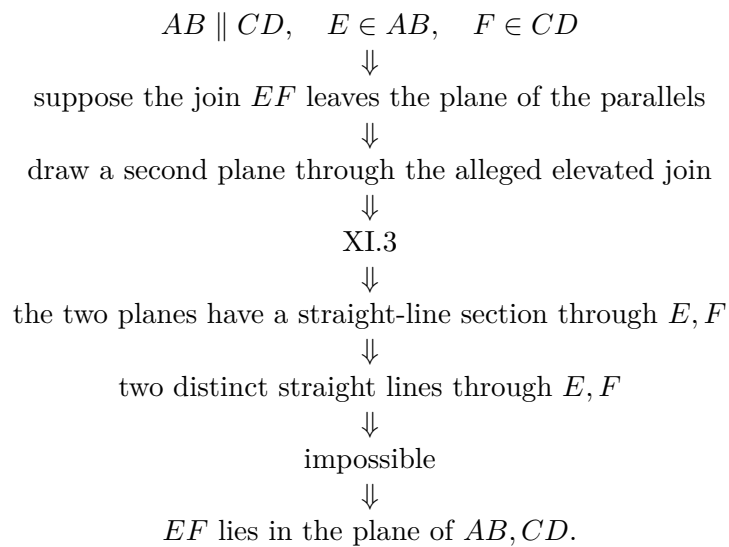
7.4 Classical Proof

Euclid argues by contradiction. Suppose that the straight line joining E and F does not lie in the plane of the parallels, but in a more elevated plane. Draw a plane through that alleged elevated straight line.

The new plane cuts the plane of the parallel lines in a straight line, by Proposition XI.3. Since both E and F belong to both planes, the common section passes through E and F . Euclid then obtains two distinct straight lines through the same two points, described in the text as two straight lines enclosing an area, which is impossible.

Hence the joining line EF cannot leave the plane of the parallels.

The local classical DAG is therefore



The only explicit Book XI proposition cited in the proof is XI.3.

7.5 What is genuinely three-dimensional here?

The genuinely spatial datum is the carrier plane of the two parallel ambient lines. In ordinary plane geometry this datum is invisible, because every line under discussion already belongs to the unique ambient plane.

In three-space, disjointness alone is insufficient for parallelism: two disjoint lines may be skew. The formal relation `HilbertSpaceLinesParallel` therefore records both

coplanarity and disjointness.

XI.7 uses the coplanarity component directly. No metric geometry is needed.

7.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_7
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (l m : Geo.Line)
  (E F : Geo.Point)
  (hParallel : HilbertSpaceLinesParallel Geo l m)
  (hEl : H.OnLine E l)
  (hFm : H.OnLine F m) :
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
  exists n : Geo.Line,
    H.OnLine E n /\
    H.OnLine F n /\
    HilbertLineInPlane Geo n sigma

```

The conclusion returns both the common carrier plane `sigma` and an explicit joining line `n` through `E` and `F`, together with the proof that `n` lies in `sigma`.

7.7 Spatial / Planar Architecture Used

XI.7 uses only the ambient incidence layer:

```

HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel

```

No induced **PlaneGeo** is constructed. There is no planar betweenness, SameRay, SameSide/OppositeSide, angle congruence, right angle, SAS, SSS, or plane-space congruence transport.

This is an important contrast with XI.4–XI.6: although XI.7 is a spatial proposition, its formal content is pure incidence.

7.8 Proof Architecture of the Reconstruction

7.8.1 Step 1: unpack spatial parallelism

From

$$\text{HilbertSpaceLinesParallel}(l, m)$$

the proof obtains a plane σ such that

$$l \subset \sigma, \quad m \subset \sigma,$$

together with

$$\text{HilbertLinesDisjoint}(l, m).$$

The definition is now part of the general **Hilbert3DInterface** rather than a proposition-local declaration.

7.8.2 Step 2: place the chosen points in the common plane

Since $E \in l$ and $l \subset \sigma$,

$$E \in \sigma.$$

Likewise $F \in m$ and $m \subset \sigma$, hence

$$F \in \sigma.$$

These are direct applications of the line-in-plane predicates supplied by the parallelism witness.

7.8.3 Step 3: prove that the chosen points are distinct

The joining line construction requires $E \neq F$. Euclid's diagram does not state this separately, but the formal proof derives it from the disjointness of the two parallel carriers.

If $E = F$, then the common point would lie on both l and m , contradicting

$$\text{HilbertLinesDisjoint}(l, m).$$

Thus

$$E \neq F.$$

7.8.4 Step 4: construct the joining line

The ordinary incidence structure supplies a line through the two distinct points:

```
HilbertPlaneIncidence.line_through E F hEF
```

This yields a line n with

$$E \in n, \quad F \in n.$$

7.8.5 Step 5: apply Hilbert I.6

Now E and F are two distinct points of n , and both lie in σ . The decisive step is:

`HilbertSpaceIncidence.line_in_plane`

which gives

$$n \subset \sigma.$$

This is the direct formal counterpart of Hilbert’s incidence axiom I.6: two points of a line lying in a plane force the whole line into that plane.

7.9 Lean Formalization

The proof DAG is extremely short:

$$\begin{array}{c}
 \text{HilbertSpaceLinesParallel } l \ m \\
 \Downarrow \\
 \exists \sigma, \quad l \subset \sigma, \quad m \subset \sigma, \quad l \cap m = \emptyset \\
 \Downarrow \\
 E \in \sigma, \quad F \in \sigma \\
 \Downarrow \\
 E \neq F \\
 \Downarrow \\
 \text{line_through } E \ F \\
 \Downarrow \\
 n \text{ through } E, F \\
 \Downarrow \\
 \text{HilbertSpaceIncidence.line_in_plane} \\
 \Downarrow \\
 n \subset \sigma \\
 \Downarrow \\
 \text{euclid_proposition_11_7.}
 \end{array}$$

There are no proposition-local helper theorems.

7.10 Hidden Formal Data

XI.7 has very little hidden data, but two obligations must be explicit.

- The arbitrary points E and F are distinct. This follows from disjointness of the two parallel carriers.
- The phrase “the straight line joining E and F ” is represented by an explicit existential line witness produced by `HilbertPlaneIncidence.line_through`.

Once those two points are available, Hilbert I.6 closes the spatial incidence problem directly.

7.11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses Proposition XI.3, because Euclid constructs a second plane and studies the straight line in which the two planes intersect.

The current Lean theorem does *not* import or invoke XI.3, XI.4, XI.5, or XI.6. Its production file imports only

```
import CGJteamLab.Hilbert3DInterface
```

This difference is structural rather than geometric. Euclid proves the plane containment by contradiction from a plane-intersection theorem. Hilbert’s incidence system takes the closure of a line under two plane points as an explicit spatial incidence axiom.

XI.6 is nevertheless conceptually adjacent. It produces `HilbertSpaceLinesParallel`, which is exactly the hypothesis consumed by XI.7. The theorem XI.7 itself, however, is independent of the proof of XI.6.

7.12 Relationship with Book I / Hilbert Infrastructure

Euclid’s definition of parallel straight lines includes coplanarity. The current formal definition makes this explicit:

```
def HilbertSpaceLinesParallel
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

For XI.7, the disjointness field is used only to derive $E \neq F$; the common-plane fields provide the main spatial information.

The decisive Hilbert source is Group I, Axiom I.6. In Hilbert’s classification, I.1–I.3 are the plane incidence axioms and I.4–I.8 are the space incidence axioms. Thus the formal proof lands exactly in the spatial incidence layer and nowhere higher in the hierarchy.

7.13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_7

#print axioms Geometry.euclid_proposition_11_7
```

and Lean reports:

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This statement must be interpreted precisely. The theorem still has explicit typeclass hypotheses, including `HilbertSpaceIncidence Geo`. The audit says that its proof term does not depend on any additional global Lean axioms or project axiom constants beyond those explicit parameters.

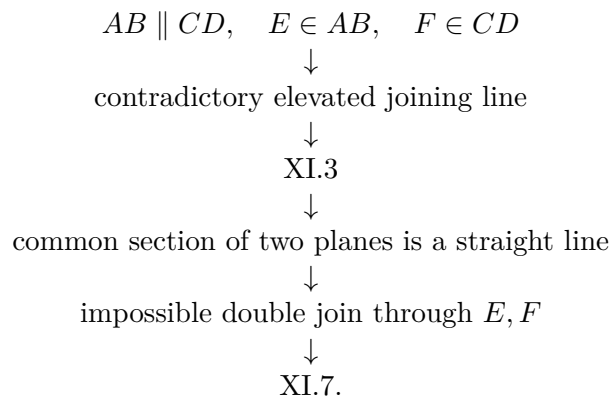
Therefore:

- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;
- spatial order: not used;
- spatial congruence: not used;
- induced `PlaneGeo`: not used;
- new proposition-local helper theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- reusable `Hilbert3DInterface` change: yes, `HilbertSpaceLinesParallel` was promoted there;
- new `PlaneGeo` bridge theorem: no;
- `sorry/admit`: no.

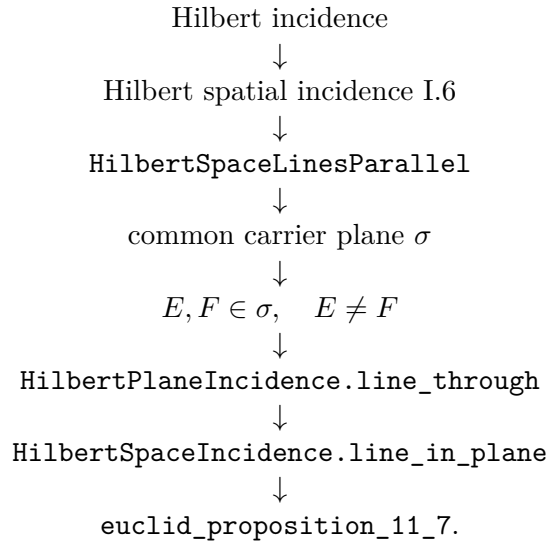
The production theorem and the full repository build were reported clean at this checkpoint.

7.14 Dependency Summary

The classical local chain is



The actual formal chain is



7.15 Conclusion

Proposition XI.7 is a useful contrast with the preceding Book XI formalizations. Its statement is spatial, but its proof requires no order, congruence, right angles, induced plane geometry, or earlier Book XI theorem.

Euclid proves the result by a reductio involving a second plane and Proposition XI.3. The Hilbert reconstruction exposes the same geometric content at the incidence level: spatial parallelism already supplies the common plane, and Axiom I.6 says that a line with two distinct points in that plane lies wholly in the plane.

The resulting Lean theorem is correspondingly small. Its axiom audit reports no global axiomatic dependencies, while the required spatial incidence assumptions remain explicit theorem parameters. No coordinates, vectors, scalar products, numerical angle measures, continuity principles, or Euclidean parallel axiom are introduced.

Chapter 8

Proposition XI.8

8.1 Introduction

Proposition XI.8 is the Euclidean converse companion to XI.6. Proposition XI.6 proved that two straight lines perpendicular to the same plane are parallel. Proposition XI.8 starts with two parallel spatial lines and assumes that one of them is perpendicular to a reference plane; the conclusion is that the second parallel is perpendicular to the same plane.

The proposition is also a structural boundary in the reconstruction. The formalizations XI.4–XI.7 did not require Hilbert Group IV. XI.8 does. The reason is already visible in Euclid’s classical proof: the decisive transversal step uses Proposition I.29, the converse direction of the neutral parallel-angle criterion.

The Lean proof preserves the synthetic geometry but makes the Euclidean dependency more explicit. Instead of calling a planar I.29 wrapper, it separates the argument into

$$\text{neutral I.27} \quad + \quad \text{uniqueness of the parallel through a point.}$$

The second component is precisely the plane-local Group IV assumption.

A second feature is genuinely three-dimensional. Euclid begins by saying that the two parallel lines meet the reference plane at points B and D . The first intersection is given by the hypothesis that the first line is perpendicular to the plane. The second intersection is not automatic in an abstract incidence space. The formal proof derives it.

8.2 Euclid’s Statement

Euclid’s Proposition XI.8 states:

If two straight lines be parallel, and one of them be at right angles to any plane, the remaining one will also be at right angles to the same plane.

Let AB and CD be parallel straight lines, and suppose that AB is perpendicular to the reference plane π at B . The claim is that CD meets π at a point D and is perpendicular to π there.

8.3 Classical Configuration

Euclid lets the parallel lines AB and CD meet the reference plane at B and D . He joins BD , draws DE in the reference plane perpendicular to BD , makes $DE = AB$, and joins BE , AE , and AD .

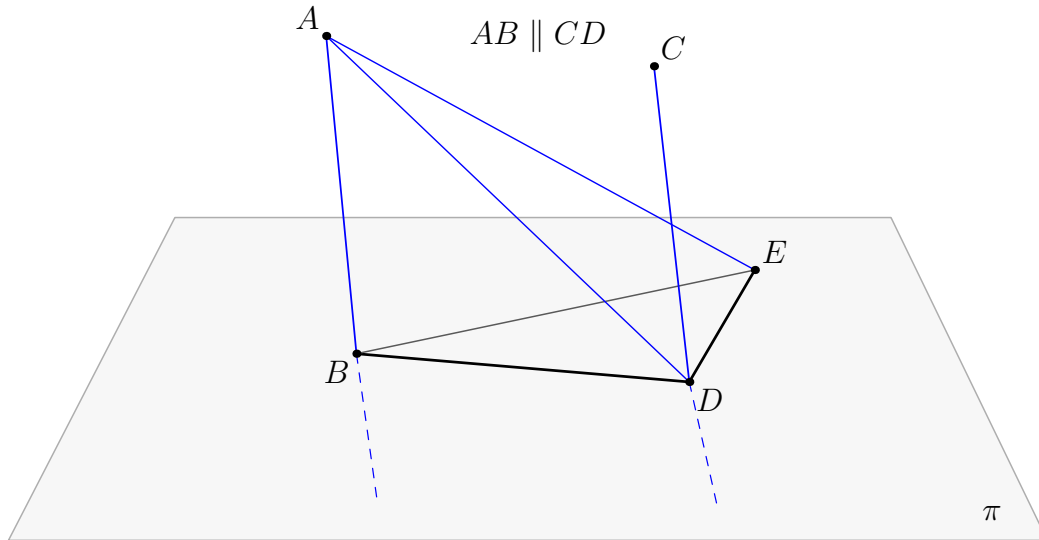


Figure 8.1: The classical configuration of Proposition XI.8. Black lines lie in the reference plane π . Blue lines or segments leave π . The hidden continuations of the two spatial parallel carriers are shown blue and dashed. The auxiliary construction reproduces the metric core already used in XI.6: $DE \perp BD$ and $DE = AB$.

The line BD plays two roles. It is a line of the reference plane and also lies in the common plane of the spatial parallels. This is the transversal on which the classical I.29 step acts.

8.4 Classical Proof

The local classical proof has five main stages.

First, since $AB \parallel CD$, Proposition XI.7 places AB , CD , and the joining line BD in one plane. The opening phrase that the parallel lines meet the reference plane at B, D is diagrammatic in the classical text.

Second, Euclid constructs DE in the reference plane with

$$DE \perp BD, \quad DE = AB.$$

Since $AB \perp \pi$, the angles

$$\angle ABD, \quad \angle ABE$$

are right.

Third, BD falls on the parallel lines AB and CD . Proposition I.29 says that the appropriate interior angles sum to two right angles. Since $\angle ABD$ is right, $\angle CDB$ is right. Hence

$$CD \perp BD.$$

Fourth, Euclid repeats the metric triangle argument familiar from XI.6. From

$$AB = DE, \quad BD = DB,$$

and equality of the included right angles, Proposition I.4 gives

$$AD = BE.$$

Then

$$AB = DE, \quad BE = AD, \quad AE = EA,$$

so Proposition I.8 gives

$$\angle ABE = \angle EDA.$$

The former is right, hence the latter is right:

$$ED \perp AD.$$

Now ED is perpendicular to the two intersecting lines DB and DA . Proposition XI.4 therefore gives

$$ED \perp \text{plane}(BDA).$$

The line CD lies in this plane, so

$$ED \perp CD.$$

Finally, in the reference plane π the two lines DB and DE meet at D , and CD is perpendicular to both:

$$CD \perp DB, \quad CD \perp DE.$$

A second application of XI.4 yields

$$CD \perp \pi.$$

The classical local dependency chain is therefore

$$\begin{array}{c}
 AB \parallel CD, \quad AB \perp \pi \\
 \Downarrow \\
 \text{XI.7} \quad \text{and the joining line } BD \\
 \Downarrow \\
 DE \perp BD, \quad DE = AB \quad (\text{I.11, I.3}) \\
 \Downarrow \\
 AB \perp \pi \Rightarrow \angle ABD, \angle ABE \text{ right} \\
 \Downarrow \\
 \text{I.29} \\
 \Downarrow \\
 CD \perp BD \\
 \Downarrow \\
 \text{I.4, I.8} \\
 \Downarrow \\
 ED \perp AD \\
 \Downarrow \\
 \text{XI.4} \\
 \Downarrow \\
 ED \perp CD \\
 \Downarrow \\
 \text{XI.4} \\
 \Downarrow \\
 CD \perp \pi.
 \end{array}$$

The incidence assertion that CD belongs to the plane through BD, DA also uses the fact that the triangle ABD is planar and that AB, CD, BD share the plane supplied by XI.7. In a formal proof this plane identification cannot be left to the printed diagram.

8.5 What is genuinely three-dimensional here?

There are two distinguished planes.

The first is the reference plane π , to which AB is perpendicular. The second is the common plane σ containing the parallel spatial lines AB and CD . Since AB is not contained in π , the planes are distinct.

The points B and D lie in both planes. Consequently their joining line

$$d = BD$$

lies in both π and σ ; in the formal proof it is the relevant line of intersection of the two plane slices.

The auxiliary line

$$e = DE$$

lies in π , while the line

$$a = DA$$

lies in σ . The proof therefore alternates between two induced plane geometries.

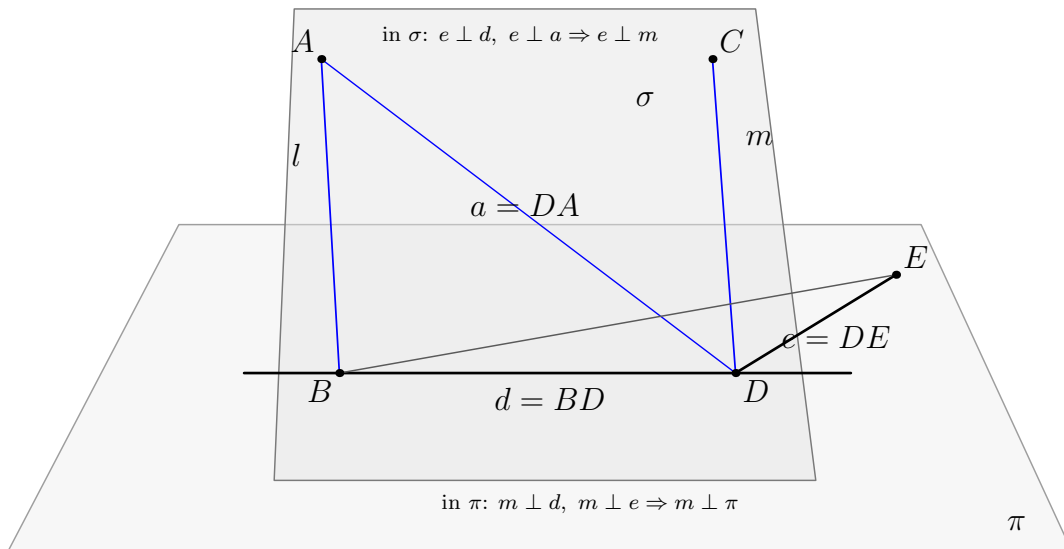


Figure 8.2: The two plane slices used by the reconstruction. The common line $d = BD$ lies in both π and σ . The first XI.4 application uses the two lines $d, a \subset \sigma$ to prove $e \perp \sigma$, hence $e \perp m$. The final XI.4 application uses $d, e \subset \pi$ to prove $m \perp \pi$.

This change of plane is not bookkeeping. It is the spatial geometry of the proposition.

8.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_8
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
```

```

[HSI : HilbertSpaceIncidence Geo]
[_HSO : HilbertSpaceOrder
  (Geo := Geo) (H := H) (S := S)]
[HSC : HilbertSpaceCongruence
  (Geo := Geo) (H := H) (S := S)]
[HSE : HilbertSpaceEuclidean Geo]
(l m : Geo.Line)
(pi : S.Plane)
(B : Geo.Point)
(hParallel : HilbertSpaceLinesParallel Geo l m)
(hPerp : HilbertLinePerpendicularPlaneAt Geo l pi B) :
exists D : Geo.Point,
  HilbertLinePerpendicularPlaneAt Geo m pi D

```

The point D is deliberately not an input. Its existence is part of the theorem. This is stronger as a formal interface than assuming in advance that the second parallel already meets the plane.

8.7 Spatial / Planar Architecture Used

The ambient spatial layer consists of

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt
HilbertLinesPerpendicularAt

```

The proof then uses induced planar geometry in two different fixed planes:

$$\text{PlaneGeo}(Geo, \sigma) \quad \text{and} \quad \text{PlaneGeo}(Geo, \pi).$$

Inside σ , the proof uses neutral planar order, right angles, and the I.27 alternate-angle criterion. Inside π , the final configuration supplies two distinct intersecting lines d, e for XI.4.

The plane-space bridges used in the argument include

```

planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_linesDisjoint_iff_ambient

```

The last bridge was added with the Group IV layer. It identifies disjointness of two lines in a fixed **PlaneGeo** with ambient disjointness of their underlying spatial carriers.

8.8 The Euclidean Layer: Group IV

XI.8 is the first proposition in the present Book XI sequence whose formal proof requires the Euclidean parallel axiom.

The spatial axiom is not installed as a global `HilbertEuclideanPlaneGeo` instance. Such an instance would incorrectly treat the ambient three-space as a plane. Instead the axiom is stated plane-locally:

```
class HilbertSpaceEuclidean
  (Geo : Geometry.Geo)
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  [HilbertSpaceOrder Geo]
  [HilbertSpaceCongruence Geo] : Prop where

parallel_unique_in_plane :
  forall pi : S.Plane,
    forall l : Geo.Line,
      HilbertLineInPlane Geo l pi ->
        forall A : Geo.Point,
          S.OnPlane A pi ->
            Not (H.OnLine A l) ->
              forall b c : Geo.Line,
                HilbertLineInPlane Geo b pi ->
                  HilbertLineInPlane Geo c pi ->
                    H.OnLine A b ->
                      HilbertLinesDisjoint Geo b l ->
                        H.OnLine A c ->
                          HilbertLinesDisjoint Geo c l ->
                            b = c
```

For every fixed ambient plane, the bridge instance

```
instance planeGeoHilbertEuclidean
  ...
  (pi : S.Plane) :
    HilbertEuclideanPlane (PlaneGeo Geo pi)
```

recovers the existing planar Group IV API. The architectural rule is therefore preserved:

Euclidean parallelism is planar, even inside three-space.

8.9 Proof Architecture of the Reconstruction

8.9.1 Step 1: derive the hidden intersection point D

The helper

```
hilbert_XI8_parallel_meets_perpendicular_plane
```

proves that the second parallel m meets the reference plane.

Unpacking

$$\text{HilbertSpaceLinesParallel}(l, m)$$

gives a common plane σ containing l, m , with the two lines disjoint. The foot B of $l \perp \pi$ lies in both π and σ .

The planes are distinct. If $\sigma = \pi$, then $l \subset \pi$, contradicting the theorem

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

The general plane-intersection theorem then supplies a line

$$q = \pi \cap \sigma$$

through B .

Suppose m did not meet q . Then both l and q would be lines through B in σ disjoint from m . Group IV uniqueness would force $l = q$, but $q \subset \pi$, again contradicting $l \not\subset \pi$.

Hence m meets q at some point D , and therefore

$$D \in m \cap \pi.$$

This is the formal proof of a fact Euclid leaves implicit at the opening of XI.8.

8.9.2 Step 2: reuse the XI.6 metric core

With D fixed, the proof reuses

```
hilbert_XI6_second_SSS_right_angle
```

from Proposition XI.6.

The helper constructs points and lines corresponding to Euclid's auxiliary configuration and returns, in particular,

$$d = BD, \quad e = DE,$$

with

$$d, e \subset \pi, \quad e \perp d,$$

and the right angle

$$\angle EDA.$$

Thus the SAS/SSS part of XI.8 is not duplicated. The earlier XI.6 development is used as reusable spatial congruence infrastructure.

8.9.3 Step 3: first application of XI.4 in σ

Let a be the line DA . The proof shows

$$d, a \subset \sigma, \quad d \neq a.$$

The inequality is geometrically meaningful: A lies on the spatial line l but outside the reference plane π , while $d \subset \pi$.

The XI.6 metric result gives

$$e \perp d, \quad e \perp a.$$

Therefore XI.4, applied with the induced geometry of σ , gives

$$e \perp \sigma.$$

Since $m \subset \sigma$ and $D \in m$,

$$e \perp m.$$

This is the formal counterpart of Euclid's first XI.4 step.

8.9.4 Step 4: replace I.29 by neutral I.27 plus Group IV

The remaining task is to prove

$$m \perp d.$$

The proof does not call a public I.29 theorem. Instead, inside $\text{PlaneGeo}(\text{Geo}, \sigma)$, it constructs a line p through D perpendicular to d .

The neutral helper

```
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
```

shows that p and l are disjoint. Its proof is an I.27 argument: two right angles are normalized to equal alternate angles and the neutral alternate-angle theorem produces nonintersection.

But m is also disjoint from l , because $l \parallel m$. Thus m and p are two lines of the same plane σ , both passing through D , both disjoint from l . Group IV uniqueness forces

$$m = p.$$

Since $p \perp d$,

$$m \perp d.$$

This factorization isolates the exact Euclidean content:

$$\underbrace{\text{right angles} \Rightarrow \text{nonintersection}}_{\text{neutral I.27}} + \underbrace{\text{unique parallel through } D}_{\text{Group IV}} \implies m \perp d.$$

It is the formal counterpart of Euclid's I.29 step.

8.9.5 Step 5: final application of XI.4 in π

At this point

$$d, e \subset \pi, \quad m \perp d, \quad m \perp e.$$

The lines d and e are distinct because $e \perp d$.

The final call to

```
euclid_proposition_11_4
```

uses d, e as the two distinct intersecting lines of the reference plane. It returns

$$m \perp \pi$$

at D , which is the required conclusion.

8.10 Lean Formalization

The proposition-local proof is organized around four named helpers:

```
hilbert_XI8_parallel_meets_perpendicular_plane
hilbert_XI8_auxiliary_line_perpendicular_to_parallel
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
hilbert_XI8_parallel_line_perpendicular_to_given_BD
```

The actual Lean DAG is

$$\begin{array}{c}
 l \parallel m, \quad l \perp \pi \text{ at } B \\
 \Downarrow \\
 \text{hilbert_XI8_parallel_meets_perpendicular_plane} \\
 \Downarrow \\
 D \in m \cap \pi \\
 \Downarrow \\
 \text{hilbert_XI6_second_SSS_right_angle} \\
 \Downarrow \\
 d = BD, \quad e = DE, \quad e \perp d, \quad \angle EDA \text{ right} \\
 \Downarrow \\
 \text{XI.4 in } \sigma \\
 \Downarrow \\
 e \perp m \\
 \Downarrow \\
 \text{I.27 in PlaneGeo}(\sigma) + \text{Group IV} \\
 \Downarrow \\
 m \perp d \\
 \Downarrow \\
 \text{XI.4 in } \pi \\
 \Downarrow \\
 \text{euclid_proposition_11_8.}
 \end{array}$$

The production file imports XI.4, XI.5, and XI.6. It does not import XI.7. The common plane that Euclid obtains through XI.7 is already contained in the formal hypothesis `HilbertSpaceLinesParallel`.

8.11 Hidden Formal Data

XI.8 contains substantially more hidden incidence information than its classical diagram suggests.

- The second parallel must be proved to meet the reference plane. The point D is an existential output, not an assumed input.
- The common plane σ of the parallels must be proved distinct from π .
- The intersection line $q = \pi \cap \sigma$ is constructed explicitly.
- The points B, D must be proved distinct from disjointness of the parallel carriers.
- The line $d = BD$ must be shown to lie both in π and in σ .
- The point A selected on l must be proved to lie outside π ; otherwise the line l would lie in π , contrary to $l \perp \pi$.
- The line $a = DA$ must be constructed and shown distinct from d .
- Right-angle and perpendicularity statements must be transported between ambient geometry and PlaneGeo.
- The neutral I.27 argument requires explicit SameSide / OppositeSide and SameRay normalization that Euclid suppresses in the diagram.
- The two disjointness orientations required by Group IV are supplied explicitly.
- Before the final XI.4 application, $d \neq e$ is derived from $e \perp d$.

Subtype projections and record fields occur in the Lean code, but they are implementation details. The mathematically significant hidden data are the plane memberships, distinctness statements, side orientations, and the derived intersection point D .

8.12 Relationship with Earlier Book XI Propositions

The classical and formal dependency graphs differ in a controlled way.

Classically, XI.7 is an explicit opening dependency: it places AB, CD, BD in one plane. XI.4 is then used twice. The classical plane-identification step also relies on the ordinary fact from XI.2 that a triangle lies in one plane.

The production Lean file imports

```
import CGJteamLab.Proposition11_4
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition11_6
```

but does not import Proposition XI.7.

XI.4 is used twice exactly as in the classical architecture: first in the common plane σ , then in the reference plane π .

XI.5 contributes established line–plane perpendicularity infrastructure, including the theorem that a line perpendicular to a plane is not contained in that plane.

XI.6 contributes the reusable metric construction and the spatial SAS/SSS right-angle machinery. This reuse is substantial: the central auxiliary configuration of XI.8 is almost the same as the configuration built for XI.6.

XI.7 is absent from the Lean import graph because `HilbertSpaceLinesParallel` already stores the common carrier plane as part of the definition of spatial parallelism. Thus the formal proof consumes the conclusion-like data that XI.7 exposes, without invoking the proposition theorem itself.

8.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical proof uses I.11 and I.3 to construct the auxiliary perpendicular and equal segment. In the Lean reconstruction these operations are already packaged in the XI.6 helper chain over the Hilbert/Book Zero construction interface.

The metric part corresponds to I.4 and I.8 and is likewise inherited from XI.6.

The parallel step is the important logical boundary. The direction represented by I.27,

$$\text{equal alternate angles} \implies \text{nonintersection},$$

belongs to the neutral congruence/order layer. The reverse use of parallelism represented by I.29 requires the Euclidean parallel axiom. The formal proof displays this by using the neutral I.27 theorem first and applying Group IV only to identify the unique line through D disjoint from l .

No coordinate system, vector space, scalar product, distance formula, trigonometric function, or numerical angle measure is used.

8.14 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_8

#print axioms Geometry.euclid_proposition_11_8
```

and Lean reports:

```
'Geometry.euclid_proposition_11_8' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

These are inherited dependencies already accepted in the Book XI development. XI.8 introduces no new proposition-specific global Lean axiom constant.

The Group IV assumption must be distinguished from this audit output. The public theorem explicitly assumes

```
[HSE : HilbertSpaceEuclidean Geo]
```

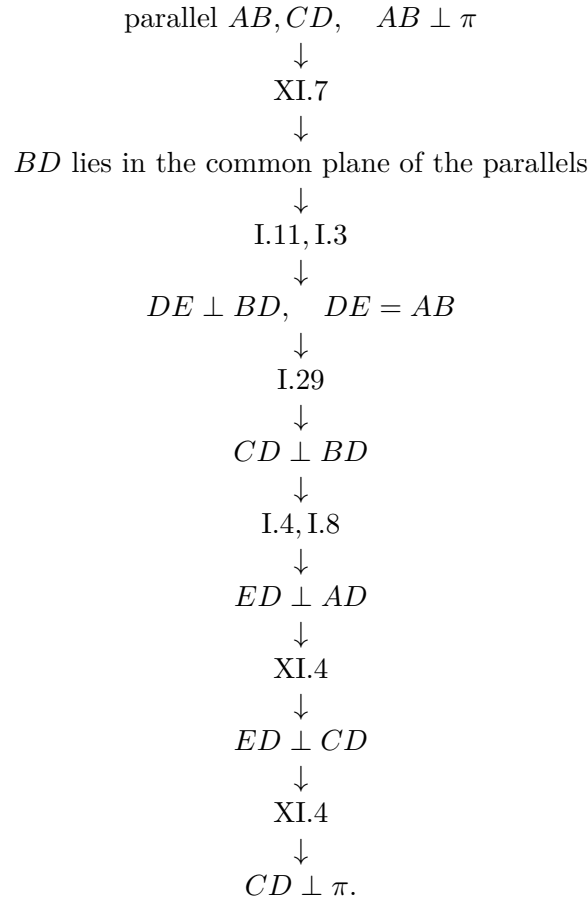
as part of its geometric hypothesis. This typeclass represents the Euclidean parallel axiom for spatial geometry, stated plane-locally. It is therefore a genuine mathematical assumption of XI.8 even though it does not appear as an additional global constant in `#print axioms`.

At this checkpoint:

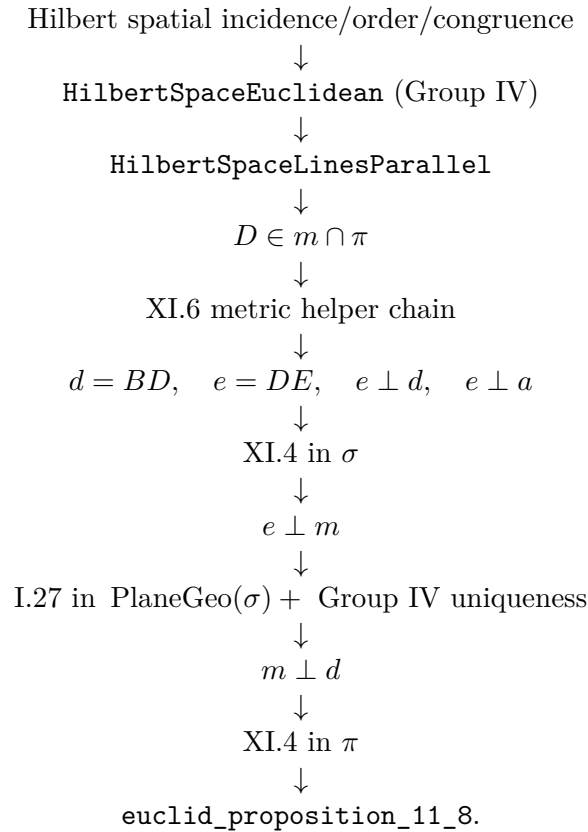
- new proposition-specific geometric axiom: no;
- new spatial axiom declaration in the general architecture: yes, `HilbertSpaceEuclidean`;
- Euclidean parallel axiom / Hilbert Group IV: used;
- new continuity assumption: no;
- new proposition-local helper theorems: yes;
- new reusable `Hilbert3DInterface` theorem: yes, `planeGeo_linesDisjoint_iff_ambient`;
- new induced-plane Euclidean bridge: yes, `planeGeoHilbertEuclidean`;
- global ambient `HilbertEuclideanPlaneGeo`: not installed;
- public theorem compiles: yes, as reported at this checkpoint;
- full repository build: clean, as reported at this checkpoint;
- `sorry/admit`: no.

8.15 Dependency Summary

The classical local chain is



The actual formal chain is



The dependency classification is therefore:

New proposition-specific global axiom	no
New spatial axiom class	yes: <code>HilbertSpaceEuclidean</code>
Continuity	no
Group IV / Euclidean parallelism	yes
Earlier Book XI production imports	XI.4, XI.5, XI.6
Classical explicit Book XI dependency	XI.7, XI.4
New PlaneGeo Euclidean bridge	yes
Coordinates / vectors / scalar product	no
sorry/admit	no

8.16 Conclusion

Proposition XI.8 is the first point in the present Book XI reconstruction where Euclidean parallel uniqueness is mathematically necessary. The formal proof makes that boundary sharper than the classical text: neutral I.27 proves nonintersection, while Group IV supplies the uniqueness needed to recover the I.29 direction.

At the same time, the proof remains fully synthetic. It reuses the metric construction of XI.6, applies XI.4 in two different plane slices, and derives the hidden intersection point of the second parallel with the reference plane instead of assuming it from the diagram.

The result therefore tests three parts of the spatial architecture at once: plane-local Euclidean parallelism, transport between ambient three-space and induced plane geometry, and the universal line–plane perpendicularity predicate. No continuity or analytic representation is introduced.

Chapter 11

Proposition XI.11

11.1 Introduction

Proposition XI.11 is the first Book XI construction in the present reconstruction whose target is the existence of a perpendicular from an arbitrary point outside a plane. Given a plane π and an elevated point $A \notin \pi$, the theorem constructs a line through A that is perpendicular to π .

The proposition is a natural test of the architecture developed in XI.4–XI.8. Its final statement is spatial, but the construction repeatedly moves into ordinary planar Hilbert geometry on explicitly chosen plane slices. In the nontrivial branch three planes are relevant:

$$\pi, \quad \sigma = \text{plane through } A \text{ and } BC, \quad \tau = \text{plane through } AD \text{ and } DE.$$

The proof also exposes an important distinction between the theorem proved and the proof route used. The current production proof follows Euclid’s construction closely and therefore invokes XI.8. Since XI.8 uses the plane-local Euclidean parallel axiom, the public Lean theorem assumes `HilbertSpaceEuclidean`. This records the strength of the current proof, not a claim that the existence statement itself is axiomatically minimal.

11.2 Euclid’s Statement

Euclid’s Proposition XI.11 states:

From a given elevated point to draw a straight line perpendicular to a given plane.

Let A be the elevated point and let π be the plane of reference. The required construction is a line AF through A such that

$$AF \perp \pi.$$

11.3 Classical Configuration

Choose a straight line BC at random in the plane of reference. Draw AD from A perpendicular to BC . If AD is already perpendicular to the reference plane, the construction is complete.

Otherwise draw DE in the reference plane through D , with

$$DE \perp BC.$$

Then draw AF from A perpendicular to DE , and through F draw GH parallel to BC .

Thus the reference plane contains

$$BC, \quad DE, \quad GH,$$

while the two lines

$$AD, \quad AF$$

leave the reference plane and meet it at D and F , respectively.

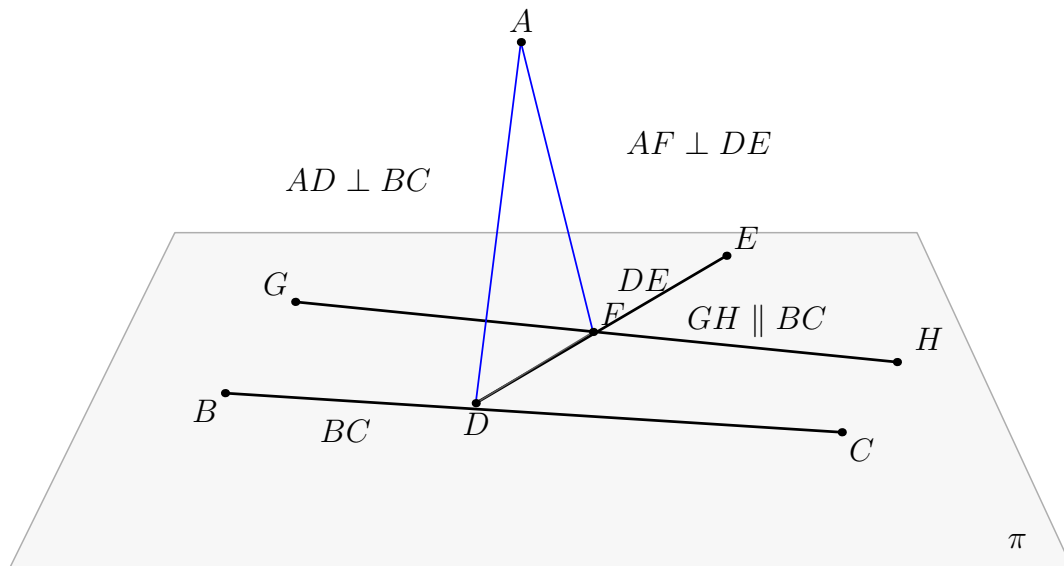


Figure 11.1: The classical configuration of Proposition XI.11. Black lines lie in the reference plane π ; blue segments leave π . The construction starts with $AD \perp BC$, then uses $DE \perp BC$, $AF \perp DE$, and $GH \parallel BC$. The desired conclusion is $AF \perp \pi$.

11.4 Classical Proof

The classical proof is short but structurally dense.

First, choose $BC \subset \pi$ and draw

$$AD \perp BC$$

from the elevated point A by Proposition I.12. Euclid immediately splits into two cases.

If $AD \perp \pi$, the required perpendicular has already been drawn.

Assume therefore that AD is not perpendicular to π . In the reference plane draw

$$DE \perp BC$$

through D by Proposition I.11. Next draw

$$AF \perp DE$$

from A by a second application of I.12. Finally draw through F

$$GH \parallel BC$$

by I.31.

Now BC is perpendicular to both DA and DE . These two lines intersect at D , so Proposition XI.4 gives

$$BC \perp \text{plane}(ED, DA).$$

Since $GH \parallel BC$, Proposition XI.8 gives

$$GH \perp \text{plane}(ED, DA).$$

By XI.Def.3, a line perpendicular to a plane is perpendicular to every line of that plane through the foot. The line AF belongs to $\text{plane}(ED, DA)$ and meets GH at F . Hence

$$GH \perp AF,$$

and therefore also

$$AF \perp GH.$$

But $AF \perp DE$ by construction. Thus AF is perpendicular at F to the two intersecting lines DE and GH of the reference plane. A second application of XI.4 yields

$$AF \perp \text{plane}(DE, GH).$$

The plane through DE and GH is the plane of reference, hence

$$AF \perp \pi.$$

The local classical dependency chain is therefore

$$\begin{array}{c}
A \notin \pi, \quad BC \subset \pi \\
\Downarrow \\
\text{I.12 : } AD \perp BC \\
\Downarrow \\
AD \perp \pi \quad \text{or} \quad AD \not\perp \pi \\
\Downarrow \\
\text{I.11 : } DE \perp BC \\
\Downarrow \\
\text{I.12 : } AF \perp DE \\
\Downarrow \\
\text{I.31 : } GH \parallel BC \\
\Downarrow \\
\text{XI.4 : } BC \perp \text{plane}(ED, DA) \\
\Downarrow \\
\text{XI.8 : } GH \perp \text{plane}(ED, DA) \\
\Downarrow \\
\text{XI.Def.3 : } GH \perp AF \\
\Downarrow \\
AF \perp DE, \quad AF \perp GH \\
\Downarrow \\
\text{XI.4 : } AF \perp \pi.
\end{array}$$

No use of XI.9 or XI.10 occurs in this classical chain. The Book XI dependencies needed locally are XI.4 and XI.8.

11.5 What is genuinely three-dimensional here?

The proof is not simply a planar perpendicular construction repeated in space. The essential issue is selecting the correct plane for each planar construction and then transporting its conclusion back to ambient space.

The first auxiliary plane is

$$\sigma = \text{plane}(A, BC).$$

It exists because $A \notin BC$. Inside σ , the point A and the line BC are ordinary planar objects, so I.12 can be applied in the induced geometry $\text{PlaneGeo}(\sigma)$.

After the foot D is obtained, the line DE is constructed inside the reference plane π .

The second auxiliary plane is

$$\tau = \text{plane}(AD, DE).$$

The lines AD and DE intersect at D and are distinct. Inside $\text{PlaneGeo}(\tau)$, the second I.12 construction produces the line $AF \perp DE$.

The three plane slices have a particularly transparent intersection pattern:

$$\pi \cap \sigma = BC, \quad \sigma \cap \tau = AD, \quad \pi \cap \tau = DE,$$

at the level of the carriers used in the construction.

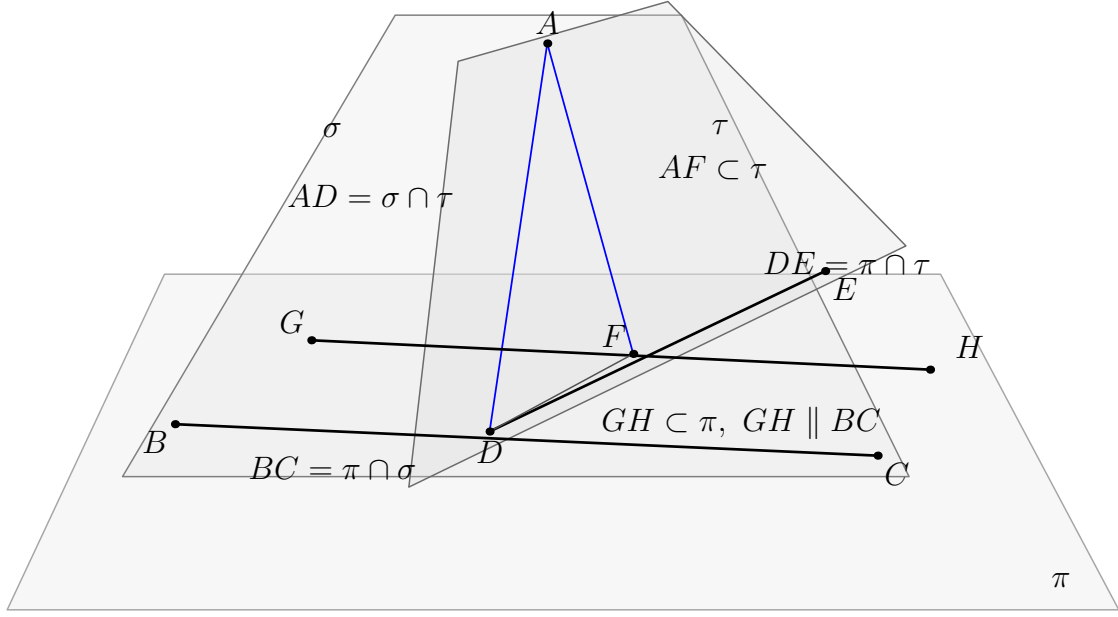


Figure 11.2: The three plane slices used by the reconstruction. The reference plane π contains BC, DE, GH . The plane σ carries the first planar perpendicular construction and shares BC with π . The plane τ carries AD, DE, AF ; it shares AD with σ and DE with π . The final argument transfers perpendicularity from BC to GH through XI.8 and then applies XI.4 in π .

The figure should not be interpreted as a coordinate model. It is only a perspective rendering of incidence and plane membership. The Lean proof uses no coordinates, vectors, scalar products, normals, or numerical angle measure.

11.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_11
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi : S.Plane)
  (A : Geo.Point)
  (hApi : Not (S.OnPlane A pi)) :
  exists l F,
    H.OnLine A l /\
    HilbertLinePerpendicularPlaneAt Geo l pi F

```

The conclusion returns both the perpendicular carrier l and its foot F . It asserts that A lies on l and that l is perpendicular to π at F .

The predicate

```
HilbertLinePerpendicularPlaneAt Geo l pi F
```

is the spatial target relation established for XI.4. It expresses XI.Def.3 synthetically: F lies on the line and in the plane, and the line is perpendicular to every line of the plane through F .

11.7 Spatial / Planar Architecture Used

11.7.1 Ambient spatial layer

The ambient construction uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

There is no global ambient instance

```
HilbertOrder Geo
```

and no global ambient instance

```
HilbertCongruence Geo
```

installed merely to make planar theorems available in space.

11.7.2 Induced planar geometry

Planar constructions are performed in explicitly named plane slices using

```
PlanePoint
PlaneLine
PlaneGeo
```

The proof uses three instances of induced planar geometry:

$$\text{PlaneGeo}(\sigma), \quad \text{PlaneGeo}(\pi), \quad \text{PlaneGeo}(\tau).$$

The first I.12 construction is performed in $\text{PlaneGeo}(\sigma)$. The construction of DE and the I.31

parallel construction are performed in $\text{PlaneGeo}(\pi)$. The second I.12 construction is performed in $\text{PlaneGeo}(\tau)$.

11.7.3 Plane-space bridges

The principal bridges used by the proof include

```
planeGeo_rightAngle_iff_ambient
planeGeo_linesDisjoint_iff_ambient
HilbertSpaceIncidence.line_in_plane
HilbertLinePerpendicularPlaneAt.incidence
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

The proof also uses the ambient symmetry normalizer

```
hilbert_space_linesPerpendicularAt_symm
```

when XI.4 requires the perpendicularity relation in the opposite carrier orientation.

11.8 Proof Architecture of the Reconstruction

The formal proof follows the classical route but expands the incidence and plane-selection steps that the printed diagram leaves implicit.

11.8.1 1. Choosing the reference line

The proof first obtains three noncollinear points of the reference plane:

```
hilbert_three_noncollinear_on_plane
```

Two of them, B, C , determine the line bc . The theorem

```
HilbertSpaceIncidence.line_in_plane
```

then proves

$$bc \subset \pi.$$

Since $A \notin \pi$, it follows immediately that

$$A \notin bc.$$

11.8.2 2. Constructing the first auxiliary plane

The helper

```
hilbert_plane_through_line_and_external_point
```

constructs the plane σ containing bc and A .

This step supplies exactly the missing incidence premise needed before a planar I.12 theorem can be applied. The classical phrase “draw AD from A perpendicular to BC ” presupposes that A and BC have first been put in a common plane.

11.8.3 3. First I.12 construction in σ

Inside

PlaneGeo(σ)

the theorem

```
hilbert_perpendicular_from_point_exists
```

constructs the foot D on bc and gives a right-angle witness. The ambient carrier ad through D, A is then constructed explicitly and proved to lie in σ .

The right-angle data are packaged as

```
HilbertLinesPerpendicularAt Geo bc ad D
```

using the bridge

```
planeGeo_rightAngle_iff_ambient
```

and explicit noncollinearity.

11.8.4 4. Euclid’s case split

The formal proof uses the explicit split

```
by_cases hADpi :  
  HilbertLinePerpendicularPlaneAt Geo ad pi D
```

If this proposition holds, the proof returns ad, D immediately.

This is not a Lean-specific mathematical invention. It is exactly Euclid’s sentence: if AD is also perpendicular to the plane of reference, the construction is already complete.

11.8.5 5. Constructing DE in the reference plane

In the nontrivial branch the point D is known to lie in π . The formal proof constructs opposite points on bc in $\text{PlaneGeo}(\pi)$ and then uses

```
hilbert_right_angle_exists_nondegenerate
```

to obtain a nondegenerate right-angle direction through D . The resulting carrier de satisfies

$$de \subset \pi, \quad bc \perp de \text{ at } D.$$

Thus the formal proof realizes the role of Euclid I.11 without importing a separate I.11 proposition wrapper in the production file.

11.8.6 6. Constructing the second auxiliary plane

The proof establishes

$$ad \neq de.$$

Indeed, if $ad = de$, then A would lie on $de \subset \pi$, contrary to $A \notin \pi$.

The theorem

```
hilbert_plane_through_two_intersecting_lines
```

then constructs the plane τ containing ad and de .

11.8.7 7. Second I.12 construction in τ

Inside

$$\text{PlaneGeo}(\tau)$$

the proof applies

```
hilbert_perpendicular_from_point_exists
```

a second time, now from A to de . This yields the foot F and the carrier af , with

$$af \subset \tau, \quad de \perp af \text{ at } F.$$

Because $de \subset \pi$, the point F also lies in the reference plane.

11.8.8 8. The hidden fact $F \notin BC$

Before I.31 can be applied through F , the formal proof must establish that F does not lie on bc .

Assume $F \in bc$. Since D, F lie on both bc and de , line uniqueness forces $F = D$. The two lines ad and af then contain the two distinct points D, A , so line uniqueness forces

$$ad = af.$$

The proof already knows

$$ad \perp bc, \quad ad \perp de$$

at D . Since $bc, de \subset \pi$, XI.4 would imply

$$ad \perp \pi,$$

contradicting the nontrivial branch assumption.

Therefore

$$F \notin bc.$$

This is one of the main pieces of hidden formal data in XI.11.

11.8.9 9. I.31 in the reference plane

Now B, C, F are noncollinear in $\text{PlaneGeo}(\pi)$. The proof applies

```
euclid_proposition_31
```

to obtain a point G and a line gh through F, G parallel to bc in the induced plane.

The result of I.31 is initially planar point-pair parallelism. The proof converts it to disjointness of the actual carriers using

```
hilbert_mem_pointLine_iff_onLine
```

and then transports planar disjointness back to ambient space by

```
planeGeo_linesDisjoint_iff_ambient
```

Finally it packages

```
HilbertSpaceLinesParallel Geo bc gh
```

with the reference plane π as the explicit common carrier plane.

11.8.10 10. First XI.4: $BC \perp \tau$

In the plane τ , the lines ad and de are distinct and meet at D . The proof has

$$bc \perp ad, \quad bc \perp de.$$

Therefore

```
euclid_proposition_11_4
```

gives

$$bc \perp \tau.$$

This is the first use of XI.4 in the construction.

11.8.11 11. XI.8: transfer to the parallel GH

The proof now has

$$bc \parallel gh, \quad bc \perp \tau.$$

It applies

```
euclid_proposition_11_8
```

and obtains a point K such that

$$gh \perp \tau \text{ at } K.$$

This is exactly the classical XI.8 step, but the formal theorem returns the foot existentially. Euclid's diagram already identifies the relevant intersection as F ; Lean requires that identification to be proved.

11.8.12 12. Normalizing the XI.8 foot: $K = F$

The proof knows from the XI.8 conclusion that

$$K \in gh, \quad K \in \tau,$$

and from the construction that

$$F \in gh, \quad F \in \tau.$$

Suppose $K \neq F$. Since the two distinct points K, F lie on gh and in τ , spatial incidence gives

$$gh \subset \tau.$$

But $gh \perp \tau$ at K . Applying the universal line–plane-perpendicularity condition to the line gh itself would give

$$gh \perp gh.$$

The theorem

```
hilbert_linesPerpendicularAt_ne
```

then yields the absurd conclusion $gh \neq gh$.

Hence

$$K = F.$$

This argument is purely synthetic and uses the nondegeneracy already built into `HilbertLinesPerpendicularAt`.

11.8.13 13. XI.Def.3 gives $AF \perp GH$

After replacing K by F , the proof has

$$gh \perp \tau \text{ at } F.$$

Since

$$af \subset \tau, \quad F \in af,$$

the universal line condition is unpacked by

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

to obtain

$$gh \perp af.$$

The symmetry helper

```
hilbert_space_linesPerpendicularAt_symm
```

normalizes this to

$$af \perp gh.$$

The same helper normalizes the previously constructed $de \perp af$ to

$$af \perp de.$$

11.8.14 14. Proving $DE \neq GH$

The final XI.4 application requires two distinct lines of the reference plane. This distinctness is not left to the diagram.

Suppose

$$de = gh.$$

Since $D \in de$, it follows that $D \in gh$. But $D \in bc$, while the spatial parallelism package $bc \parallel gh$ contains disjointness of these two ambient carriers. Contradiction.

Thus

$$de \neq gh.$$

11.8.15 15. Final XI.4 in the reference plane

Both de and gh lie in π , meet at F , and are distinct. The proof has

$$af \perp de, \quad af \perp gh.$$

A final call to

```
euclid_proposition_11_4
```

therefore gives

$$af \perp \pi \text{ at } F.$$

The returned witnesses are af and F .

11.9 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition12
import CGJteamLab.Proposition31
```

The direct imports therefore expose XI.8, the Book I perpendicular-from- external-point construction used as I.12, and I.31.

XI.4 is invoked twice by the proof and is available transitively through the established Book XI import chain. The production theorem does not import or invoke XI.9 or XI.10.

The principal reusable declarations consumed by XI.11 are:

```

hilbert_three_noncollinear_on_plane
hilbert_plane_through_line_and_external_point
hilbert_perpendicular_from_point_exists
planeGeo_rightAngle_iff_ambient
planeGeo_opposite_points_on_line_congruent_to
hilbert_right_angle_exists_nondegenerate
hilbert_plane_through_two_intersecting_lines
hilbert_space_linesPerpendicularAt_symm
euclid_proposition_31
hilbert_mem_pointLine_iff_onLine
planeGeo_linesDisjoint_iff_ambient
euclid_proposition_11_4
euclid_proposition_11_8
HilbertLinePerpendicularPlaneAt.incidence
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
hilbert_linesPerpendicularAt_ne

```

No XI.11-specific helper theorem is exported. The full construction is contained in the proof of `euclid_proposition_11_11`.

11.10 Hidden Formal Data

The classical diagram suppresses several obligations that are explicit in the Lean proof.

11.10.1 The first I.12 needs a plane

The statement $A \notin \pi$ does not by itself put A and the chosen line $BC \subset \pi$ in a common planar geometry. The plane σ through A and BC must be constructed before planar I.12 can be invoked.

11.10.2 Every constructed line needs a carrier proof

After constructing ambient lines ad , de , af , and gh , the proof records explicitly which plane contains each line. These are the data that permit later calls to XI.4, XI.8, and induced-plane theorems.

11.10.3 The two feet are not automatically the same

XI.8 returns a foot K of gh on τ . The earlier construction already supplied the point $F \in gh \cap \tau$. The equality $K = F$ is a real incidence theorem, not a definitional simplification.

11.10.4 Distinctness for the final XI.4

The statement $de \neq gh$ follows from the disjointness part of $bc \parallel gh$, but it must be proved explicitly before the two lines can be packaged as distinct `PlaneLines` of π .

11.10.5 Orientation of perpendicularity

The project relation `HilbertLinesPerpendicularAt` carries ordered line arguments. Several classical sentences silently use symmetry of perpendicularity. The proof normalizes these orientations explicitly with `hilbert_space_linesPerpendicularAt_symm`.

11.11 Relationship with Earlier Book XI Propositions

The only earlier Book XI propositions invoked by the mathematical proof are XI.4 and XI.8.

XI.4 is used twice:

$$bc \perp ad, \quad bc \perp de \implies bc \perp \tau,$$

and later

$$af \perp de, \quad af \perp gh \implies af \perp \pi.$$

XI.8 is used once:

$$bc \parallel gh, \quad bc \perp \tau \implies gh \perp \tau.$$

The source audit is important here because the current working sequence skips XI.9 and XI.10. Heath's proof of XI.11 does not cite either proposition. Nor does the production Lean source import or invoke them. They are therefore genuinely absent from both the classical local DAG and the actual Lean DAG for XI.11.

XI.7 is also not a local dependency of XI.11. The spatial parallelism used here is created directly by I.31 inside the reference plane and then packaged as `HilbertSpaceLinesParallel`.

11.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical Book I dependencies are:

$$\text{I.12, I.11, I.12, I.31.}$$

The formal proof calls the established I.12 theorem

```
hilbert_perpendicular_from_point_exists
```

twice, once in σ and once in τ .

The role of I.11 is reconstructed from the lower-level right-angle construction interface in $\text{PlaneGeo}(\pi)$. The proof does not import a separate proposition-I.11 wrapper.

The I.31 step is represented directly by

```
euclid_proposition_31
```

inside $\text{PlaneGeo}(\pi)$.

Book Zero enters transitively through the established planar Hilbert construction and right-angle infrastructure. XI.11 introduces no new Book Zero declaration.

The ambient three-space remains separated from planar order and congruence. All ordinary planar constructions are carried out in a named **PlaneGeo**; the spatial classes provide only the incidence, congruence transport, and plane-local Euclidean data required to connect those slices.

11.13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_11

#print axioms Geometry.euclid_proposition_11_11
```

and Lean reports:

```
'Geometry.euclid_proposition_11_11' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

This is the same inherited global axiom profile already accepted at XI.8. There is no new XI.11-specific global axiom constant and no **sorry/admit**.

The explicit theorem signature nevertheless contains

```
[HSE : HilbertSpaceEuclidean Geo]
```

because the proof invokes XI.8. Thus Group IV is a genuine hypothesis of the current production theorem.

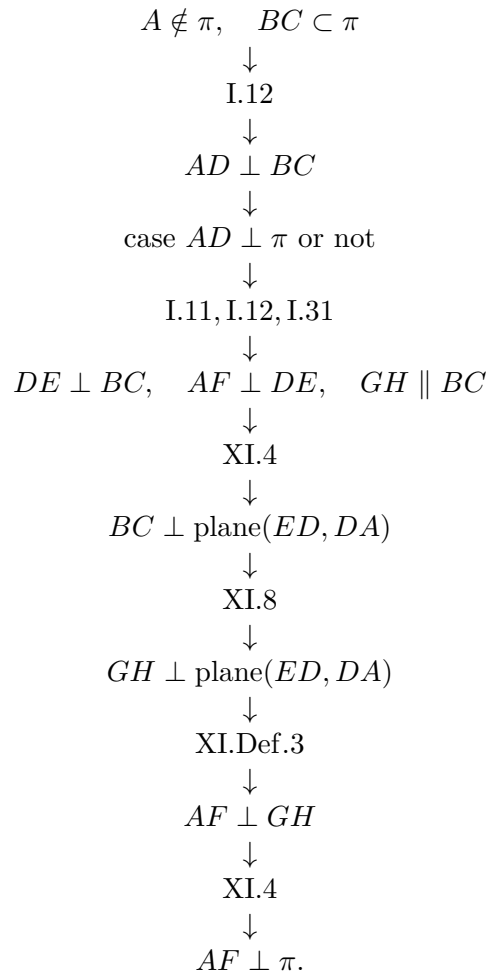
This must not be confused with a claim of logical necessity for the target statement. The present theorem records an Euclid-faithful construction through XI.8. The source comparison in the Borsuk-Szmielew appendix shows that a stronger, axiomatically more economical existence-and-uniqueness theorem is available in a pre-Euclidean congruence development. Therefore a future neutral strengthening of the Lean result is a legitimate separate project question, not part of the present XI.11 production milestone.

At this checkpoint:

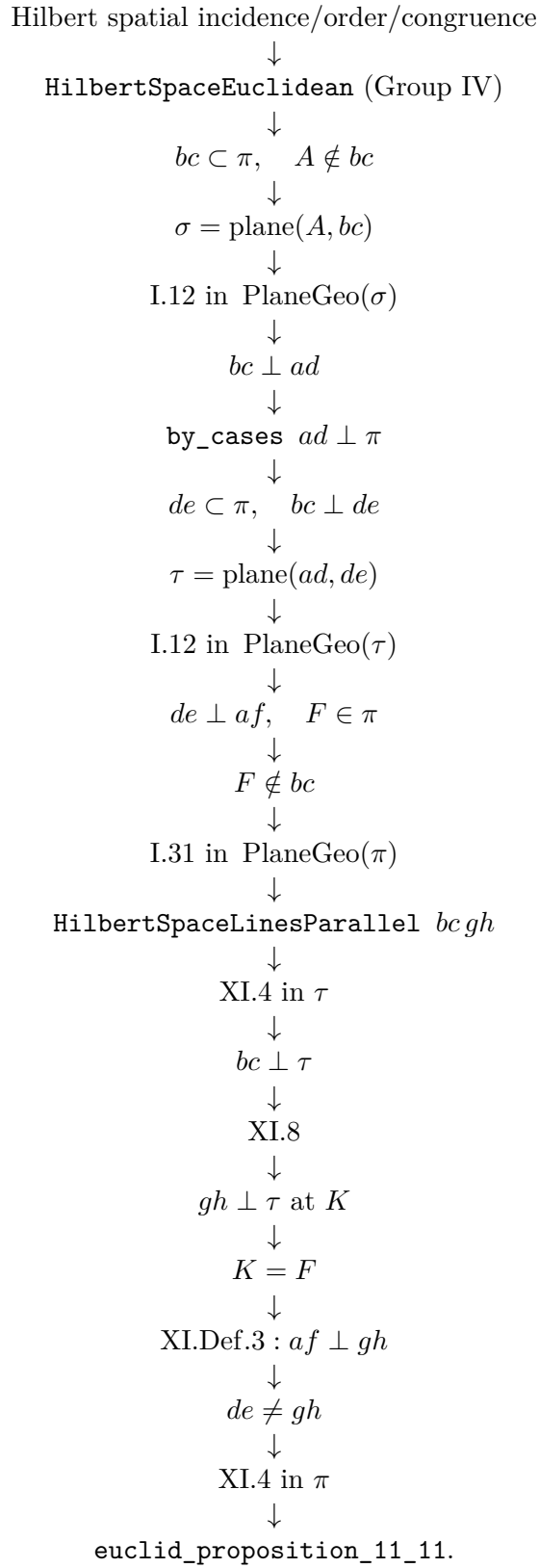
- new proposition-specific geometric axiom: no;
- new spatial axiom declaration: no;
- Euclidean parallel axiom / Hilbert Group IV: used, via XI.8;
- new continuity assumption: no;
- new proposition-local helper theorem: no;
- new reusable `Hilbert3DInterface` theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- new `PlaneGeo` bridge theorem: no;
- global ambient `HilbertOrderGeo`: not installed;
- global ambient `HilbertCongruenceGeo`: not installed;
- public theorem compiles: yes, as reported at this checkpoint;
- full repository build: clean, as reported at this checkpoint;
- axiom audit: completed;
- `sorry/admit`: no.

11.14 Dependency Summary

The classical local chain is



The actual formal chain is



The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	yes, via XI.8
Direct production imports	XI.8, I.12 module, I.31
Classical explicit Book XI dependencies	XI.4, XI.8
XI.9 / XI.10	not used
New proposition-local helper theorem	no
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
sorry/admit	no

11.15 Conclusion

Proposition XI.11 is a compact classical construction whose formal reconstruction exercises almost the entire plane-space interface developed so far. The key mathematical idea remains Euclid's: construct two successive planar perpendiculars, introduce a parallel in the reference plane, transfer a line-plane perpendicularity by XI.8, and close with XI.4.

Lean makes the spatial bookkeeping visible. The first I.12 needs the plane σ ; the second needs the plane τ ; the I.31 construction must be converted from planar parallelism to ambient spatial parallelism; the foot returned by XI.8 must be proved equal to F ; and the final two reference-plane carriers must be proved distinct.

No analytic geometry enters. The production theorem is fully synthetic, compiles in the current project, and has the inherited axiom audit

```
[propext, Classical.choice, Geometry.bookZero_nullSegment1, Quot.sound].
```

Its present Group IV hypothesis is inherited from the Euclid-faithful use of XI.8. The independent source comparison shows that reduction of this hypothesis is a possible later strengthening, not an unresolved defect in the completed XI.11 milestone.

Chapter 12

Proposition XI.12

12.1 Introduction

Proposition XI.12 is the companion construction to XI.11. Proposition XI.11 constructs a perpendicular to a plane from a point outside the plane; XI.12 starts with a point A already lying in the reference plane π and constructs a line through A perpendicular to π .

The classical proof is exceptionally short. Euclid chooses an elevated point, uses XI.11 to obtain one perpendicular to the reference plane, draws through A a parallel to that perpendicular by I.31, and then uses XI.8 to transfer the line–plane perpendicularity to the new line.

The Lean reconstruction follows this classical route, but two incidence obligations that are invisible in the printed proof become explicit. First, the perpendicular obtained from XI.11 may already meet the reference plane at A . In that case it is itself the required line, and there is no new parallel to construct through A . Second, XI.8 returns the foot of the new perpendicular existentially; Lean must prove that this foot is the already specified point A .

These two obligations are handled by the proposition-local theorem `hilbert_XI12_perpendicular_foot_unique`. The other proposition-local helper, `hilbert_XI12_parallel_through_point_in_plane`, isolates the planar I.31 construction and packages its result as ambient spatial parallelism.

12.2 Euclid’s Statement

Euclid’s Proposition XI.12 states:

To set up a straight line at right angles to a given plane from a given point in it.

Let A be the given point of the reference plane π . The required construction is a straight line through A satisfying

$$AD \perp \pi.$$

12.3 Classical Configuration

Let B be an arbitrary elevated point. By XI.11 draw BC perpendicular to the reference plane, with $C \in \pi$. Through the given point $A \in \pi$, draw AD parallel to BC .

The generic classical configuration is shown in Figure 12.1. The two blue lines leave the reference plane. Their hidden continuations below the plane are dashed, following the standing Book XI figure convention.

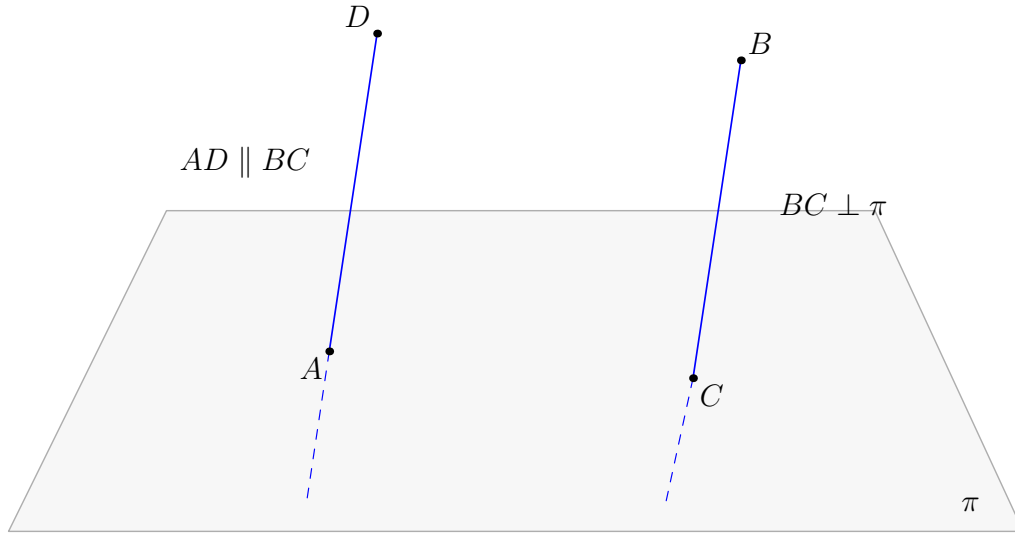


Figure 12.1: The classical configuration of Proposition XI.12. The points A, C lie in the reference plane π . XI.11 gives $BC \perp \pi$, and I.31 draws $AD \parallel BC$ through A . XI.8 then gives $AD \perp \pi$. The figure shows the generic case $A \neq C$; if $A = C$, the line BC already solves the construction.

12.4 Classical Proof

Let A be the given point in the reference plane. Choose any elevated point B . By Proposition XI.11 draw

$$BC \perp \pi.$$

Through A , draw

$$AD \parallel BC$$

by Proposition I.31. Since AD and BC are parallel and BC is perpendicular to the reference plane, Proposition XI.8 gives

$$AD \perp \pi.$$

Thus AD is the required straight line.

The local classical dependency chain is therefore

$$\begin{array}{c}
A \in \pi \\
\Downarrow \\
\text{choose an elevated point } B \\
\Downarrow \\
\text{XI.11} \\
\Downarrow \\
BC \perp \pi \\
\Downarrow \\
\text{I.31} \\
\Downarrow \\
AD \parallel BC, \quad A \in AD \\
\Downarrow \\
\text{XI.8} \\
\Downarrow \\
AD \perp \pi.
\end{array}$$

Neither XI.9 nor XI.10 occurs in this local proof. The only earlier Book XI propositions used explicitly are XI.11 and XI.8.

12.5 What is genuinely three-dimensional here?

The I.31 step is planar, but the input line BC is not a line of the reference plane: it is perpendicular to that plane and meets it only at its foot C . To apply a planar parallel theorem through A , the formal reconstruction first has to place the spatial line and the point A in a common auxiliary plane.

In the nontrivial branch this plane is

$$\sigma = \text{plane}(BC, A).$$

Inside $\text{PlaneGeo}(\sigma)$, I.31 becomes an ordinary planar construction. Its output must then be transported back to ambient three-space and packaged as

$$\text{HilbertSpaceLinesParallel}(BC, AD).$$

Only after this packaging can XI.8 be applied.

The other genuinely spatial ingredient is the meaning of a line perpendicular to a plane. The predicate

```
HilbertLinePerpendicularPlaneAt Geo l pi A
```

contains the incidence of the foot with both line and plane and the universal assertion that the line is perpendicular to every line of the plane through the foot. This universal structure is what makes the foot-uniqueness argument possible.

12.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_12
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi : S.Plane)
  (A : Geo.Point)
  (hApi : S.OnPlane A pi) :
  exists l : Geo.Line,
    HilbertLinePerpendicularPlaneAt Geo l pi A

```

The theorem therefore returns an ambient line l that is perpendicular to the given plane exactly at the prescribed point A .

12.7 Spatial / Planar Architecture Used

12.7.1 Ambient spatial layer

The ambient proof uses

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt

```

As throughout the current Book XI architecture, there is no global ambient

```

HilbertOrder Geo
HilbertCongruence Geo

```

installed to make plane geometry available in space.

12.7.2 Induced planar geometry

The only new plane slice constructed specifically by XI.12 is σ , the plane through the XI.11 perpendicular and the point A , in the branch where A does not already lie on that line.

The I.31 construction is then carried out in

PlaneGeo(σ).

The helper `hilbert_XI12_parallel_through_point_in_plane` selects two points of the ambient line, views them together with A as `PlanePoints` of σ , applies the established planar parallel-existence theorem, and converts the result back to an ambient line.

12.7.3 Plane-space bridges

The principal bridge operations used locally are

```
planeGeo_primCollinear_to_ambient
hilbert_on_line_of_primCollinear_with_two_on_line
hilbert_mem_pointLine_iff_onLine
HilbertSpaceIncidence.line_in_plane
```

The planar I.31 output is not identified with ambient spatial parallelism by definition. The proof explicitly establishes ambient disjointness and supplies σ as the common carrier plane.

12.8 Proof Architecture of the Reconstruction

12.8.1 1. Choosing an elevated point and invoking XI.11

The proof first calls

```
hilbert_point_off_plane
```

to choose a point $B \notin \pi$. It then invokes

```
euclid_proposition_11_11
```

and obtains an ambient line l and a point F such that

$$B \in l, \quad l \perp \pi \text{ at } F.$$

At this point the formal proof has exactly the spatial object that Euclid denotes by BC .

12.8.2 2. The exceptional incidence branch

The classical text immediately asks for a parallel through A . Formally, however, one must first decide whether A already lies on the XI.11 line l :

```
by_cases hA1 : H.OnLine A l
```

If $A \in l$, then both A and the perpendicular foot F lie in the reference plane and on l . The helper

```
hilbert_XI12_perpendicular_foot_unique
```

proves

$$A = F.$$

Hence $l \perp \pi$ at A , and the theorem is complete in this branch. No I.31 construction is needed.

12.8.3 3. Why the perpendicular foot is unique

The helper has the normalized signature

```
theorem hilbert_XI12_perpendicular_foot_unique
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (pi : S.Plane)
  (l : Geo.Line)
  (C A : Geo.Point)
  (hPerp :
    HilbertLinePerpendicularPlaneAt Geo l pi C)
  (hA1 : H.OnLine A l)
  (hApi : S.OnPlane A pi) :
  A = C
```

Suppose $A \neq C$. Since the two distinct points A, C lie both on l and in π , the spatial incidence axiom `line_in_plane` forces

$$l \subset \pi.$$

But $l \perp \pi$ at C . The universal defining clause of line–plane perpendicularity can therefore be applied to l itself, giving

$$l \perp l.$$

The established theorem `hilbert_linesPerpendicularAt_ne` then yields $l \neq l$, a contradiction. Thus the common point of a perpendicular line and its plane is unique.

12.8.4 4. Constructing the auxiliary carrier plane

In the second branch $A \notin l$. The theorem

```
hilbert_plane_through_line_and_external_point
```

constructs a plane σ satisfying

$$l \subset \sigma, \quad A \in \sigma.$$

This is the precise formal replacement for the plane that the classical diagram leaves implicit when it draws a line through A parallel to the spatial line BC .

12.8.5 5. I.31 inside the induced plane

The helper

```
hilbert_XI12_parallel_through_point_in_plane
```

has the normalized signature

```
theorem hilbert_XI12_parallel_through_point_in_plane
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (sigma : S.Plane)
  (l : Geo.Line)
  (A : Geo.Point)
  (hlsigma : HilbertLineInPlane Geo l sigma)
  (hAsigma : S.OnPlane A sigma)
  (hA1 : Not (H.OnLine A l)) :
  exists m : Geo.Line,
    H.OnLine A m /\
    HilbertSpaceLinesParallel Geo l m
```

The proof chooses two distinct points B, C on l , proves that B, C, A are noncollinear in $\text{PlaneGeo}(\sigma)$, and then calls

```
hilbert_parallel_through_point_exists
```

inside the induced plane.

This is the formal I.31 step. The production module imports `Proposition31`, but the local helper calls the established lower-level parallel-existence theorem directly.

12.8.6 6. Packaging planar parallelism as spatial parallelism

The planar theorem initially gives disjoint `PointLine` carriers in $\text{PlaneGeo}(\sigma)$. The proof constructs the actual ambient carrier m , proves that it lies in σ , and shows that an ambient intersection point of l and m would contradict the planar disjointness.

It can therefore package

$$l \parallel m$$

in the spatial sense

```
HilbertSpaceLinesParallel Geo 1 m
```

with σ as the explicit common carrier plane. This step excludes skew lines by construction.

12.8.7 7. XI.8 transfers perpendicularity

The proof now has

$$l \parallel m, \quad l \perp \pi.$$

It applies

```
euclid_proposition_11_8
```

and obtains a point D such that

$$m \perp \pi \text{ at } D.$$

This is the exact classical XI.8 step. The only extra issue is that the formal theorem returns D existentially, whereas the target of XI.12 requires the foot to be the prescribed point A .

12.8.8 8. The second foot normalization

The constructed line m passes through A , and the original hypothesis gives $A \in \pi$. Applying `hilbert_XI12_perpendicular_foot_unique` to the XI.8 conclusion therefore gives

$$A = D.$$

After this substitution the conclusion is exactly

$$m \perp \pi \text{ at } A.$$

This closes the nontrivial branch and completes XI.12.

12.9 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_11
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition31
```

The two proposition-local declarations introduced by XI.12 are

```
hilbert_XI12_perpendicular_foot_unique
hilbert_XI12_parallel_through_point_in_plane
```

and the public theorem is

```
euclid_proposition_11_12
```

The first helper is a general uniqueness fact about a perpendicular line and its plane. The second is a general plane-slice packaging fact for parallel construction. Both are currently located in the proposition module; their signatures are sufficiently general that later promotion to `Hilbert3DInterface` may be considered if subsequent Book XI proofs reuse them.

12.10 Hidden Formal Data

12.10.1 The classical parallel construction has an exceptional case

Euclid chooses an arbitrary elevated point B , drops the XI.11 perpendicular, and then draws through A a parallel to it. The printed proof does not discuss the possibility that the foot of the XI.11 perpendicular is already A .

The formal proof cannot ignore this case. If $A \in l$, foot uniqueness shows that A is the foot and l is already the required perpendicular. Only when $A \notin l$ is the I.31 construction invoked.

12.10.2 I.31 needs an explicit plane slice

A planar parallel theorem cannot be applied directly to an arbitrary ambient spatial line and a point. The carrier plane σ through l and A is constructed first, and the entire I.31 step is performed inside `PlaneGeo( $\sigma$ )`.

12.10.3 Planar and spatial parallelism are different data

The I.31 result is initially a planar point-pair parallel relation. XI.8 expects `HilbertSpaceLinesParallel`, which includes an explicit common plane and ambient disjointness. The helper proves and packages these data rather than treating the two relations as definitionally identical.

12.10.4 XI.8 returns a new foot

XI.8 proves that the parallel line is perpendicular to π , but its formal conclusion contains an existentially returned foot D . The equality $D = A$ is a genuine incidence theorem and is

discharged by the same foot-uniqueness helper used in the first branch.

12.11 Relationship with Earlier Book XI Propositions

The classical local Book XI dependencies are exactly XI.11 and XI.8.

XI.11 supplies a line perpendicular to the reference plane from an elevated point:

$$B \notin \pi \implies \exists l, F, \quad B \in l, \quad l \perp \pi \text{ at } F.$$

XI.8 supplies the transfer along spatial parallel lines:

$$l \parallel m, \quad l \perp \pi \implies m \perp \pi.$$

The production file imports both propositions directly. XI.9 and XI.10 are neither classical local dependencies nor Lean dependencies of XI.12.

XI.4 is not called directly by XI.12, but it remains in the transitive proof history through XI.11 and XI.8.

12.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The only explicit classical Book I proposition in XI.12 is I.31. In the formal proof its role is implemented by

```
hilbert_parallel_through_point_exists
```

inside `PlaneGeo( $\sigma$ )`.

The XI.12-specific parallel helper does not itself require `HilbertSpaceEuclidean`; its theorem signature stops at spatial order and congruence. The Euclidean Group IV assumption of the public XI.12 theorem is inherited from the calls to XI.11 and XI.8.

Book Zero contributes no new XI.12 declaration. The final public axiom profile contains the already accepted inherited dependency `bookZero_nullSegment1`, whereas both new proposition-local helpers audit without that dependency.

The ambient three-space continues to obey the Book XI instance discipline: ordinary order and congruence are used only inside explicit `PlaneGeo` slices, while the ambient layer uses the spatial classes.

12.13 Axiomatic Status

The audit command for the public theorem is

```
import CGJteamLab.Proposition11_12
```

```
#print axioms Geometry.euclid_proposition_11_12
```

and Lean reports

```
'Geometry.euclid_proposition_11_12' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

The two new helpers were audited separately:

```
'Geometry.hilbert_XI12_perpendicular_foot_unique'
depends on axioms: [propext, Classical.choice, Quot.sound]

'Geometry.hilbert_XI12_parallel_through_point_in_plane'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

Thus XI.12 introduces no new proposition-specific global axiom constant. In particular, the two new helpers do not introduce `bookZero_nullSegment1`; that dependency is inherited by the public theorem through the established construction chain.

The explicit signature nevertheless contains

```
[HSE : HilbertSpaceEuclidean Geo]
```

so Hilbert Group IV is a genuine hypothesis of the current production theorem. It is consumed through XI.11 and XI.8. No continuity assumption is used.

The current theorem records the strength of the Euclid-faithful proof route. It should not be read as a minimality claim about the abstract existence theorem.

At this checkpoint:

- new proposition-specific geometric axiom: no;
- new spatial axiom declaration: no;
- Euclidean parallel axiom / Hilbert Group IV: used, via XI.11 and XI.8;
- new continuity assumption: no;
- new proposition-local helper theorem: yes, two;
- new reusable `Hilbert3DInterface` theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- new `PlaneGeo` bridge theorem: no;
- global ambient `HilbertOrderGeo`: not installed;
- global ambient `HilbertCongruenceGeo`: not installed;

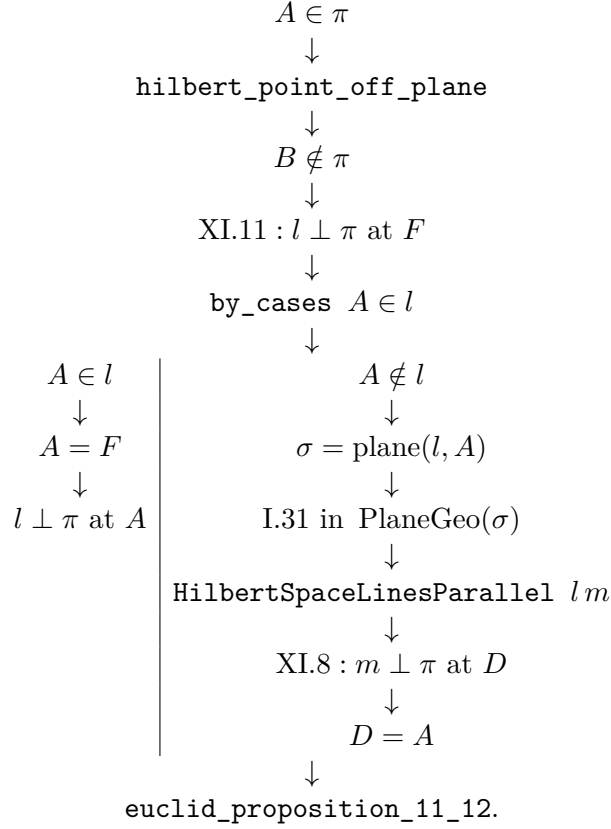
- public theorem compiles: yes, as reported at this checkpoint;
- full repository build: clean, as reported at this checkpoint;
- axiom audit: completed;
- sorry/admit: no.

12.14 Dependency Summary

The classical local chain is

$$\begin{array}{c}
 A \in \pi \\
 \downarrow \\
 B \text{ elevated} \\
 \downarrow \\
 \text{XI.11} \\
 \downarrow \\
 BC \perp \pi \\
 \downarrow \\
 \text{I.31} \\
 \downarrow \\
 AD \parallel BC \\
 \downarrow \\
 \text{XI.8} \\
 \downarrow \\
 AD \perp \pi.
 \end{array}$$

The actual formal chain is



The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	yes, via XI.11 and XI.8
Direct production imports	XI.11, XI.8, I.31 module
Classical explicit Book XI dependencies	XI.11, XI.8
XI.9 / XI.10	not used
New proposition-local helper theorem	yes, two
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
sorry/admit	no

12.15 Conclusion

Proposition XI.12 has a minimal classical proof but a revealing formal shape. Euclid's three-step route remains visible:

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8}.$$

Lean adds no alternative geometry. It makes explicit the carrier plane needed for the planar parallel construction, the exceptional case in which the XI.11 perpendicular already passes through the prescribed point, the conversion from planar to spatial parallelism, and the identification of the foot returned by XI.8.

The two new helpers are axiomatically clean relative to the inherited Book Zero dependency. The public theorem has the same global axiom profile as XI.11 and XI.8:

`[propext, Classical.choice, Geometry.bookZero_nullSegment1, Quot.sound]`.

The current result remains fully synthetic and uses no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure. Its Group IV hypothesis belongs to the chosen Euclid-faithful route through XI.11 and XI.8; it is not a claim that the abstract perpendicular-existence statement intrinsically requires Euclidean parallelism.

Chapter 13

Proposition XI.13

13.1 Introduction

Proposition XI.13 is the uniqueness theorem that follows the two perpendicular-construction propositions XI.11 and XI.12. Euclid now assumes that two straight lines have been erected from the same point at right angles to the same plane and proves that, on the same side of the plane, the configuration is impossible.

The classical proof is short but genuinely spatial. The two supposed perpendiculars determine an auxiliary plane. That plane intersects the reference plane in a straight line through the common foot. Each of the two perpendiculars is therefore perpendicular to the same intersection line. Euclid then reaches a planar impossibility inside the auxiliary plane.

The formal reconstruction keeps the same spatial core but packages the last planar uniqueness argument differently. Since `Geo.Line` is an unoriented extensional carrier, the phrase “on the same side” has no independent content at the level of line equality: opposite rays through the common point belong to the same ambient line. The formal target is therefore equality of the two line carriers.

Rather than reconstructing a separate proposition-local theorem for uniqueness of a perpendicular ray in the auxiliary plane, the Lean proof reuses XI.4. If the intersection line of the two planes were perpendicular to two distinct lines of the auxiliary plane, XI.4 would make that intersection line perpendicular to the whole auxiliary plane. But the intersection line itself lies in that plane, which is impossible.

13.2 Euclid’s Statement

Heath gives Proposition XI.13 as follows:

From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

Assume, for contradiction, that AB and AC are two such straight lines erected from the common point A of the reference plane π .

13.3 Classical Configuration

Draw the plane through AB and AC . Denote it by σ . By XI.3 its section with the reference plane is a straight line through A ; Euclid writes this line as DAE .

Thus the classical configuration contains

$$AB, AC, DAE \subset \sigma, \quad DAE \subset \pi, \quad A \in \pi \cap \sigma.$$

The configuration is shown in Figure 13.1. The reference plane π is the light gray plane. The auxiliary plane σ contains the two blue spatial lines and intersects π along the black line DAE .

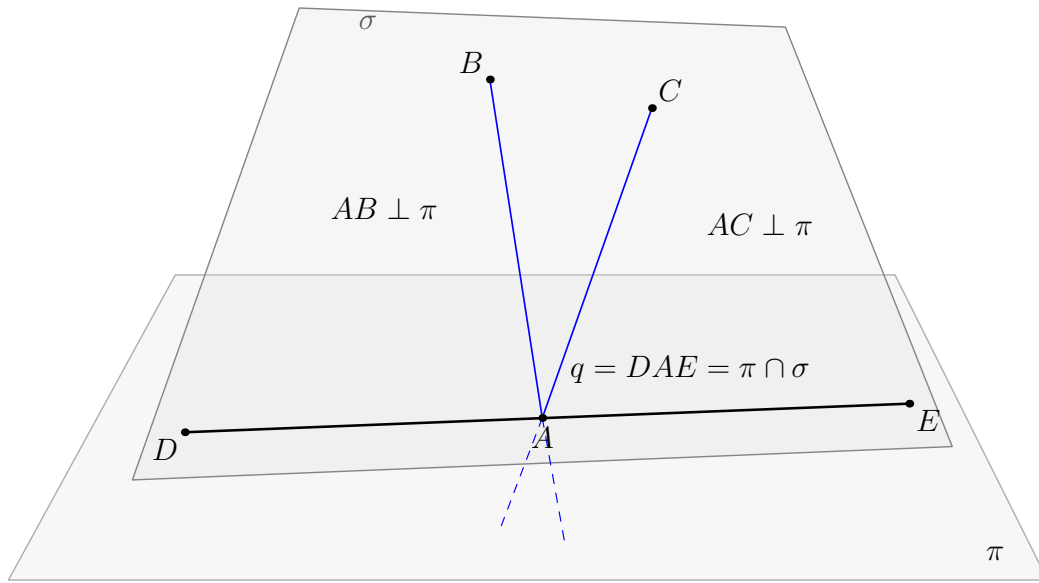


Figure 13.1: The classical configuration of Proposition XI.13. The two supposed perpendiculars AB and AC meet the reference plane π at the same point A and lie in the auxiliary plane σ . By XI.3 the planes π and σ intersect in the straight line DAE through A . Since both spatial lines are perpendicular to π , each is perpendicular to DAE .

13.4 Classical Proof

Suppose that AB and AC are distinct straight lines erected from A at right angles to the same plane π , on the same side.

Draw the plane σ through AB and AC . By XI.3 the intersection of σ with π is a straight line through A ; write it as DAE .

Since $AC \perp \pi$, Definition XI.3 says that AC is perpendicular to every straight line in π meeting it at the foot. Hence

$$AC \perp DAE.$$

For the same reason,

$$AB \perp DAE.$$

Therefore both $\angle CAE$ and $\angle BAE$ are right angles and hence equal. The two angles lie in the same plane σ , share the same side AE , and the remaining sides AC and AB are assumed to lie on the same side. Euclid concludes that this is impossible.

The local classical dependency chain is therefore

$$\begin{array}{c}
 AB \perp \pi, \quad AC \perp \pi, \quad A \in \pi \\
 \Downarrow \\
 \sigma = \text{plane}(AB, AC) \\
 \Downarrow \\
 \text{XI.3} \\
 \Downarrow \\
 DAE = \pi \cap \sigma \\
 \Downarrow \\
 \text{XI.Def.3} \\
 \Downarrow \\
 AB \perp DAE, \quad AC \perp DAE \\
 \Downarrow \\
 \angle BAE, \angle CAE \text{ are equal right angles in } \sigma \\
 \Downarrow \\
 \text{planar uniqueness / contradiction} \\
 \Downarrow \\
 AB = AC \text{ as a line carrier.}
 \end{array}$$

Neither XI.9 nor XI.10 occurs in this proof. XI.11 and XI.12 are also not dependencies of XI.13: they establish existence, whereas XI.13 is a uniqueness statement.

13.5 What is genuinely three-dimensional here?

The proof has one essential spatial move and one spatial intersection move.

First, the two supposed distinct perpendicular lines determine the auxiliary plane

$$\sigma = \text{plane}(l, m).$$

Second, this plane must be distinguished from the reference plane π , after which their common section is a line

$$q = \pi \cap \sigma$$

through the common point A .

Once q has been obtained, the classical argument is planar inside σ . The formal proof does not need to enter $\text{PlaneGeo}(\sigma)$ explicitly, because the relevant planar conclusion has already been packaged synthetically by XI.4.

The point is therefore not that XI.13 is “really planar”. The existence of the correct carrier plane, the distinctness of the two planes, and the common intersection line are genuine three-dimensional proof obligations.

13.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_13
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (l m : Geo.Line)
  (A : Geo.Point)
  (hLperp :
    HilbertLinePerpendicularPlaneAt Geo l pi A)
  (hMperp :
    HilbertLinePerpendicularPlaneAt Geo m pi A) :
  l = m

```

There is no `HilbertSpaceEuclidean` parameter. Thus the current formal XI.13 does not use Hilbert Group IV.

The difference between Euclid’s phrase “on the same side” and the formal conclusion $l = m$ is representational. A `Geo.Line` is an unoriented carrier. Reversing the ray through A does not create a second line object, so the side qualification disappears when the target is equality of line carriers.

13.7 Spatial / Planar Architecture Used

13.7.1 Ambient spatial layer

The proof uses the ambient notions

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt

```

together with the general incidence constructors

```

hilbert_plane_through_two_intersecting_lines
hilbert_plane_intersection_line

```

and the universal elimination theorem

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

As in the preceding Book XI development, no global ambient instances

```
HilbertOrder Geo
HilbertCongruence Geo
```

are installed.

13.7.2 Induced planar geometry

The final production proof makes no direct theorem call in $\text{PlaneGeo}(\sigma)$. This is a genuine simplification of the formal DAG relative to the classical exposition.

The planar content has not disappeared. It is already contained in the earlier theorem

```
euclid_proposition_11_4
```

which states synthetically that a line perpendicular to two distinct intersecting lines of a plane is perpendicular to the entire plane.

13.7.3 Plane–space bridges

XI.13 introduces no new plane–space bridge theorem. The proof stays in ambient predicates and uses the already established spatial perpendicularity symmetry

```
hilbert_space_linesPerpendicularAt_symm
```

before applying XI.4.

13.8 Proof Architecture of the Reconstruction

13.8.1 1. Contradiction hypothesis

Lean begins with

```
by_contra hlm
```

so the working assumption is

$$l \neq m.$$

The incidence components of the two line–plane perpendicularities give

$$A \in l, \quad A \in m, \quad A \in \pi.$$

13.8.2 2. Constructing the auxiliary plane

Since l and m are distinct and meet at A , the theorem

```
hilbert_plane_through_two_intersecting_lines
```

produces a plane σ such that

$$l \subset \sigma, \quad m \subset \sigma, \quad A \in \sigma.$$

This is the formal version of Euclid's instruction to draw the plane through the two supposed perpendiculars.

13.8.3 3. Proving that the planes are distinct

The intersection theorem for two planes requires an explicit proof that

$$\sigma \neq \pi.$$

Suppose instead that $\sigma = \pi$. Since $l \subset \sigma$, this would imply $l \subset \pi$. But the already established theorem

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

says that a line perpendicular to a plane cannot itself lie in that plane. Hence $\sigma \neq \pi$.

This is one of the principal hidden spatial obligations of XI.13.

13.8.4 4. Constructing the intersection line

The theorem

```
hilbert_plane_intersection_line
```

is applied to π, σ at the common point A . It returns an ambient line q with

$$A \in q, \quad q \subset \pi, \quad q \subset \sigma.$$

This is the formal structural counterpart of Euclid's XI.3 step producing the line DAE .

13.8.5 5. Both candidate lines are perpendicular to the intersection

Since $q \subset \pi$ and $A \in q$, the universal clause of

```
HilbertLinePerpendicularPlaneAt
```

gives

$$l \perp q \text{ at } A, \quad m \perp q \text{ at } A.$$

The proof then applies

```
hilbert_space_linesPerpendicularAt_symm
```

to reverse the two relations:

$$q \perp l \text{ at } A, \quad q \perp m \text{ at } A.$$

13.8.6 6. XI.4 replaces the terminal planar uniqueness step

The lines l, m are now packaged as two distinct `PlaneLines` of σ , and A as a `PlanePoint` of σ . Proposition XI.4 applies:

$$q \perp l, \quad q \perp m, \quad l \neq m, \quad l, m \subset \sigma \implies q \perp \sigma.$$

This is the central difference between the classical and Lean DAGs. Euclid compares two right angles in the auxiliary plane and invokes their impossibility as distinct same-side angles. Lean reuses the already proved universal line–plane theorem XI.4.

The method remains synthetic: no metric formula, coordinate argument, or vector normal is introduced.

13.8.7 7. Final contradiction

The intersection construction preserved

$$q \subset \sigma.$$

But the XI.4 conclusion says

$$q \perp \sigma.$$

Applying

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

to q and σ gives the contradiction. Hence the assumption $l \neq m$ is false, and therefore

$$l = m.$$

13.9 Lean Formalization

The production module has the single direct import

```
import CGJteamLab.Proposition11_12
```

and introduces no proposition-local helper theorem before the public result. The public declaration is

```
euclid_proposition_11_13
```

The proof consumes general infrastructure and earlier Book XI theorems available transitively through the current production chain. In particular, the substantive reused declarations are

```
hilbert_plane_through_two_intersecting_lines
hilbert_plane_intersection_line
hilbert_linePerpendicularPlaneAt_not_in_plane
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
hilbert_space_linesPerpendicularAt_symm
euclid_proposition_11_4
```

XI.12 itself is not called in the proof. Its module is the immediate production import because XI.13 follows XI.12 in the current Book XI module sequence.

13.10 Hidden Formal Data

13.10.1 The auxiliary plane must be explicit

The classical diagram makes it visually immediate that the two supposed perpendiculars lie in a common plane. Lean constructs this plane from the two distinct intersecting line carriers and keeps both containment proofs.

13.10.2 The two planes must be proved distinct

Euclid immediately speaks of their section. The formal `hilbert_plane_intersection_line` theorem requires $\sigma \neq \pi$. This is proved by showing that equality of the planes would put a perpendicular line inside its reference plane.

13.10.3 The intersection carrier must survive

The proof must preserve all three facts returned with q :

$$A \in q, \quad q \subset \pi, \quad q \subset \sigma.$$

The first two are needed to extract the two line–line perpendicularities. The last is needed only at the final contradiction.

13.10.4 The “same side” phrase is not a missing formal hypothesis

The formal theorem does not introduce a relation saying that two lines are on the same side of a plane. This is not an omitted diagrammatic condition. Euclid’s condition distinguishes erected rays. The formal objects $l, m : \text{Geo.Line}$ are unoriented carriers, so opposite rays through the foot determine the same line.

13.11 Relationship with Earlier Book XI Propositions

The classical local dependency is XI.3 together with Definition XI.3. Propositions XI.9–XI.12 are absent from the classical proof.

The Lean proof reorganizes the dependency graph:

- the content of the XI.3 plane-section step is supplied by the general theorem `hilbert_plane_intersection_line`;
- XI.4 is used directly to package the final planar impossibility;
- the helper `hilbert_linePerpendicularPlaneAt_not_in_plane`, introduced in the XI.5 production module, proves both the distinctness of the planes and the final contradiction;
- the spatial symmetry theorem `hilbert_space_linesPerpendicularAt_symm`, introduced in the XI.6 production module, normalizes the two perpendicular relations before XI.4.

Thus XI.13 is formally dependent on earlier XI.4–XI.6 infrastructure, but not on the existence propositions XI.11 and XI.12 as theorem calls.

13.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by XI.13. The planar right-angle machinery needed by XI.4 is inherited through the already proved Book XI infrastructure.

The current theorem uses spatial incidence, order, and congruence, but its signature has no

HilbertSpaceEuclidean Geo

parameter. Hence Group IV is absent from XI.13.

Book Zero contributes no new XI.13 declaration. The final axiom profile contains the already accepted inherited dependency `bookZero_nullSegment1`; XI.13 introduces no new global axiom constant.

13.13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_13

#print axioms Geometry.euclid_proposition_11_13
```

and Lean reports

```
'Geometry.euclid_proposition_11_13' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

The full repository build was also reported clean after XI.13 was added to the root import.

The theorem signature does not contain

```
HilbertSpaceEuclidean Geo
```

so the Euclidean parallel axiom / Hilbert Group IV is not used by the current XI.13 proof. No continuity assumption is present.

At this checkpoint:

- new proposition-specific geometric axiom: no;
- new spatial axiom declaration: no;
- Euclidean parallel axiom / Hilbert Group IV: no;
- new continuity assumption: no;
- new proposition-local helper theorem: no;
- new reusable `Hilbert3DInterface` theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- new `PlaneGeo` bridge theorem: no;
- global ambient `HilbertOrderGeo`: not installed;
- global ambient `HilbertCongruenceGeo`: not installed;
- public theorem compiles: yes, as reported at this checkpoint;

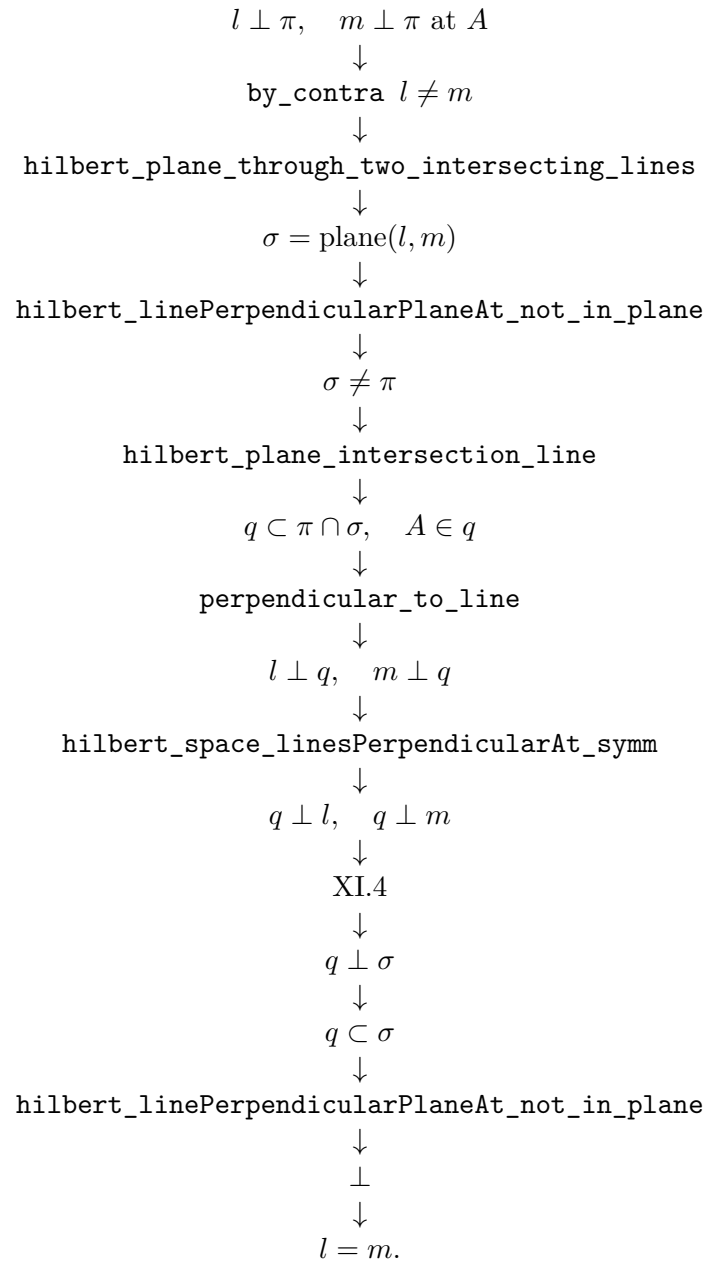
- full repository build: clean, as reported at this checkpoint;
- axiom audit: completed;
- sorry/admit: no.

13.14 Dependency Summary

The classical local chain is

$$\begin{array}{c}
 l \perp \pi, \quad m \perp \pi, \quad A \in l \cap m \cap \pi \\
 \downarrow \\
 \sigma = \text{plane}(l, m) \\
 \downarrow \\
 \text{XI.3} \\
 \downarrow \\
 q = \pi \cap \sigma \\
 \downarrow \\
 \text{XI.Def.3} \\
 \downarrow \\
 l \perp q, \quad m \perp q \\
 \downarrow \\
 \text{equal right angles in one plane} \\
 \downarrow \\
 \text{contradiction.}
 \end{array}$$

The actual formal chain is



The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	no
Direct production import	XI.12 module
Classical explicit Book XI dependency	XI.3
Classical definition dependency	XI.Def.3
XI.9 / XI.10	not used
XI.11 / XI.12 theorem calls	not used
Earlier formal Book XI theorem reused	XI.4
New proposition-local helper theorem	no
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
sorry/admit	no

13.15 Conclusion

Proposition XI.13 is a compact uniqueness theorem with a particularly clean formal architecture. The classical proof uses an auxiliary plane, its intersection with the reference plane, and the impossibility of two distinct same-side perpendicular rays in one plane.

The Lean proof preserves the first two spatial steps exactly but reuses XI.4 for the terminal planar contradiction. The key hidden obligation is the proof that the auxiliary plane differs from the reference plane. Once the intersection line is retained with both carrier-plane proofs, the final contradiction is immediate from the fact that a line perpendicular to a plane cannot itself lie in that plane.

The theorem introduces no new helper, bridge, spatial axiom, Group IV assumption, or continuity principle. Its audited global axiom profile is

`[propext, Classical.choice, Geometry.bookZero_nullSegment1, Quot.sound].`

The result is fully synthetic and uses no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure.

Appendix A

Euclid in Heath's Edition: Source Audit

A.1 Scope

This appendix records the Book XI source audit against Euclid's *Elements* in Heath's edition through Proposition XI.13 [1]. The purpose is to control the classical statement, construction, and local proposition dependencies. Lean-specific incidence, subtype, and plane-transport obligations are documented in the proposition chapters and are not attributed to Euclid or Heath.

A.2 Proposition XI.4

Proposition XI.4 states that if a straight line is set up at right angles to each of two straight lines at their point of intersection, then it is also at right angles to the plane through those two lines.

The classical proof chooses equal segments on the two given lines and introduces an arbitrary line of the plane through the common point. Triangle-congruence arguments are then used to prove that the candidate spatial line makes a right angle with that arbitrary transversal.

For the formal reconstruction the significant classical features are:

- the target is universal over every line of the plane through the common point;
- the arbitrary transversal is part of the proposition's logical content, not merely a feature of the diagram;
- the proof is synthetic and based on incidence, right angles, equal segments, and triangle congruence.

A.3 Proposition XI.5

Proposition XI.5 says that if one straight line is perpendicular at a common point to three straight lines meeting there, then those three straight lines lie in one plane.

Euclid's contradiction argument uses the plane through two of the three candidate coplanar lines and another plane through the common perpendicular and the remaining line. The intersection

line of the two planes produces, inside one plane, two distinct lines through the same point both perpendicular to the common perpendicular.

The local classical dependency is therefore governed by the preceding spatial propositions: the intersection-of-planes configuration and Proposition XI.4 supply the contradiction. The current Lean proof organizes this material differently, but preserves the same spatial core: perpendicularity to a plane forces the required intersection line to be perpendicular to the common normal.

A.4 Proposition XI.6

Heath gives the statement:

If two straight lines be at right angles to the same plane, the straight lines will be parallel.

Let AB and CD be perpendicular to the reference plane at B and D . Euclid joins BD , draws DE in the reference plane perpendicular to BD , makes

$$DE = AB,$$

and joins BE , AE , and AD .

Since AB is perpendicular to the reference plane,

$$\angle ABD, \quad \angle ABE$$

are right. Since CD is also perpendicular to the same plane, the corresponding lines of the plane through D are perpendicular to CD .

The first triangle comparison uses

$$AB = DE, \quad BD = DB,$$

and equality of the included right angles. Proposition I.4 gives

$$AD = BE.$$

The second comparison uses

$$DE = BA, \quad EA = AE, \quad DA = BE.$$

Proposition I.8 therefore gives

$$\angle EDA = \angle ABE.$$

Hence $\angle EDA$ is right and

$$ED \perp DA.$$

At D , the line ED is now perpendicular to

$$DB, \quad DA, \quad DC.$$

Proposition XI.5 places these three lines in one plane. Proposition XI.2 supplies the corresponding triangle-plane incidence, bringing AB into that same plane. The two right angles made with the transversal BD then give the final parallelism by Proposition I.28.

The local classical DAG can be recorded as

$$\begin{array}{c}
 \text{construct } DE \perp BD, \quad DE = AB \\
 \Downarrow \\
 \text{I.4} \\
 \Downarrow \\
 AD = BE \\
 \Downarrow \\
 \text{I.8} \\
 \Downarrow \\
 ED \perp DA \\
 \Downarrow \\
 \text{XI.5} \\
 \Downarrow \\
 DB, DA, DC \text{ coplanar} \\
 \Downarrow \\
 \text{XI.2} \\
 \Downarrow \\
 AB, BD, DC \text{ coplanar} \\
 \Downarrow \\
 \text{I.28} \\
 \Downarrow \\
 AB \parallel CD.
 \end{array}$$

A.5 Comparison with the Lean reconstruction at XI.6

The Lean proof preserves the central metric sequence of the classical argument:

$$\text{auxiliary } DE \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5}.$$

The formal expansion is concentrated in the data suppressed by the classical diagram:

- each planar construction is performed in an explicit induced plane geometry;
- cross-plane segment and angle comparisons use the spatial congruence layer;
- noncollinearity and point-in-plane obligations are explicit;
- the common plane of the two final normals is produced as a witness;
- spatial parallelism records both coplanarity and disjointness, so skew lines are excluded;
- the last planar disjointness proof uses the established neutral alternate-angle infrastructure rather than a direct call to the public I.28 wrapper.

The last point is a dependency-graph difference, not a change in the geometric target. Euclid closes the configuration by I.28; the current Lean source imports the Proposition I.27 module and invokes the lower alternate-angle criterion in the induced common plane.

A.6 Checkpoint XI.6

Through XI.6 the reconstruction remains synthetic. No coordinate, vector, scalar-product, trigonometric, or numerical-angle representation is used in the proposition proofs.

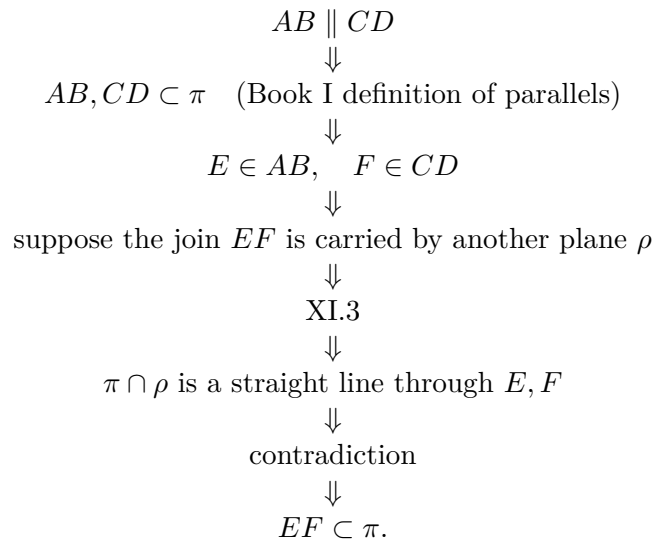
The source comparison also isolates the new structural point of XI.6: in three-space the proof of parallelism has to establish coplanarity before planar nonintersection can be invoked. Euclid obtains this by XI.5 and XI.2. The Lean theorem records the same distinction directly in the target relation `HilbertSpaceLinesParallel`, which consists of an explicit common plane together with disjointness of the two ambient carriers.

A.7 Proposition XI.7

Proposition XI.7 concerns the plane determined by two parallel straight lines. In Book I the definition of parallel lines already requires the two lines to be in one plane. Thus the reference plane is part of the hypothesis, not an additional conclusion of XI.7.

Choose a point on each of the two parallel lines and join the chosen points. Euclid argues by contradiction. He supposes that the joining straight line is carried by another, more elevated plane. A plane through that alleged join is then intersected with the reference plane. Proposition XI.3 makes the common section a straight line. Since both chosen points lie in both planes, that section passes through both of them. The contradiction is expressed in the classical text by the impossibility of two straight lines enclosing an area. The join must therefore lie in the original plane of the parallels.

The local classical DAG is



No Proposition XI.4, XI.5, or XI.6 occurs in this local proof. The explicit Book XI dependency is XI.3.

A.8 Comparison with the Lean reconstruction at XI.7

The Lean proof has a shorter incidence DAG. The formal relation `HilbertSpaceLinesParallel` already contains a witness plane σ for the two parallel carriers, together with their disjointness.

The selected points therefore lie in σ . Their distinctness follows from disjointness of the parallel carriers. After the joining carrier is constructed, `HilbertSpaceIncidence.line_in_plane` places the whole join in σ .

This final call represents the Hilbert incidence principle that two distinct points of a line lying in a plane force the entire line to lie in that plane. Consequently the formal proof does not reproduce Euclid's auxiliary second plane or his use of XI.3. This is a difference in foundational packaging, not in the geometric conclusion.

A.9 Checkpoint XI.7

Through XI.7 the reconstruction remains entirely synthetic. Proposition XI.7 uses only incidence: no order, congruence, angle theory, continuity, coordinate representation, vector formalism, or scalar product enters its proof.

The source comparison now exhibits two distinct ways to express the same closure principle. Euclid derives it by a plane-intersection contradiction; the Hilbert-style formalization uses an explicit common-plane witness for spatial parallelism and a general line-in-plane incidence axiom.

A.10 Proposition XI.8

Proposition XI.8 states that if two straight lines are parallel and one of them is perpendicular to a plane, then the other is perpendicular to the same plane.

Let $AB \parallel CD$, with AB perpendicular to the reference plane at B . Euclid lets CD meet the plane at D , joins BD , draws DE in the reference plane perpendicular to BD , makes $DE = AB$, and joins BE , AE , and AD .

The local classical dependencies are unusually informative because they cross the Euclidean parallel boundary.

- XI.7 places AB , CD , and BD in one plane.
- I.11 and I.3 supply the auxiliary perpendicular and equal segment.
- The definition of line-plane perpendicularity makes $\angle ABD$ and $\angle ABE$ right.
- I.29 converts the given parallelism into the right angle $\angle CDB$, hence $CD \perp BD$.
- I.4 and I.8 give $AD = BE$ and then show $\angle EDA$ right.
- XI.4 applied in the plane through BD, DA gives $ED \perp CD$.
- XI.4 applied again in the reference plane, to DB, DE , gives $CD \perp \pi$.

The incidence step placing CD in the plane through BD, DA uses the planarity of triangle ABD together with the common plane of AB, CD, BD . The printed diagram suppresses this plane identification.

The local classical DAG is

$$\begin{array}{c}
AB \parallel CD, \quad AB \perp \pi \\
\Downarrow \\
\text{XI.7} \\
\Downarrow \\
DE \perp BD, \quad DE = AB \\
\Downarrow \\
\text{I.29} \\
\Downarrow \\
CD \perp BD \\
\Downarrow \\
\text{I.4, I.8} \\
\Downarrow \\
ED \perp AD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
ED \perp CD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
CD \perp \pi.
\end{array}$$

A.11 Comparison with the Lean reconstruction at XI.8

The Lean proof preserves the synthetic metric core but reorganizes two classical steps.

First, the point D where the second parallel meets the reference plane is not assumed. It is derived from the common carrier plane of the parallels, the intersection of that plane with the reference plane, and Group IV uniqueness.

Second, the proof does not call I.29 directly. It constructs through D a line perpendicular to BD , proves by the neutral I.27 alternate-angle criterion that this line is disjoint from the first parallel, and then uses Group IV uniqueness to identify it with the second parallel.

Thus the formal factorization is

$$\text{neutral I.27} + \text{Euclidean uniqueness of parallels} \implies \text{the I.29-type conclusion needed in XI.8.}$$

The two applications of XI.4 remain visible in the production proof. The first is made in the common plane of the parallel lines; the second is made in the reference plane.

A.12 Checkpoint XI.8

Through XI.8 the reconstruction remains synthetic: no coordinates, vectors, scalar product, trigonometric functions, or numerical angle measure are introduced.

Unlike XI.4–XI.7, however, XI.8 is genuinely Euclidean in the axiomatic sense. The formal proof requires plane-local parallel uniqueness, represented by `HilbertSpaceEuclidean`. No new continuity assumption is used.

A.13 Proposition XI.11

Heath gives the statement:

From a given elevated point to draw a straight line perpendicular to a given plane.

Let A be the elevated point and let π be the plane of reference. Euclid draws an arbitrary line $BC \subset \pi$, then draws AD from A perpendicular to BC by I.12. If $AD \perp \pi$, the problem is finished.

In the nontrivial branch he constructs

$$DE \perp BC \quad (\text{I.11}), \quad AF \perp DE \quad (\text{I.12}), \quad GH \parallel BC \quad (\text{I.31}).$$

Since BC is perpendicular to both DA and DE , XI.4 gives

$$BC \perp \text{plane}(ED, DA).$$

Then XI.8 transfers this perpendicularity to the parallel line GH :

$$GH \perp \text{plane}(ED, DA).$$

By XI.Def.3, GH is perpendicular to AF , which meets it and lies in that plane. Hence AF is perpendicular to both GH and DE . A second application of XI.4 gives

$$AF \perp \text{plane}(ED, GH) = \pi.$$

Thus the exact local classical dependency chain is

$$\begin{array}{c} \text{I.12} \\ \Downarrow \\ \text{I.11, I.12, I.31} \\ \Downarrow \\ \text{XI.4} \\ \Downarrow \\ \text{XI.8} \\ \Downarrow \\ \text{XI.Def.3} \\ \Downarrow \\ \text{XI.4} \\ \Downarrow \\ \text{XI.11.} \end{array}$$

Neither XI.9 nor XI.10 is cited in Heath's proof, and neither is needed transitively for the local construction described there. This verifies that the current reconstruction may pass directly from XI.8 to XI.11 without silently skipping a classical dependency.

One incidence point is implicit in the classical prose. Before the first I.12 construction can be interpreted as a planar construction, the elevated point A and the chosen line BC must be placed in a common plane. The formal reconstruction supplies this plane explicitly.

A.14 Comparison with the Lean reconstruction at XI.11

The production proof follows the same construction and retains Euclid's initial case split. Its additional work is spatial bookkeeping rather than a change of method.

The principal formal expansions are:

- construction of the plane σ through A and BC before the first planar I.12;
- construction of the plane τ through AD and DE before the second planar I.12;
- proof that the second foot F does not lie on BC , needed before I.31 is applied in the reference plane;
- conversion of planar I.31 parallelism into the ambient `HilbertSpaceLinesParallel` relation;
- proof that the foot K returned existentially by XI.8 equals the already constructed point F ;
- proof that $DE \neq GH$ before the final XI.4 application.

The two XI.4 applications and the single XI.8 application remain exactly visible in the formal proof architecture.

A.15 Checkpoint XI.11

Through XI.11 the reconstruction remains synthetic: no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure are introduced.

The classical proof of XI.11 explicitly uses XI.8 and therefore inherits the Euclidean parallel content of that proposition. No continuity principle is cited. XI.9 and XI.10 are absent from the local source DAG.

A.16 Proposition XI.12

Heath gives the statement:

To set up a straight line at right angles to a given plane from a given point in it.

Let A be the given point in the plane of reference. Euclid lets any elevated point B be conceived and applies XI.11 to draw BC perpendicular to the reference plane. Through A he then draws

$$AD \parallel BC$$

by I.31. Proposition XI.8 immediately transfers the line–plane perpendicularity:

$$BC \perp \pi, \quad AD \parallel BC \implies AD \perp \pi.$$

The complete local classical dependency chain is therefore

$$\begin{array}{c}
 A \in \pi \\
 \Downarrow \\
 B \text{ elevated} \\
 \Downarrow \\
 \text{XI.11} \\
 \Downarrow \\
 BC \perp \pi \\
 \Downarrow \\
 \text{I.31} \\
 \Downarrow \\
 AD \parallel BC \\
 \Downarrow \\
 \text{XI.8} \\
 \Downarrow \\
 AD \perp \pi.
 \end{array}$$

Neither XI.9 nor XI.10 is cited or required in this local proof.

There is one exceptional incidence configuration that the classical prose does not discuss. If the foot C of the XI.11 perpendicular happens to equal the prescribed point A , then the line BC itself already solves XI.12. In that configuration I.31 should not be read as producing a distinct parallel through a point lying on BC . The formal proof therefore separates this completed case before carrying out the generic parallel construction.

A.17 Comparison with the Lean reconstruction at XI.12

The production proof preserves the source-level chain

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8},$$

but expands three pieces of hidden incidence data.

First, it explicitly tests whether the prescribed point A already lies on the perpendicular returned by XI.11. A proposition-local foot-uniqueness theorem then identifies A with the XI.11 foot and closes that branch.

Second, in the nontrivial branch the spatial line and the point A must be placed in a common plane σ before I.31 can be used. The parallel construction is performed inside $\text{PlaneGeo}(\sigma)$, after which its planar disjointness data are packaged as ambient `HilbertSpaceLinesParallel`.

Third, XI.8 returns the new perpendicular foot existentially. The same foot-uniqueness theorem proves that this returned point is exactly the prescribed point A .

These are formal expansions of incidence and carrier-plane information; they do not replace Euclid's geometric method.

A.18 Checkpoint XI.12

Through XI.12 the reconstruction remains synthetic: no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure enter the construction.

The classical proof explicitly uses XI.11, I.31, and XI.8. Hence the present Euclid-faithful route inherits the Euclidean parallel content already present in XI.8 and XI.11. No continuity principle is cited, and XI.9 and XI.10 remain absent from the local source DAG.

A.19 Proposition XI.13

Heath gives the statement:

From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

Assume that AB and AC are two such straight lines erected from the common point A . Euclid draws the plane through BA and AC . By XI.3 its section with the reference plane is a straight line through A , written DAE .

Definition XI.3 then applies to each supposed perpendicular. Since DAE lies in the reference plane and meets both lines at A ,

$$AC \perp DAE, \quad AB \perp DAE.$$

Hence $\angle CAE$ and $\angle BAE$ are both right and therefore equal. They lie in the same plane, share the side AE , and the other two sides are assumed to be erected on the same side. Euclid concludes that this is impossible.

The complete local source DAG is therefore

$$\begin{array}{c}
 AB \perp \pi, \quad AC \perp \pi \\
 \Downarrow \\
 \text{plane through } AB, AC \\
 \Downarrow \\
 \text{XI.3} \\
 \Downarrow \\
 DAE = \pi \cap \text{plane}(AB, AC) \\
 \Downarrow \\
 \text{XI.Def.3} \\
 \Downarrow \\
 AB \perp DAE, \quad AC \perp DAE \\
 \Downarrow \\
 \text{two equal right angles in one plane on the same side} \\
 \Downarrow \\
 \text{contradiction.}
 \end{array}$$

Neither XI.9 nor XI.10 occurs in the proof. XI.11 and XI.12 are also not source dependencies: they are existence constructions, while XI.13 proves uniqueness.

A.20 Comparison with the Lean reconstruction at XI.13

The formal proof preserves Euclid's two genuinely spatial steps:

1. construct the plane σ through the two supposed distinct perpendicular carriers;
2. construct the intersection line $q = \pi \cap \sigma$ through the common foot.

The production proof makes one hidden condition explicit before the second step: it proves $\sigma \neq \pi$. If the planes were equal, one of the perpendicular lines would lie in the reference plane, which is impossible.

The terminal planar contradiction is then packaged differently. From Definition XI.3, represented by the universal clause of `HilbertLinePerpendicularPlaneAt`, the proof gets

$$l \perp q, \quad m \perp q.$$

After symmetry, XI.4 is applied in the auxiliary plane σ . If $l \neq m$, it yields $q \perp \sigma$. But the plane-intersection construction already gives $q \subset \sigma$, producing a contradiction.

Thus the Lean DAG replaces Euclid's final same-side angle-uniqueness sentence by an earlier Book XI theorem, but it does not replace the synthetic method by coordinates or metric formulas.

A.21 Checkpoint XI.13

Through XI.13 the source DAG remains independent of XI.9 and XI.10. The classical dependencies specific to the proposition are XI.3 and Definition XI.3.

The proposition is also independent of XI.11 and XI.12 as source dependencies: those propositions establish the two existence cases, whereas XI.13 supplies the corresponding uniqueness statement at a point of the plane.

The current reconstruction remains synthetic and introduces no coordinate, vector, scalar-product, trigonometric, or numerical-angle argument.

Appendix B

Hilbert: Spatial Axiomatic Architecture

B.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book XI at the current checkpoint, Proposition XI.13. The purpose is architectural: identify which layers of incidence, order, congruence, and Euclidean parallelism enter at each theorem, and distinguish them from continuity assumptions.

B.2 Spatial architecture

The formal project separates the spatial hierarchy into

```
HilbertSpacePrimitive  
HilbertSpaceIncidence  
HilbertSpaceOrder  
HilbertSpaceCongruence  
HilbertSpaceEuclidean
```

and derives ordinary planar Hilbert geometry on each fixed plane via

```
PlanePoint  
PlaneLine  
PlaneGeo
```

with explicit bridges for incidence, betweenness, congruence, same rays, angles, and right angles.

This separation is essential at XI.4. There is deliberately no global `HilbertCongruence Geo` or `HilbertOrder Geo` instance installed on the ambient three-space. Planar order arguments are carried out inside a specified `PlaneGeo`; spatial congruence transports only the data needed between such plane slices.

B.3 Neutral status of XI.4 in the current hierarchy

The public theorem assumes the spatial incidence/order/congruence hierarchy but no `HilbertEuclideanPlane` instance. In particular, Hilbert Group IV – the Euclidean parallel axiom – does not occur in the proof of XI.4.

The final axiom audit reported by Lean is

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[propept,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this checkpoint. XI.4 introduces no proposition-specific geometric axiom.

B.4 What XI.4 adds to the interface

The theorem forced one useful correction to the general 3D API: `HilbertLinesPerpendicularAt` now carries an explicit noncollinearity witness. Consequently distinctness of perpendicular carriers is derived by `hilbert_linesPerpendicularAt_ne` rather than supplied as a separate hypothesis.

The universal target is represented by

```
HilbertLinePerpendicularPlaneAt
```

which quantifies over every incidence line of the plane through the foot. This is a direct synthetic representation and introduces no metric structure.

B.5 Checkpoint XI.4

At XI.4 Hilbert's role is therefore not a one-to-one proposition-number match. It is the axiomatic architecture in which Euclid's spatial proof can be made explicit. The most significant result of the audit is the boundary: XI.4 is completed in the current incidence/order/congruence 3D hierarchy without Group IV and without a new continuity assumption.

B.6 Incidence axiom I.6 and Proposition XI.7

Proposition XI.7 isolates a much smaller part of the spatial hierarchy than XI.4–XI.6. The decisive Hilbert incidence principle is represented in the current formal interface by

```
HilbertSpaceIncidence.line_in_plane
```

Its geometric content is: if two distinct points of a line lie in a plane, then the whole line lies in that plane. In the numbering adopted by the current project this is Hilbert incidence axiom I.6.

The public XI.7 theorem assumes only

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
```

It does not assume `HilbertSpaceOrder`, `HilbertSpaceCongruence`, or an induced `PlaneGeo` instance. The Euclidean parallel axiom, Group IV, and continuity are therefore absent from the proof architecture.

B.7 Spatial parallelism as reusable incidence data

The relation

```
HilbertSpaceLinesParallel
```

is now part of `Hilbert3DInterface`. It packages two assertions: a common carrier plane and disjointness of the two ambient lines. The common plane is essential because ambient disjointness alone would also admit skew lines.

This notion was introduced during the XI.6 development and promoted to the general 3D interface before XI.7. Proposition XI.7 is the first subsequent Book XI theorem that consumes the common-plane witness directly.

B.8 Axiomatic status of XI.7

The final audit command is

```
import CGJteamLab.Proposition11_7

#print axioms Geometry.euclid_proposition_11_7
```

and Lean reports

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This result concerns global Lean axiom constants. It does not remove the explicit Hilbert incidence structures from the theorem signature: those structures remain hypotheses of the theorem. What the audit establishes is that no additional globally declared axiom constant occurs in the proof term.

B.9 Checkpoint XI.7

At XI.7 the spatial reconstruction reaches a pure incidence theorem. The formal proof needs neither Group IV nor continuity, and it introduces no new proposition-specific geometric or spatial axiom. The decisive spatial content is exactly the combination of a common-plane witness for parallel lines and Hilbert’s line-in-plane incidence principle.

B.10 Group IV becomes active at Proposition XI.8

Proposition XI.8 is the first theorem in the present Book XI sequence whose production signature assumes

```
[HSE : HilbertSpaceEuclidean Geo]
```

The class is the spatial representation of Hilbert Group IV. Its parallel-uniqueness field is explicitly restricted to one ambient plane: through a point of that plane outside a given line, two lines in the same plane that are disjoint from the given line must coincide.

This formulation is essential in three-space. The project does not install a global

```
HilbertEuclideanPlane Geo
```

instance on the ambient spatial geometry. Instead, every fixed plane inherits the planar Euclidean interface through

```
planeGeo_linesDisjoint_iff_ambient
planeGeoHilbertEuclidean
```

The first theorem identifies planar disjointness in a fixed `PlaneGeo` with ambient disjointness of the underlying spatial lines. The second packages the plane-local Group IV field as the existing planar `HilbertEuclideanPlane` instance.

B.11 Neutral I.27 and Euclidean uniqueness in XI.8

The formal replacement for Euclid’s I.29 step exposes the logical boundary between neutral and Euclidean geometry.

Inside the common plane σ of the parallel lines, a line p is constructed through D perpendicular to the transversal $d = BD$. The established I.27 alternate-angle theorem shows that p is disjoint from the first parallel l . This part uses order and congruence but not Group IV.

The given second parallel m is also disjoint from l . Both p and m pass through D and lie in σ . Only now is Group IV used: uniqueness forces $p = m$. Therefore $m \perp d$.

The formal proof consequently separates

existence/nonintersection from Euclidean uniqueness.

B.12 Axiomatic status of XI.8

The final audit command is

```
import CGJteamLab.Proposition11_8

#print axioms Geometry.euclid_proposition_11_8
```

and Lean reports

```
'Geometry.euclid_proposition_11_8' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

These are inherited global Lean/project dependencies. No new proposition-specific global axiom constant is introduced.

This audit output must be distinguished from the theorem’s explicit geometric hypotheses. The typeclass `HilbertSpaceEuclidean Geo` is a genuine mathematical assumption of XI.8: it represents Group IV. It is passed as a theorem parameter, rather than appearing as an additional global Lean axiom constant.

No continuity assumption is used.

B.13 Checkpoint XI.8

At XI.8 the spatial hierarchy first crosses from the neutral incidence/order/congruence layers into the Euclidean parallel layer. The transition is plane-local: Group IV is applied only inside a specified ambient plane and is inherited by its induced `PlaneGeo` geometry.

The proof remains synthetic. Its Euclidean content is exactly the uniqueness step required to turn neutral nonintersection into the I.29-type perpendicularity conclusion.

B.14 Proposition XI.11 in the current Hilbert architecture

The production theorem XI.11 assumes the same full spatial hierarchy as XI.8:

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

The proof does not install ambient `HilbertOrder Geo` or ambient `HilbertCongruence Geo`. Instead, Book I perpendicular and parallel constructions are applied inside explicit induced

planes $\text{PlaneGeo}(\sigma)$, $\text{PlaneGeo}(\pi)$, and $\text{PlaneGeo}(\tau)$.

The Group IV hypothesis enters through the call to

```
euclid_proposition_11_8
```

which transfers perpendicularity from BC to the constructed parallel GH . No independent Group IV argument is added by XI.11 itself.

The two calls to

```
euclid_proposition_11_4
```

remain in the incidence/order/congruence spatial layer. The first proves $BC \perp \tau$; the second proves the final $AF \perp \pi$.

B.15 Axiomatic status of XI.11

The final audit command is

```
import CGJteamLab.Proposition11_11

#print axioms Geometry.euclid_proposition_11_11
```

and Lean reports

```
'Geometry.euclid_proposition_11_11' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

These are inherited global Lean/project dependencies. XI.11 introduces no proposition-specific global axiom constant, no new spatial axiom declaration, and no continuity assumption.

The explicit parameter

```
[HSE : HilbertSpaceEuclidean Geo]
```

is nevertheless a genuine hypothesis of the current theorem because XI.8 is used in the proof. As at XI.8, this theorem parameter does not appear as an extra global constant in the output of `#print axioms`.

B.16 Checkpoint XI.11

At XI.11 the existing plane-local Group IV layer is consumed but not extended. The proposition introduces no new general 3D interface theorem. Its main architectural role is to coordinate

three induced plane slices and to return a spatial line–plane perpendicularity statement.

The current formal proof is therefore Euclid-faithful with respect to its parallel-theoretic step: it inherits the Euclidean strength of XI.8. The audit does not claim that this hypothesis is minimal for the abstract existence statement.

B.17 Proposition XI.12 in the current Hilbert architecture

The public XI.12 theorem assumes the same full spatial hierarchy as XI.11 and XI.8:

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

The proof again avoids global ambient `HilbertOrder Geo` and `HilbertCongruence Geo`. The only new planar construction is the parallel through the prescribed point. It is carried out in an explicit auxiliary plane σ via `PlaneGeo( $\sigma$ )`.

XI.12 introduces two proposition-local theorems.

The first,

```
hilbert_XI12_perpendicular_foot_unique
```

is an incidence consequence of the universal definition of line–plane perpendicularity. If two distinct points of the perpendicular line lay in the plane, Hilbert spatial incidence would force the whole line into the plane; the universal perpendicularity clause would then make the line perpendicular to itself.

The second,

```
hilbert_XI12_parallel_through_point_in_plane
```

performs the I.31 construction inside a fixed induced plane and packages the resulting planar disjointness as `HilbertSpaceLinesParallel`. Its signature does not assume `HilbertSpaceEuclidean`; the helper itself remains below Group IV.

The public theorem nevertheless assumes Group IV because it consumes XI.11 and XI.8. No additional Euclidean uniqueness argument is introduced by the two new XI.12 helpers.

B.18 Axiomatic status of XI.12

The public audit command is

```
import CGJteamLab.Proposition11_12
```

```
#print axioms Geometry.euclid_proposition_11_12
```

and Lean reports

```
'Geometry.euclid_proposition_11_12' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

The two local helpers were audited separately:

```
'Geometry.hilbert_XI12_perpendicular_foot_unique'
depends on axioms: [propext, Classical.choice, Quot.sound]

'Geometry.hilbert_XI12_parallel_through_point_in_plane'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

Thus the XI.12-specific incidence and plane-slice packaging steps add no new global geometric axiom constant and do not themselves depend on `bookZero_nullSegment1`. The public theorem inherits that existing project dependency through the established earlier construction chain.

The explicit parameter

```
[HSE : HilbertSpaceEuclidean Geo]
```

remains a genuine geometric hypothesis of the current theorem. It enters through XI.11 and XI.8, not through a new XI.12-specific Group IV step.

No new continuity assumption, spatial axiom class, or `sorry/admit` is introduced.

B.19 Checkpoint XI.12

XI.12 adds no new primitive 3D structure and no new plane-space bridge declaration. Its two local helpers expose reusable consequences of the existing architecture: uniqueness of the perpendicular foot and the packaging of a planar parallel construction as spatial parallelism.

The current public proof is Euclid-faithful and therefore retains the Group IV strength inherited from XI.11 and XI.8. The axiom audit distinguishes that explicit theorem hypothesis from the global Lean/project axiom constants appearing in the proof term.

B.20 Proposition XI.13 in the current Hilbert architecture

The production XI.13 theorem assumes

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
```

```
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
```

and, unlike XI.8, XI.11, and XI.12, does not assume

```
HilbertSpaceEuclidean Geo
```

Thus XI.13 returns to the neutral spatial incidence/order/congruence layer of the current hierarchy.

The proof needs no new induced-plane construction theorem and no new plane–space bridge. Its principal spatial ingredients are:

- the plane through two distinct intersecting lines;
- the line of intersection of two distinct planes;
- the universal clause of line–plane perpendicularity;
- symmetry of line–line perpendicularity;
- XI.4;
- the fact that a line perpendicular to a plane cannot itself lie in that plane.

The explicit proof that the auxiliary plane differs from the reference plane is an incidence consequence of the existing line–plane perpendicularity interface. No parallel uniqueness principle is needed.

B.21 Axiomatic status of XI.13

The audit command is

```
import CGJteamLab.Proposition11_13

#print axioms Geometry.euclid_proposition_11_13
```

and Lean reports

```
'Geometry.euclid_proposition_11_13' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

These are the accepted inherited global Lean/project dependencies. XI.13 introduces no proposition-specific global axiom constant.

The absence of `HilbertSpaceEuclidean` from the theorem signature is a genuine geometric distinction from XI.8, XI.11, and XI.12. Group IV is not used in the current XI.13 proof. No continuity assumption, new spatial axiom declaration, or `sorry/admit` is introduced.

B.22 Checkpoint XI.13

At XI.13 the current reconstruction isolates uniqueness of the normal at a point without crossing the Euclidean parallel boundary. The proof uses the existing spatial incidence/order/congruence hierarchy and reuses XI.4 as the synthetic closure theorem.

No global ambient `HilbertOrder Geo` or `HilbertCongruence Geo` instance is installed, and no new `Hilbert3DInterface` declaration is required.

Appendix C

Borsuk–Szmielw: Axiomatic Layers and Spatial Geometry

C.1 Scope

This appendix records the Book XI audit against Karol Borsuk and Wanda Szmielw, *Foundations of Geometry*, through Proposition XI.13. Their primitive language includes points, lines, planes, betweenness, and equidistance. Their axioms are organized into incidence, order, congruence, continuity, and the Euclidean axiom.

C.2 Congruence results relevant to XI.4

The Book I audit already identifies the triangle-congruence results used in the XI.4 argument within the pre-Euclidean congruence layer. In particular:

- the SSS theorem corresponding to Euclid I.8 occurs as Borsuk–Szmielw Theorem 21;
- right-angle and perpendicular constructions belong to their synthetic congruence development and are not formulated by numerical angle measure.

The formal XI.4 proof likewise treats SAS/SSS and right-angle transport as congruence infrastructure rather than as consequences of the Euclidean parallel axiom.

C.3 Spatial geometry

Borsuk–Szmielw develop three-dimensional geometry with points, lines, planes, perpendicular lines, perpendicular projection, and later an absolute coordinate system in space. In that spatial development a line perpendicular to a plane is part of the geometric language before coordinates are introduced as a later representation.

At the current checkpoint no theorem number has been identified in the audited source whose statement is exactly Euclid XI.4. Accordingly the Lean theorem is not attributed here to a specific Borsuk–Szmielw theorem.

C.4 Axiomatic comparison

For XI.4 the comparison is:

$$\begin{array}{c}
 \text{incidence and order in planes} \\
 + \\
 \text{synthetic segment and angle congruence} \\
 + \\
 \text{spatial incidence/congruence transport} \\
 \Downarrow \\
 \text{line-plane perpendicularity.}
 \end{array}$$

The formal proof does not use the Euclidean parallel axiom. This agrees with the placement of the triangle-congruence ingredients below the Euclidean layer in the Borsuk–Szczepański organization.

C.5 Checkpoint XI.4

The current audit records no exact Borsuk–Szczepański counterpart for XI.4 itself. Exact theorem-number correspondence should be added only if it is located in the source.

C.6 Proposition XI.7 and incidence Axiom I7

For XI.7 the Borsuk–Szczepański source contains an exact structural counterpart of the decisive Hilbert incidence clause.

Their Axiom I7 states that for a line L and a plane P , if there exist two distinct points a, b belonging both to L and to P , then

$$L \subset P.$$

This is exactly the line-in-plane principle used in the formal proof of XI.7. Borsuk–Szczepański also classify Axioms I1–I4 as plane axioms and Axioms I5–I9 as space axioms. Hence I7 belongs to the spatial incidence layer.

Their general organization is especially transparent here: the first group consists of incidence axioms concerning points, lines, and planes; order, congruence, continuity, and the Euclidean axiom are introduced only in later groups. Proposition XI.7 therefore lies strictly below all of those stronger layers.

The correspondence with the Lean proof is

$$\begin{array}{c}
 E, F \in \sigma, \quad E \neq F \\
 \Downarrow \\
 \text{line } n \text{ through } E, F \\
 \Downarrow \\
 \text{Borsuk–Szczepański I7 / Hilbert I.6} \\
 \Downarrow \\
 n \subset \sigma.
 \end{array}$$

No Borsuk–Szczepański theorem number is needed here: the relevant source item is the incidence axiom itself.

C.7 Checkpoint XI.7

At XI.7 the comparison becomes sharper than at XI.4. The source contains an explicit spatial-incidence axiom whose content matches the decisive formal step exactly.

Thus XI.7 requires only incidence structure in the Borsuk–Szmielew hierarchy. It does not require their axioms of order, congruence, continuity, or the Euclidean axiom. This agrees with the Lean theorem, which assumes only the ambient incidence classes and whose axiom audit reports no additional global axiom constants.

C.8 Proposition XI.8 and the Euclidean parallel boundary

No exact Borsuk–Szmielew theorem number matching the full statement of Euclid XI.8 has been identified in the audited source. The source does, however, give a precise axiomatic placement for the parallel-theoretic step on which XI.8 depends.

The Book I source audit places the I.27-type implication

$$\text{appropriate angle equality} \implies \text{nonintersection}$$

in the congruence development before continuity and before the Euclidean axiom.

After the Euclidean axiom E is introduced, the source obtains uniqueness of the parallel and develops the converse Euclidean parallel-angle theory. In particular, the material corresponding to the I.29 direction lies on the Euclidean side of this boundary.

This is exactly the split exposed by the Lean proof of XI.8. The proposition-local helper first uses the neutral alternate-angle theorem to construct a nonintersecting line. The spatial Group IV class then supplies uniqueness in the fixed plane and identifies that line with the given parallel.

Thus the source-layer comparison is

$$\begin{array}{c} \text{incidence} + \text{order} + \text{congruence} \\ \Downarrow \\ \text{I.27-type nonintersection} \\ \Downarrow \\ + \text{Euclidean axiom } E / \text{Group IV} \\ \Downarrow \\ \text{unique parallel and I.29-type converse} \\ \Downarrow \\ \text{XI.8 perpendicularity transport.} \end{array}$$

The metric SAS/SSS and right-angle ingredients used around this step remain in the pre-Euclidean congruence layer. The new axiomatic input is parallel uniqueness, not a new metric or continuity principle.

C.9 Checkpoint XI.8

At XI.8 the Borsuk–Szmielew comparison records the first Book XI checkpoint in this reconstruction that requires the Euclidean axiom layer. No new continuity assumption is needed.

The source does not justify assigning a one-to-one Borsuk–Szmielew theorem number to XI.8 itself. The durable source observation is instead the same boundary made explicit by the formal API: neutral nonintersection precedes E , while the converse parallel-angle information requires Euclidean uniqueness.

C.10 Proposition XI.11 and Borsuk–Szmielew Theorem 89

At XI.11 the source comparison becomes stronger than a general architectural analogy. In Chapter II, Section 31, Borsuk–Szmielew prove Theorem 89: through a given point there exists exactly one line perpendicular to a given plane.

Their theorem covers both cases:

- the given point lies in the plane;
- the given point does not lie in the plane.

The second case contains the existence statement corresponding directly to Euclid XI.11, while the full theorem also includes uniqueness and the on-plane case later treated by Euclid XI.12.

The construction in their off-plane case is recognizably related to Euclid’s synthetic strategy. Starting from a line in the plane, they use perpendicular projections and a second in-plane perpendicular, then invoke their two-line criterion for perpendicularity to a plane (Theorem 87). Their proof is organized inside the congruence chapter, before the later continuity and Euclidean-axiom layers.

This yields an important comparison with the current Lean theorem. The production proof of XI.11 deliberately follows Euclid’s route through XI.8, and therefore assumes `HilbertSpaceEuclidean`. Borsuk–Szmielew Theorem 89 shows that the existence-and-uniqueness result itself admits a synthetic treatment below the Euclidean parallel layer in their axiomatic organization.

Consequently the present Group IV hypothesis should be read as a property of the chosen Euclid-faithful Lean proof, not as evidence that line–plane perpendicular construction is intrinsically Euclidean.

C.11 Comparison of the two proof architectures at XI.11

The comparison can be summarized as

Euclid / current Lean route	Borsuk–Szmielew route
two successive line perpendiculars	successive perpendicular projections
parallel $GH \parallel BC$	no Euclidean parallel step required
XI.8 / Group IV	Theorem 87 in congruence layer
XI.4	Theorem 87
existence	existence + uniqueness

This does not invalidate the current formalization. It identifies a possible later strengthening: a neutral theorem proving the existence (and potentially uniqueness) of the perpendicular through a point could be added independently, after its proof is reconstructed in the project’s current Hilbert spatial API.

C.12 Checkpoint XI.11

Borsuk–Szmielew Theorem 89 is the first exact numbered source result in this appendix whose statement directly subsumes the target of Euclid XI.11. It also sharpens the axiomatic audit: the current Lean proof uses the Euclidean layer because it reuses XI.8, whereas the target theorem has a pre-Euclidean synthetic proof in the Borsuk–Szmielew development.

No continuity assumption is suggested by this comparison, and no new axiom should be added to the Lean project on its basis.

C.13 Proposition XI.12 and Theorem 89: the on-plane case

Theorem 89 already identified at the XI.11 checkpoint is stronger than the separate Euclidean construction problems XI.11 and XI.12. It states that through a given point there exists exactly one line perpendicular to a given plane, and its proof treats both positions of the point.

For XI.12 the relevant branch is precisely

$$A \in \pi.$$

Thus the on-plane existence statement proved by Euclid XI.12 is contained directly in Borsuk–Szmielew Theorem 89, while the source also supplies uniqueness. The theorem remains in Chapter II, Section 31, in the synthetic congruence development before the later continuity and Euclidean axiom layers.

This sharpens the comparison with the current Lean proof. The production XI.12 deliberately follows Euclid:

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8}.$$

Consequently it assumes `HilbertSpaceEuclidean`, because both XI.11 and XI.8 in the present library use the plane-local Group IV layer. Borsuk–Szmielew Theorem 89 shows that this Euclidean strength belongs to the chosen proof route rather than to the abstract existence-and-uniqueness result.

The source also clarifies the status of the new Lean foot-uniqueness helper. Euclid XI.12 asks only for existence, while Theorem 89 packages existence and uniqueness together. The formal helper is therefore not an ad hoc strengthening foreign to the synthetic theory; it isolates one incidence consequence needed to make the Euclid-faithful construction total over all configurations.

C.14 Checkpoint XI.12

At XI.12 the Borsuk–Szmielew comparison is now exact for both Euclidean perpendicular-construction problems: the off-plane case of Theorem 89 subsumes XI.11, and the on-plane case subsumes XI.12.

The source therefore provides a precise future strengthening target for the formal library: a neutral existence-and-uniqueness theorem for a perpendicular through an arbitrary point. No new axiom should be added to obtain such a theorem; its reconstruction would belong to the existing synthetic incidence/order/congruence architecture.

No continuity assumption is suggested by this source comparison.

C.15 Proposition XI.13 and the uniqueness part of Theorem 89

Theorem 89, already identified at the XI.11–XI.12 checkpoints, states that through a given point there exists exactly one line perpendicular to a given plane. Proposition XI.13 corresponds directly to the uniqueness part of this result when the given point lies in the plane.

Thus the relationship among the three Euclidean propositions and Borsuk–Szmielew Theorem 89 is

Euclid	Theorem 89
XI.11	existence, point outside the plane
XI.12	existence, point in the plane
XI.13	uniqueness of the perpendicular carrier

Borsuk–Szmielew formulate the theorem for an arbitrary given point, so their uniqueness statement is not restricted to the on-plane position of XI.13.

Theorem 89 belongs to Chapter II, Section 31, within the synthetic congruence development before the later continuity and Euclidean-axiom layers. This source placement agrees with the current Lean XI.13 signature, which uses spatial incidence/order/congruence but no `HilbertSpaceEuclidean` parameter.

The two proof architectures are not identical. The current Lean XI.13 follows Euclid’s auxiliary-plane geometry and closes the contradiction through XI.4. Borsuk–Szmielew package existence and uniqueness as one structural perpendicular-to-plane theorem. The source comparison therefore identifies an exact theorem-level counterpart without forcing a line-by-line proof correspondence.

C.16 Checkpoint XI.13

At XI.13 the comparison with Borsuk–Szmielew completes the three-part Euclidean sequence XI.11–XI.13. Theorem 89 subsumes both existence cases and the uniqueness conclusion.

The source also confirms the axiomatic placement exposed by the final Lean theorem: uniqueness of the perpendicular carrier is available below continuity and below the Euclidean parallel axiom. No new axiom should be introduced for XI.13.

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